

9

Field management

This chapter describes the field management capabilities of Eclipse and covers the following:

- [Group production rules](#)
- [Gas lift optimization](#)
- [Gas field operations model](#)
- [Reservoir coupling](#)
- [Network option](#)
- [Gas calorific value controls](#)

Group production rules

x	Eclipse 100
	Eclipse 300

This chapter gives details of the implementation of the [PRORDER](#) keyword actions, which are performed by Eclipse 100 whenever a group fails to meet its target production rate set in keyword [GCONPROD](#). This feature is currently not available in Eclipse 300.

The available actions are:

1. Drill new wells from the drilling queue
2. Open new connections on AUTO in existing wells
3. Reduce each well's THP limit to a specified lower value
4. Change VFP table numbers (for example to model a retubing operation)
5. Change well ALQ values (that is apply a fixed amount of artificial lift)
6. Switch on, or increase power of, compressors linked to the group. (This option is only available if the Extended Network model is in use - see "[The extended network model](#)").

The order in which the actions are performed is dictated by the order in which their respective mnemonics are entered in record 1 of the [PRORDER](#) keyword. Whenever a group fails to meet its target production rate set in keyword [GCONPROD](#), Eclipse first looks for a well on which to perform the first action. If no suitable well can be found, or if rig availability constraints prevent this action from being performed at that time, Eclipse looks for a well on which to perform the second action in the list. If it cannot perform the second action on a well, it looks for a well on which to perform the third action, and so on. The implementation of each of the five actions is described in detail below.

Drilling new wells

A drilling queue must be set up either with keyword [QDRILL](#) (for a sequential drilling queue) or keyword [WDRILPRI](#) (for a prioritized drilling queue), containing production wells declared as [SHUT](#) in keyword [WCONPROD](#). In a sequential drilling queue, wells are considered for opening in the sequence in which they were placed in the queue. In a prioritized drilling queue, wells are considered for opening in decreasing order of their drilling priority. Well drilling priorities may be set manually in keyword [WDRILPRI](#) or calculated from their current potentials according to a function defined with keyword [DRILPRI](#). Restrictions on the rate at which wells can be drilled may be imposed by declaring the time taken to drill each well (keyword [WDRILTIM](#) item 2) and defining drilling rigs (keyword [GRUPRIG](#)). When a group cannot achieve its target rate of the required phase, Eclipse scans the drilling queue for a suitable well to open. A well is rejected if:

- it is not subordinate to the group that is unable to achieve its production target
- it is subordinate to an intermediate level group that is already operating at its target or limiting rate for the required phase, and so cannot increase its flow further
- it is subordinate to a prioritization group (keyword [GCONPRI](#)). The drilling queue logic is not activated under prioritization group control
- the well takes a non-zero time to drill (keyword [WDRILTIM](#) item 2) and all drilling rigs available to its group (keyword [GRUPRIG](#)) are already occupied for the whole of the current timestep
- its group, or any superior group in the hierarchy, already has its maximum allowed number of open wells (keyword [GECON](#) item 9)

- the oil saturation in all its completed grid blocks is below a specified minimum value (keyword [WELSOMIN](#))
- the well has a connection within a grid block that also contains a connection belonging to another open well, and keyword [WDRILRES](#) has been entered to prevent more than one well being completed in the same block.

The first producer in the drilling queue that survives all the above tests is drilled.

Opening new connections (re-perforating)

Well connections may be placed on “automatic opening”, by specifying [AUTO](#) in item 6 of keyword [COMPDAT](#) (or in keyword [WELOPEN](#) or [COMPDATL](#) as appropriate). Wells will be worked over in turn to open these connections. Restrictions on the rate at which wells can be worked over may be imposed by declaring the time taken to perform a workover (keyword [WORKLIM](#)) and defining workover rigs (keyword [GRUPRIG](#)). When a group cannot achieve its target rate of the required phase, Eclipse examines each open production well in turn, rejecting the well if:

- it has no connections in its [AUTO](#) opening queue
- it is already operating at its own target or limiting rate for the required phase, and so cannot increase its flow further
- it is not subordinate to the group that is unable to achieve its production target
- it is subordinate to an intermediate level group that is already operating at its target or limiting rate for the required phase, and so cannot increase its flow further
- workovers take a non-zero time (keyword [WORKLIM](#)) and all workover rigs available to its group (keyword [GRUPRIG](#)) are already occupied for the whole of the current timestep.

For each well that survives all these tests, Eclipse determines $\sum T_{wj} M_{pj}$, the sum of the connection factors multiplied by the mobility of the required phase over all the connections in its [AUTO](#) queue. Eclipse then selects the well having the largest value of this sum, and opens its next connection in its [AUTO](#) queue. Within each well, connections on [AUTO](#) are opened in the order in which they are first declared with the keyword [COMPDAT](#).

This scheme allows you to dictate the order in which the new connections are opened in any given well (for example from the bottom up). However, Eclipse decides for itself which well to re-perforate, on the basis of the combined productivity of the connections remaining on [AUTO](#). Note that the procedure does not necessarily work over the 'best' connection each time. Consider a well whose first connection on [AUTO](#) is in a layer with poor mobility but which has additional connections on [AUTO](#) in highly productive layers. The procedure may choose this well on the strength of its productive [AUTO](#) connections, but opens the unproductive connection first. Thus the unproductive connection does not 'block' the opening of the productive ones further down the queue. In fact, opening the unproductive connection may not be sufficient to achieve the group's rate target, and another connection may have to be opened also. The time taken to perform a workover is assumed to be independent of the number of connections opened. Thus, if a rig has been allocated to the well to open a connection, opening a second connection in the well during the same timestep will not take up any additional rig time.

There are two additional options that relate to the opening of connections on [AUTO](#). Firstly, connections can be lumped together into 'completions' (see keyword [COMPLUMP](#)). During automatic workovers, all connections belonging to the same completion are opened or closed together. When a well has been selected for re-perforating, its completions are examined in increasing order of their completion number, and the first one found having connections on [AUTO](#) will have all its connections opened together. Secondly, a minimum oil saturation can be specified with keyword [WELSOMIN](#). Any connection or

completion on AUTO with a grid block oil saturation below this minimum value is not opened. In completions, the connection factor weighted average of the grid block saturations is used.

Note that a connection on AUTO is also opened automatically when another connection in the same well is closed in a workover at the end of the timestep, which could be triggered for example by the violation of an economic limit (keywords [WECON](#), [GECON](#)). Also, if the production rate of a well falls below its minimum economic limit (keyword [WECON](#) items 2 and 3), its next connection on AUTO is opened. If it has no connections remaining on AUTO, the well is shut or stopped. These actions result in AUTO connections being opened according to the behavior of the wells individually rather than as a group, and takes place irrespective of whether or not group targets are being met.

Reducing the THP limit

Each production well can be given a secondary THP limit in item 8 of keyword [WLIFT](#) (which should be lower than the limit currently specified in [WCONPROD](#) or [WELTARG](#)). A decrement value may also be set in item 11 of [WLIFT](#), in which case Eclipse reduces the THP to its secondary limit in stages, a decrement at a time. When a group cannot achieve its target rate of the required phase, Eclipse examines each open production well in turn, rejecting the well if:

- it has no secondary THP limit, or the THP has already been reduced to its secondary limit
- it is currently operating under a control mode other than THP control
- it is not subordinate to the group that is unable to achieve its production target
- it is subordinate to an intermediate level group that is already operating at its target or limiting rate for the required phase, and so cannot increase its flow further.

For each well that survives all these tests, Eclipse obtains an estimate of its possible increase in production rate resulting from a reduction in THP by its next decrement (or to its secondary limit if no decrements are specified), using the current values of the phase mobilities at its connections. The well with the highest estimated increase multiplied by its efficiency factor will have its THP reduced by the next decrement. If no decrements are specified for the well, or if its next decrement would cross the secondary THP limit, the THP is reduced to its secondary limit.

Thus the wells have their THPs reduced in decreasing order of the resulting increase in production rate. The THP switching process is treated as instantaneous, and there is no limit on the rate at which wells can be switched.

Secondary THP limits should not be set for wells whose group is part of a network (see "[Network option](#)"), as the wells' THP limits are set by their group's nodal pressure in the network. THP switching will not be performed on any such wells.

Note that wells also switch to their secondary THP limits if their production rate falls below a specified value (keyword [WLIFT](#) item 2), or if their water cut or gas-liquid ratio exceeds a specified limit (keyword [WLIFT](#) items 7 and 9), or if the well dies under THP control. This triggers the switch according to the behavior of the individual wells, irrespective of whether or not group targets are being met. If you wish the secondary THP values to be applied only using the group production rules to help meet a group target, or if the well dies under THP control, then you must give the rate, water cut and GLR limits in items 2, 7 and 9 of keyword [WLIFT](#) a default, zero or negative value.

Changing VFP table numbers (retubing)

Each production well can be given a secondary VFP table number in item 4 of keyword `WLIFT` (which should be a table with lower tubing pressure loss). When a group cannot achieve its target rate of the required phase, Eclipse examines each open production well in turn, rejecting the well if:

- it has no secondary VFP table number, or it has already switched to its secondary table
- it is currently operating under a control mode other than THP control
- it is not subordinate to the group that is unable to achieve its production target
- it is subordinate to an intermediate level group that is already operating at its target or limiting rate for the required phase, and so cannot increase its flow further.

For each well that survives all these tests, Eclipse obtains an estimate of the possible increase in production rate resulting from the switch to its new VFP table, using the current values of the phase mobilities at its connections. The well with the highest estimated increase multiplied by its efficiency factor is switched to its secondary VFP table.

Thus the wells are switched to their secondary VFP tables in decreasing order of the resulting increase in production rate. The switching process is treated as instantaneous, and there is no limit on the rate at which wells can be switched.

Note that wells can also be made to switch to their secondary VFP tables if their production rate falls below a specified value (keyword `WLIFT` item 2), or if their water cut or gas-liquid ratio exceeds a specified limit (keyword `WLIFT` items 7 and 9), or if the well dies under THP control. This triggers the switch according to the behavior of the individual wells, irrespective of whether or not group targets are being met. If you wish the secondary VFP tables to be applied only through the group production rules to help meet a group target, or if the well dies under THP control, then you must give the rate, water cut and GLR limits in items 2, 7 and 9 of keyword `WLIFT` a default, zero or negative value.

Changing well ALQ values (applying lift)

Each production well can be given a secondary ALQ value in item 5 of keyword `WLIFT`, which represents the application of a fixed quantity of artificial lift (for example, a fixed rate of lift gas injection). An increment value may also be set in item 10 of `WLIFT`, in which case Eclipse increases the ALQ to its secondary value in stages, an increment at a time. A new well efficiency factor can also be defined in item 6 of keyword `WLIFT` if required, which comes into effect at the same time as the secondary ALQ value (or the first ALQ increment) is applied. There is an option to restrict the total amount of lift applied within a particular group, and the number of wells on lift within a group (keyword `GLIFTLIM`). When a group cannot achieve its target rate of the required phase, Eclipse examines each open production well in turn, rejecting the well if:

- it has no secondary ALQ value set in keyword `WLIFT`, or it has already switched fully to this ALQ value
- it is currently operating under a control mode other than THP control
- it is not subordinate to the group that is unable to achieve its production target
- it is subordinate to an intermediate level group that is already operating at its target or limiting rate for the required phase, and so cannot increase its flow further
- its group, or any superior group, would exceed its limit on total lift or the number of wells on lift, set in keyword `GLIFTLIM`.

For each well that survives all these tests, Eclipse obtains an estimate of the possible increase in production rate resulting from an increase in ALQ by its next increment (or to its secondary value if no increments are specified), using the current values of the phase mobilities at its connections. The effects of a new efficiency factor, if specified, are also taken into account. The well with the highest estimated increase will have its ALQ value increased by the next increment. If no increments are specified for the well, or if its next increment would cross the secondary ALQ value, the ALQ will be increased to its secondary value.

Thus the wells are given lift in decreasing order of the resulting increase in production rate. The lift switching process is treated as instantaneous, and there is no limit on the rate at which wells can be switched.

If the Network option is in use, the estimate of production increase will not take account of the increased pressure losses in the network; if these are important, you should use the Gas Lift Optimization facility instead of `WLIFT` and `GLIFTLIM` (see "[Gas lift optimization](#)"). This offers a much more sophisticated means of optimizing the distribution of lift gas to the individual wells. The facility automatically allocates lift gas to the wells until group production targets are met or other operating limits are reached. Gas lift allocations are calculated at specified intervals, or during the first `NUPCOL` Newton iterations of each timestep, depending on the instructions in keyword `LIFTOPT` (regardless of the presence or absence of the `LIFT` mnemonic in `PRORDER`). By default, all the requested production rule actions will be attempted **before** gas lift is allocated. However, if `LIFT` is entered in the `PRORDER` keyword, gas lift allocation takes precedence over any production rule actions whose mnemonics are entered after `LIFT`. Thus, for example, it is possible to specify that drilling new wells should only be attempted if group targets cannot be met by applying gas lift. Essentially, all production rule actions entered after `LIFT` are only performed immediately after a gas lift optimization calculation, and not at any other time or Newton iteration, if a group production target still cannot be met. In this case it is best to configure `LIFTOPT` so that gas lift is optimized during each of the first `NUPCOL` iterations of every timestep.

Note that wells can also be made to switch on lift if their production rate falls below a specified value (keyword `WLIFT` item 2), or if their water cut or gas-liquid ratio exceeds a specified limit (keyword `WLIFT` items 7 and 9), or if the well dies under THP control. This triggers the switch according to the behavior of the individual wells, irrespective of whether or not group targets are being met. If you wish lift to be applied only through the group production rules to help meet a group target, or if the well dies under THP control, then you must give the rate, water cut and GLR limits in items 2, 7 and 9 of keyword `WLIFT` a default, zero or negative value.

Activating compressors

This option is only available if the extended network model is in use (see "[The extended network model](#)"). The extended network model includes a special type of compressor that is turned on automatically when a nominated group fails to meet its production rate target. These compressors are defined with the keyword `NETCOMPA`. Each compressor is located in a specified branch of the network, and its operation is modeled by changing the ALQ value used to look up the VFP table associated with the branch. Compressors may be single-level or multilevel. Single-level compressors switch directly to the new ALQ value when they are turned on. Multilevel compressors change their ALQ value in a user-defined number of increments until they are fully on at their maximum ALQ value.

It is possible for a group to have more than one compressor responding to it. The compressors can be located anywhere in the network, but of course they should be positioned to boost the production of their nominated group. If two or more compressors respond to the same group, the order in which they are activated is controlled by their 'switching sequence number' (item 12 in keyword `NETCOMPA`).

Compressors responding to the same group are activated in increasing order of their sequence number. If more compression is still required when the first compressor is fully on, the next compressor that responds to the group will be turned on (incrementally, if it is a multistage compressor).

When a group cannot achieve its target rate, Eclipse scans the network for a compressor that responds to the group. Compressors that are already fully on are ignored. If more than one compressor responds to the group, the compressor with the lowest switching sequence number is selected first. If it is a single-level compressor, it is turned fully on. If it is a multilevel compressor, its ALQ value is increased one increment at a time until either the group's rate target is met or the compressor is fully on.

The effect of closing wells during drilling and workovers

The rate at which new wells can be drilled and existing wells worked over can be controlled by specifying non-zero durations for these processes (see keywords [WDRILTIM](#) and [WORKLIM](#)) and defining drilling and workover rigs (see keyword [GRUPRIG](#)). A drilling or workover process will occupy the appropriate rig for the specified time. A well cannot be drilled (or worked over) in a given timestep if all available drilling (or workover) rigs are occupied for the whole timestep.

By default, a well starts flowing at the beginning of the timestep in which its drilling operation starts, and it continues flowing while it is being worked over. However, there is an option to treat wells as shut for the duration of their drilling and workover processes (see keyword [WDRILTIM](#) item 3). This is achieved by giving the wells temporary reduced efficiency factors, which reflect the proportion of the timestep in which they are unable to flow. But while this option gives more realistic behavior, it can cause a problem in the operation of the group production rules.

When a well is treated as shut for the duration of its drilling or workover process, the full benefit of this action is not realized until the timestep after the process has been completed. In fact the immediate effect of a 'REPERF' action may be a temporary reduction in production rate, if the workover takes a non-zero time and the well is closed for its duration. Consequently, the immediate effect of a 'DRILL' or 'REPERF' action may not be sufficient to enable the group to reach its target rate, and Eclipse attempts more actions from the production rule list, even if these would not have been necessary once the drilling or workover process is complete.

If you wish to treat wells as shut while being drilled or worked over, it is advisable to set the item in record 2 of the [PRORDER](#) keyword to 'NO'. This prevents the performance of any further production rule actions for the same group following a 'DRILL' or 'REPERF' action until the benefits of this action are fully realized.

Using production rules with the network option

When the Network option is in use (see "[Network option](#)"), by default all actions except for compressor switching are performed in any network balancing iteration when a flow target is not met. Thus it is possible, for example, that a well may be opened from the drilling queue prematurely because an intermediate network balancing iteration has high nodal pressures and consequentially low well productivity. The exception is the compressor switching action, which is only performed if more flow is required once the network has converged.

It is possible to instruct Eclipse to prevent any of the Production Rule actions from being performed before the network balancing iterations have converged, by using the [WAITBAL](#) keyword. When this option is selected, all production rule actions are suppressed while the network is being balanced. Once the network balancing iterations have converged, if more flow is still required to meet a group flow target an appropriate Production Rule action is performed and the network has to be balanced again. Using this option therefore requires more network balancing iterations in total.

Gas lift optimization

x	Eclipse 100
x	Eclipse 300

The gas lift optimization facility is a special extension to both Eclipse 100 and Eclipse 300. It determines how much lift gas to allocate to each well in order to meet well, group or field production targets. If production targets cannot be met, it determines how to make the best use of the existing lift gas resources by allocating lift gas preferentially to the wells that can make the best use of it.

The gas lift optimization facility can solve the following problems:

- Optimizing gas lift to individual wells

How much lift gas does each well need in order to achieve its individual production target? If a well cannot achieve its production target, what is the optimum amount of lift gas to allocate to it?

- Optimizing gas lift within a group of wells

How much lift gas does a group of wells need in order to achieve its group production target? If there is only a limited amount of lift gas available to the group, how should it best be distributed among the individual wells?

Eclipse 100

- Optimizing gas lift within a simple network

If the wells flow into a subsea manifold connected to the surface by a flowline, how do the pressure losses in the flowline affect the optimum distribution of lift gas to the wells?

Eclipse optimizes the distribution of lift gas at each timestep, or after user-defined intervals of time. The optimization is performed by dividing the total supply of lift gas into discrete user-defined increments, and allocating increments preferentially to the wells that can gain the most benefit from them.

The effects of gas lift are modeled by VFP tables (keyword [VFPPROD](#)). The tables must be prepared in advance with a suitable range of lift gas injection rates. The lift gas injection rate is equated with the Artificial Lift Quantity (ALQ value) in the tables. If the tables are generated by the VFP *i* program, it is essential to select the definition GRAT in the ALQ keyword in the input data file for VFP *i*. However, the VFP *i* program is also able to transform tables in which the ALQ is either IGLR (injected gas liquid ratio) or TGLR (total gas liquid ratio) into a table where the ALQ has the required definition of GRAT (lift gas injection rate). See the "VFPI User Guide" for more details.

Optimizing gas lift to individual wells

For a well under THP control, as the supply of lift gas to it is gradually increased, the production rate at first increases due to the reduced density of the mixture in the tubing. However, as the lift gas supply is increased further, friction pressure losses in the tubing become more important, and the production rate peaks and then starts to decrease (see figure 9.1). For an individual well with no constraints other than a THP limit, with an unlimited free supply of lift gas, the optimum lift gas injection rate is the value at the peak (point A).

In reality, of course, lift gas injection is never free. Compression costs can be expressed as a cost per unit rate of lift gas injection (for example, dollars/day per Mscf/day). This must be balanced against the value of the extra amount of oil produced. Thus there is a "minimum economic gradient" of oil production rate versus lift gas injection rate, at which the value of the extra amount of oil produced by a small increase in the lift gas injection rate is equal to the cost of supplying the extra amount of lift gas. The optimum lift gas injection rate is then somewhat lower than the peak value, at the point on the curve where its gradient equals the minimum economic gradient (point B).

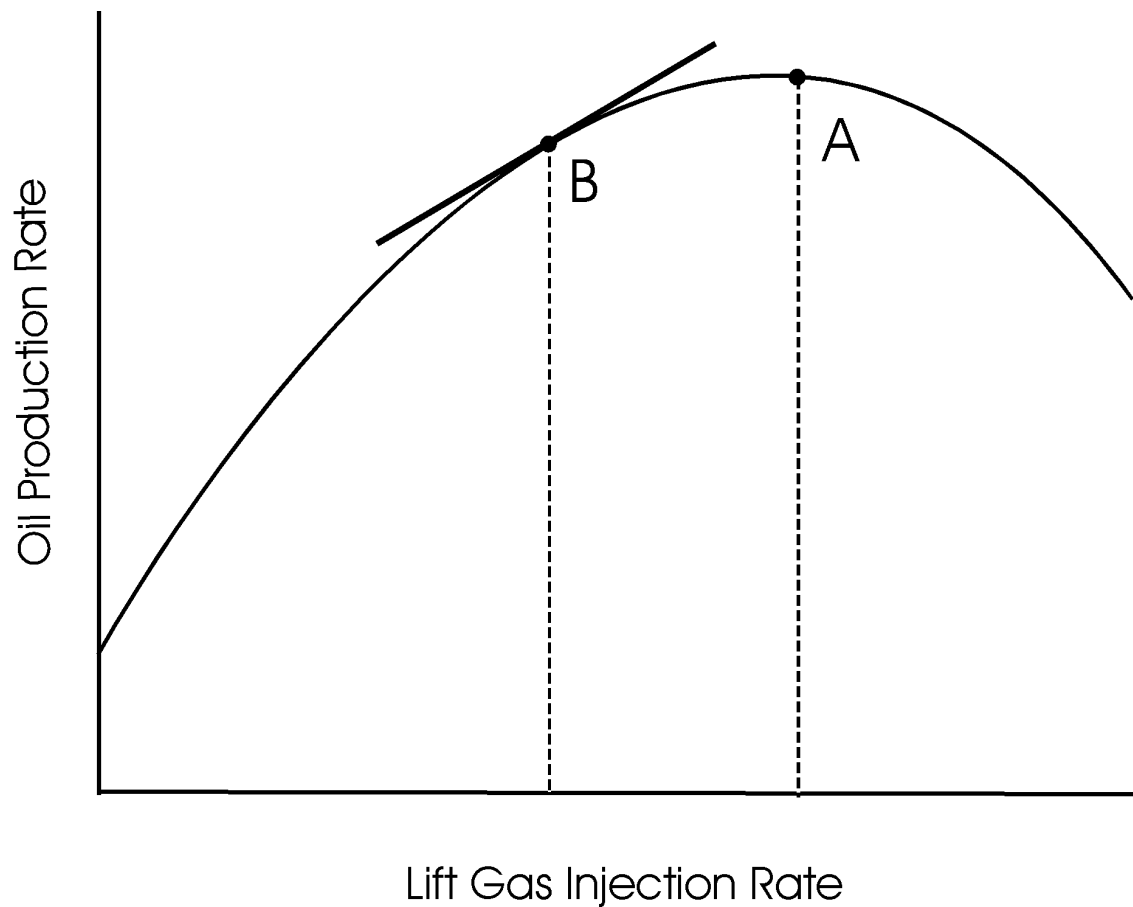


Figure 9.1. Optimum Lift Gas Injection Rate

For flexibility, each well can be assigned a “weighting factor” (which defaults to 1.0 if not specified). The local gradient of the oil production rate versus lift gas injection rate curve is multiplied by this weighting factor before being compared with the minimum economic gradient.

If the well is subject to additional constraints, the optimum lift gas injection rate may be lower than that indicated by the minimum economic gradient. For example, if the well is subject to an oil rate target that is achievable with gas lift, it is pointless increasing the lift gas supply beyond the value that maintains the rate target.

To calculate the optimum lift gas injection rate, it is first necessary to divide the lift gas supply into discrete increments of uniform size. This is accomplished using the keyword `LIFTOPT`. A typical increment size could be a twentieth of the maximum allowed lift gas injection rate for each well. A smaller increment size would determine the optimum lift gas injection rate to a finer resolution, but at the expense of a longer computing time. (In general, however, the computing time for gas lift optimization is only an important consideration when the Network option is being used.)

Each time Eclipse performs a gas lift optimization, it must decide whether lift gas increments must be added to, or subtracted from, each well. Taking the current lift gas injection rate of a well, Eclipse calculates what the well would produce if the lift gas injection rate were increased by one increment, and also what it would produce if the rate were reduced by one increment. The production rates are adjusted if necessary to obey any rate or BHP limits that the well may be subject to. From this information, Eclipse calculates the well’s “weighted incremental gradient” and its “weighted decremental gradient” according to the formula:

$$Gradient = \left\{ \frac{\alpha_w \cdot \Delta Q_o}{(\Delta Q_{gli} + \alpha_g \cdot \Delta Q_g)} \right\} \quad \text{Eq. 9.1}$$

where:

α_w is the weighting factor for the preferential allocation of lift gas.

α_g is the gas production rate weighting factor.

ΔQ_o is the increment or decrement in oil production rate.

ΔQ_g is the increment or decrement in gas production rate.

ΔQ_{gli} is the gas lift increment or decrement.

By default, the gas production rate weighting factor α_g is defined to be 0.0 such that, in general, the gradients are calculated according to the reduced formula:

$$Gradient = \left\{ \frac{\alpha_w \cdot \Delta Q_o}{\Delta Q_{gli}} \right\} \quad \text{Eq. 9.2}$$

However, a positive value of α_g may be employed to discourage the allocation of lift gas to wells which are producing more gas. The gas rate term in the denominator is subject to the constraint $\alpha_g \cdot \Delta Q_g \geq 0.0$. If $\alpha_g \cdot \Delta Q_g \gg \Delta Q_{gli}$ the gradient becomes essentially an inverse GOR term. The weighting factors α_w and α_g are specified via items 4 and 6 respectively of the **WLIFTOPT** keyword.

These gradients are updated each time an increment is added or removed. Eclipse adds lift gas increments to the well as long as its weighted incremental gradient exceeds the minimum economic gradient, subject to any limits on the lift gas supply.

Optimizing gas lift within a group

If a group has any production rate constraints, and/or a limit on its total rate of lift gas supply, Eclipse allocates lift gas preferentially to the wells that gain the most benefit from it. Lift gas increments are allocated in turn to the well that currently has the largest weighted incremental gradient. The procedure takes account of any limits on the group production rate or lift gas supply applied to any level of group, including the FIELD level group.

The optimization procedure loops over all the groups on each level, firstly updating the distribution of the lift gas increments currently allocated to the group, then removing any surplus increments from the group. The currently allocated increments are redistributed by comparing the largest weighted incremental gradient and the smallest weighted decremental gradient of the subordinate wells. Consider, for example, that the largest weighted incremental gradient belongs to well W1, and exceeds the smallest weighted decremental gradient, which belongs to well W2. It would be more profitable to subtract an increment of lift gas from well W2 and allocate it to well W1. After the exchange, the incremental and decremental gradients of wells W1 and W2 are recalculated. The exchange of increments continues until no weighted incremental gradient exceeds a weighted decremental gradient, at which point any surplus lift gas increments are removed from the group. The group has surplus lift gas if it exceeds any production rate limits or a lift gas supply limit, or contains any wells that have a weighted decremental gradient less than the minimum economic gradient. Lift gas increments are removed in turn from the well that currently has the smallest weighted decremental gradient, until there is no surplus lift gas in the group.

The final stage of the optimization process involves looking to see whether any lift gas increments could profitably be added anywhere in the field. Eclipse looks for the well with the largest weighted incremental gradient that does not exceed its own lift gas supply limit, and does not belong to a group that exceeds a rate limit or a lift gas supply limit. (Note that this means that, in general, no further lift gas will be allocated once a field rate target is achieved. However, selected wells may be allocated further lift gas in order to continue optimizing the oil rate if item 7 of the [WLIFTOPT](#) keyword is set to YES, for wells which belong to groups which are under gas target control.) If the well's weighted incremental gradient exceeds the minimum economic gradient, it receives another increment of lift gas. The process continues until no well can profitably receive any more lift gas increments.

The whole optimization process is efficient because it starts from the previous optimized distribution of lift gas. If the previous lift gas distribution is still the optimum, then no lift gas increments need be added or subtracted. This is more efficient than building the lift gas distribution up from zero at each timestep.

If wells have efficiency factors to take account of regular downtime (see keyword [WEFAC](#)), their lift gas injection rates are multiplied by their efficiency factors when they are added to the group lift gas supply rate. This is consistent with the summation of flow rates for wells with downtime, and preserves the ratio of production rate to lift gas injection rate.

If groups have efficiency factors to model synchronized downtime of their subordinate wells (see keyword [GEFAC](#)), their lift gas injection rates are multiplied by their efficiency factors when they are added to the lift gas supply rate of the parent group.

Optimizing gas lift within a network

Eclipse 100

The problem becomes much more complicated when the group belongs to a production network, with a flowline to a higher level group (see "[Network option](#)"). Here, the wells' THP limits increase as the group's production rate increases. Adding a small amount of lift gas to a particular well may increase its flow rate, but the consequent increase in the THP of all the other wells in the group will reduce the overall benefit of the extra lift gas. If the lift gas is required to flow through the flowline together with the produced fluids, the extra friction pressure drop it creates could more than cancel out the extra production from the well. (Lift gas can be included in the network branch flows by setting item 6 in keyword [GRUPNET](#) or item 4 in keyword [NODEPROP](#).)

The optimization procedure in a network is essentially the same as that described in the previous section, except that whenever a lift gas increment is added or subtracted, the incremental and decremental gradients must be recalculated for all the wells in the field. This is because a change in a well's production rate will affect the THP limits of the other wells in the network. Moreover, each time an incremental or decremental gradient is calculated, the whole network must be rebalanced with the appropriate lift gas rate in order to work out the consequent change in the field's oil production rate. This makes the computation time increase sharply according to the square of the number of wells in the field multiplied by the number of lift gas increments added and subtracted.

gas lift optimization should therefore be used with care in conjunction with the Network option. It will be slow if applied to full-field studies with many wells. It is best applied to simple network systems, such as optimizing gas lift to a group of wells flowing into a common subsea manifold connected to the surface by a flowline. The choice of increment size is important also. A small increment size increases the required computation time because more increments need to be added and subtracted. Moreover, a small increment size may produce inaccurate estimates of the incremental and decremental gradients if the magnitude of the resulting change in the flow rate is comparable with the specified tolerance in the network balancing iterations. It is recommended that the [NETBALAN](#) keyword is used to decrease the network convergence tolerance by a factor of 10 (item 2), and to increase the number of iterations in the balancing calculation (item 3).

Treatment of wells that can only flow under gas lift

Later in the life of the field, it may happen that some wells can only flow under gas lift. If Eclipse decides it is not economic to allocate lift gas to such a well, it dies. Once a well has died, it remains shut or stopped and is not considered any longer as a candidate for gas lift. However, to retain the well as a candidate for gas lift, if required, with the possibility of allocating lift gas to it at a later time if it becomes economical to do so, use the **WTEST** keyword to revive the well periodically for testing. When the well is revived, the optimization process considers it along with the other wells for allocation of lift gas increments. If it is not economic to allocate any lift gas to the well it will die again but, if it is economic to allocate lift gas to the well, it will resume production.

To give the well a fair chance of revival, Eclipse works out the number of lift gas increments that would give the well the largest ratio of oil production rate to lift gas injection rate. For example, the well may require three increments simply to initiate flow whereas five increments would provide the best ratio of oil production to lift gas injection (see figure 9.2). The well's incremental gradient would be set equal to this ratio (line O - A), to be compared with the incremental and decremental gradients of the other wells and, if this gradient is large enough, the well is given five increments of lift gas to start it flowing (position A).

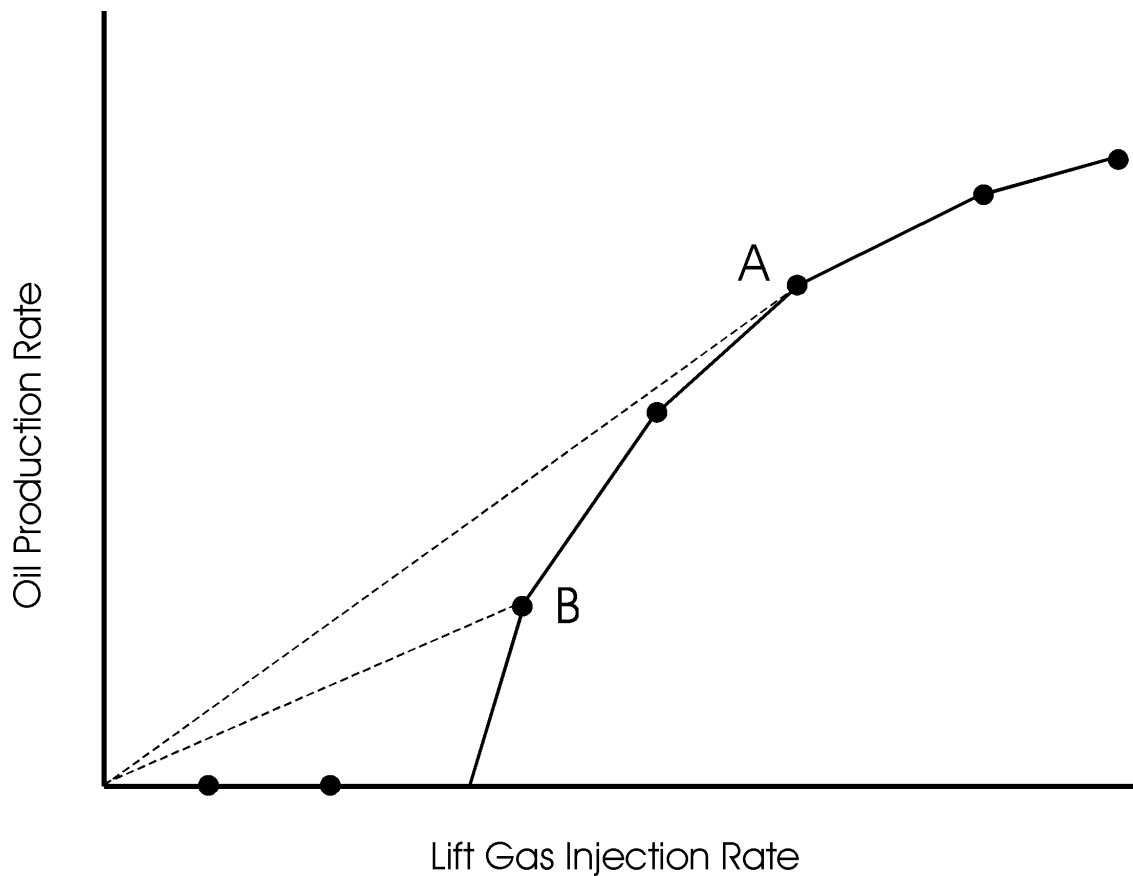


Figure 9.2. Incremental gradient calculation

Thereafter the well is treated as a normal well for optimization, with single increments being added or subtracted as necessary. However, if the well is only just able to flow at a particular rate of gas lift, that is, if the removal of another increment would cause the well to die, the decremental gradient is calculated in a special way. In the current example the well can flow with three lift gas increments, but not with two. If one increment is subtracted the well does not flow but it has two remaining increments allocated which are wasted. Hence if the lift gas rate is decreased it is expedient to remove all three increments at once. The

well's decremental gradient at that point is calculated as its oil rate divided by three increments of lift gas (line O - B) and, if this gradient compares unfavorably with the other wells, all three increments are removed and the well dies.

Keeping wells alive with a minimum lift gas rate

Wells can be prevented from dying by ensuring they are always allocated at least enough lift gas to keep them flowing. This can be done by setting a minimum lift gas injection rate for each well in keyword `WLIFTOPT`. If a positive value is specified, the well is allocated at least that amount of lift gas, unless it is unable to flow with that rate of lift gas injection or unless it can already meet one of its own rate limits before receiving its minimum lift gas rate. Alternatively a negative number can be specified for the minimum lift gas rate. In this case, Eclipse allocates at least enough lift gas to enable the well to flow, provided that it can flow at a lift gas injection rate within the maximum value specified for the well. If there is not enough lift gas available to satisfy all the wells' minimum requirements, the wells are allocated their minimum requirements in decreasing order of their weighting factor.

If a well with a minimum lift gas requirement belongs to a group (including the FIELD level group) having a production rate target that can be achieved without any lift gas, by default no lift gas is allocated to the well, regardless of any minimum requirement. However, if the well is given a weighting factor greater than 1.0, the well will receive its minimum lift gas requirement even if the group could achieve its production rate target without any gas lift. Thus, to allocate lift gas to keep a well flowing, if required, regardless of whether lift gas is needed to achieve a group or field production rate target, assign a weighting factor greater than 1.0 (for example 1.01) together with a negative value for the minimum lift gas rate.

Note: There is a possible conflict when an economic limit is applied to a well (for example using the `WECON` keyword) that has been instructed to be kept flowing in the manner described above using gas lift optimization. When the well's flow becomes too low, it may be shut and hence will not be available to take part in the allocation of gas lift. Take care to avoid conflicting economic limits if a well is to be kept alive by means of gas lift optimization.

Using the gas lift optimization facility

The VFP tables (keyword `VFPPROD`) must first be prepared for the wells (and network branches, if required). The effects of lift gas injection are handled by the Artificial Lift Quantity (ALQ value), which is equated with the lift gas injection rate. If the tables are generated by the VFP *i* program, it is essential to select the definition `GRAT` (lift gas injection rate) for the artificial lift quantity. However, the VFP *i* program is also able to transform tables in which the ALQ is either `IGLR` (injected gas liquid ratio) or `TGLR` (total gas liquid ratio) into a table where the ALQ has the required definition of `GRAT`. See the "VFPi User Guide" for more details.

Gas lift effects are modeled by interpolating the VFP table with an ALQ value equal to the current lift gas injection rate. The ALQ values in each table must span the expected range of lift gas injection rates for the well, as extrapolation of the tables can give unrealistic behavior.

In Eclipse 100, lift gas injection rates lying in between tabulated ALQ values are handled by linear interpolation, by default, like the other parameters in the table. However, there is an option to apply cubic spline interpolation to the ALQ values. This has the advantage of making the gradient of well performance with respect to the lift gas injection rate vary smoothly across the range of ALQ values. With linear interpolation, the gradient can change abruptly at the tabulated ALQ points, which may bias the optimized lift gas distribution towards these values. Spline interpolation is enabled with the keyword `VFPTABL`.

Eclipse 100

LIFTOPT keyword

The gas lift optimization facility is activated with the keyword **LIFTOPT**, where the lift gas increment size and the minimum economic gradient are specified. The keyword can also be used to adjust the frequency with which the gas lift is optimized. The default is every timestep. In fact, when there is no network, the optimization is generally fast enough to be performed in each of the first NUPCOL iterations of the timestep, so that production targets and limits are observed more accurately. The value of NUPCOL is set using keyword **NUPCOL** or **GCONTOL** (see these keywords for a discussion of the use of NUPCOL in group control).

WLIFTOPT keyword

The keyword **WLIFTOPT** must be used to tell Eclipse which wells should have their lift gas rates calculated by optimization. It is possible to operate some wells on fixed lift gas rates, if required, while other wells have their lift gas rates optimized. The maximum lift gas injection rates for each well are also set here. To prevent extrapolation of the VFP tables, their maximum lift gas rates are defaulted to the highest ALQ value in their respective VFP tables, and they must not be given a higher value. The well weighting factors are also set here, and they default to 1.0. A minimum lift gas rate can be set for each well. If a negative value is specified, Eclipse allocates at least enough lift gas to keep the well flowing, if it would otherwise die without gas lift. For wells to receive their minimum lift gas allocation even if no gas lift is necessary to achieve a group or field production rate target, if required, set their weighting factors to a value greater than 1.0. A weighted gas production rate term may be included in the denominator of the gradient calculation by setting the gas rate weighting factor to a non-zero value and, if required, selected wells may be designated to continue receiving lift gas after a gas rate group target has been achieved in order to continue optimization of the oil rate.

GLIFTOPT keyword

The keyword **GLIFTOPT** can optionally be used to set limits on the lift gas supply and the total gas rate (produced gas + lift gas) for individual groups. Remember that the lift gas supply to a group equals the sum of the lift gas injection rates of its subordinate wells or groups multiplied by their efficiency factors.

Group production rate limits

Group production rate limits are set with either keyword **GCONPROD** or **GCONPRI** as usual, depending on which method of group control is required. (Note that any gas rate limits set with these keywords refer to gas produced from the reservoir, and do not include the lift gas.) Lift gas is allocated if a group or the field cannot reach its oil production rate target without it. Subject to any limits on the lift gas supply or total gas rate set in keyword **GLIFTOPT**, increments are added until the oil production rate target is just exceeded and then the group control method adjusts the well rates to meet the rate target. Increments are not added if they would cause a group rate limit for any other phase (water, gas, liquid) to be exceeded (unless increments are necessary to make up a well's minimum lift requirement and the well has a weighting factor greater than 1.0).

If gas lift is optimized in only the first iteration of a long timestep, the reservoir conditions may change sufficiently over the timestep such that rate targets and limits are no longer honored exactly. For example, a lift gas supply calculated to achieve the field's oil rate target at the beginning of the timestep may fall somewhat short of this by the end of the timestep. Conversely, a group's water rate limit honored in the first iteration may be exceeded by the final iteration of the timestep. If the action on exceeding this limit is to work over the worst offending well in the group, an unnecessary workover will be performed. It is therefore best to set the actions for rate limit violations in **GCONPROD** or **GCONPRI** to **RATE** or **NONE**. An action of **NONE** still requires the rate limit to be honored by the gas lift optimization process. However, the rate limits

are honored more exactly if optimizations are performed at each of the first `NUPCOL` iterations of every timestep; this is permissible provided that any network present is not balanced less frequently.

Use with group production rules

Eclipse 100

In Eclipse 100, gas lift optimization can be used in combination with the group production rules (see ["Group Production Rules"](#) and keyword `PRORDER`) to define a sequence of actions to be performed if a group fails to meet its production rate target set in keyword `GCONPROD`. These include drilling new wells, opening new connections in existing wells, reducing the THP limit or changing the well's VFP table number. By default, all the requested production rule actions are attempted **before** gas lift is allocated. However, if the mnemonic `LIFT` is entered in the `PRORDER` keyword, gas lift allocation takes precedence over any production rule actions whose mnemonics are entered after `LIFT`. Thus, for example, it is possible to specify that drilling new wells should only be attempted if group targets cannot be met by applying gas lift. Essentially, all production rule actions entered after `LIFT` are only performed immediately after a gas lift optimization calculation and not at any other time or Newton iteration, if a group production target still cannot be met. In this case it is best to configure the keyword `LIFTOPT` so that gas lift is optimized during each of the first `NUPCOL` iterations of every timestep.

Use with the network option

Eclipse 100

Gas lift optimization can be used with standard or extended networks in Eclipse 100 (see ["Network Option"](#)). When there is a production network, gas lift can only be optimized when the network is being balanced. If the network is only balanced in the first iteration of the timestep, gas lift can only be optimized for conditions at the beginning of each timestep. However, if the network is balanced in each of the first `NUPCOL` Newton iterations of every timestep (by setting a negative balancing interval in keyword `NETBALAN`), then the gas lift distribution can be optimized with this frequency also. Note that the computation time taken by each optimization is much greater if a network is present and increases sharply with the square of the number of wells in the simulation. In order to reduce the computation time, it is recommended that item 2 (convergence tolerance) of the `NETBALAN` keyword is decreased by an order of magnitude and item 3 (maximum number of iterations) is increased.

When setting up the network with keyword `GRUPNET`, item 6 can be used to indicate whether or not lift gas should be included in the flow along each of the branches. Lift gas from the subordinate wells can be included in one of two ways:

- The `FLO` option adds the wells' lift gas rates, multiplied by their efficiency factors, to the gas flow rate along the branch. The branch's `ALQ` value in item 4 should be set to zero. It is important to ensure that the range of gas fraction (`GFR`) values in the branch's VFP table encompasses the anticipated range of operating lift gas rates and produced gas rates in order to avoid the need for extrapolation.
- Alternatively, the `ALQ` option automatically sets the `ALQ` value of the branch to the sum of the subordinate wells' lift gas rates, multiplied by their efficiency factors. When using this option, it is important to ensure that the range of `ALQ` values in the VFP table encompasses the anticipated operating range in order to avoid the need for extrapolation.

If the extended network option is used, the network is constructed with keywords `BRANPROP` and `NODEPROP` instead of `GRUPNET`. Set item 4 of keyword `NODEPROP` to `YES` to include lift gas flows in the branch pressure drop calculation. This is equivalent to the `FLO` option in item 6 of `GRUPNET`.

Output

The current lift gas injection rates that are applied to each well, group and the entire field are output as part of the production report at each reporting time. These rates can also be written to the `SUMMARY` file by

specifying the [SUMMARY](#) section keywords WGLIR, GGLIR and FGLIR (Well/Group/Field Gas Lift Injection Rate).

The incremental gradients of selected wells (that is, the increase in oil production rate achieved by allocating an additional increment of gas lift to the well, divided by the sum of increment size and a weighted gas production rate term) can also be written to the [SUMMARY](#) file by using the keyword WOGLR (Well Oil Gas Lift Ratio).

Restrictions

Eclipse 100

Do not use the keyword [GLIFTLIM](#) or the lift switching option in keyword [WLIFT](#) (item 5). These are meant for use with the Eclipse 100 automatic lift switching facility, which is not as comprehensive nor as flexible as the gas lift optimization option. The equivalent keywords for gas lift optimization are [GLIFTOPT](#) and [WLIFTOPT](#).

Eclipse 300

Gas Lift Optimization is not compatible with networks in Eclipse 300 (see "[Network option](#)"), and will be disabled if a standard or extended network is detected.

The flux boundary conditions facility should not be used with gas lift optimization, as no account will be taken of lift gas supplied to wells outside the flux boundary.

Summary of keywords

SUMMARY

Refer to [Well and group gas flows](#) in the *Eclipse Reference Manual*.

Note: In the presence of gas lift optimization, the produced gas reported (for example using the FGPR, GGPR and WGPR summary section keywords) does not include produced lift gas, only produced formation gas.

SCHEDULE

Restriction	Keyword	Description
	GLIFTOPT	Sets limits on group lift gas supply.
	LIFTOPT	Activates gas lift optimization and sets some governing parameters.
Eclipse 100	VFPPROD	Inputs VFP tables.
Eclipse 100	VFPTABL	Selects the method of interpolating the ALQ variable in VFP tables.
	WLIFTOPT	Sets gas lift optimization data for wells.

Gas field operations model

x	Eclipse 100
x	Eclipse 300

The gas field operations model is a special extension to both Eclipse 100 and Eclipse 300. It contains a set of facilities designed to model gas field production constraints and operations. The principal features in this set are:

Seasonality profile

The required rate of gas production is not constant throughout the year, but varies from month to month to reflect the annual pattern of demand. To generate this varying production rate, you enter an annual profile made up of monthly multipliers to the mean rate or DCQ (Daily Contracted Quantity). The field's gas production rate target for each month will be set equal to the DCQ multiplied by the month's profile factor.

Adjustment of DCQ to allow for swing

A typical gas sales contract requires that the field should be able to produce at a rate, which is a given multiple of the DCQ. The given multiple is the swing factor, which for example may have a value of 1.67. A suitable value of the DCQ must be calculated that is valid over the period of each contract, depending on the deliverability of the field over the period. The DCQ may be reviewed on an annual basis (contract years) or more frequently (contract quarters, for example). Eclipse can determine a suitable value of the DCQ for each contract period by solving the period in two passes. In the first pass, Eclipse tests whether the field can produce at the rate of $\text{DCQ} \times \text{Swing Factor}$, with the DCQ set at the current value. If the field cannot produce at this rate, Eclipse works out how much the DCQ should be reduced to make this target sustainable. If requested, Eclipse may perform a number of iterations of the first pass, with different DCQ values, to determine the highest DCQ that obeys the swing requirement over the period, to a specified accuracy. In the second pass, the field is produced at the rate of $\text{DCQ (reduced if necessary)} \times \text{profile factor}$, to model the actual production profile over the contract period. Note that, because each contract period must be simulated at least twice, use of the DCQ adjustment option requires a corresponding increase in CPU time.

Control and output of sales gas rate

Sales gas is what is left over after taking account of gas consumption at each group (known as gas fuel usage in Eclipse 300). For Eclipse 100, the sales gas rate can also include gas imported at a specified rate from outside the field. Eclipse normally applies the group or field gas rate targets to the **gross** production rate (that is the gas extracted from the formation). But when the gas field operation model is in use, the targets are applied to the sales gas rate (that is the gas sent to shore). The sales gas rates and totals are included in the production reports, and can be output to the SUMMARY file.

Estimation of delivery capacity

As well as modeling the actual performance of the field at the demanded rate, it is useful to have an idea of its delivery capacity. This is defined as the sales gas production rate that would ensue immediately after the removal of all group and field gas rate targets and limits, at the current grid block conditions. You can specify whether any group limits for other phases, and any rate limits on the wells themselves, should remain in operation. The delivery capacity is calculated at the start of each timestep, and can be output to the SUMMARY file.

Automatic compressor operation

When the gas field operations model is used in conjunction with the network option, gas compressors in the pipelines can be switched on and off automatically as required. The compressors may consume gas at a specified rate for power, which will be added to the gas consumption rate whenever the compressors are operating. The simplest option has two states: either "compression on" or "compression off". A more versatile option switches on a series of compressors in a predefined sequence, and allows each compressor to have up to five levels of compression.

Description of the facilities

Seasonality profile generation

To use this facility, you enter 12 profile factors, one for each month starting with January, using the keyword `SWINGFAC`. The simulation is then advanced one contract period at a time, using the keyword `GASYEAR` or `GASPERIO` instead of `TIME`, `TSTEP` or `DATES`. The contract period is assumed to start on the current date at which `GASYEAR` or `GASPERIO` is read. Eclipse flags an error if this is not the first day of a month. During the contract period, the monthly sales gas production rate target for the field is set equal to the DCQ multiplied by the month's profile factor. At the end of the contract period, Eclipse writes an output report and dump a `RESTART` file if required.

When the profile factor, and hence the production rate, changes from one month to the next, Eclipse automatically limits the timestep to the end of the month. Thus Eclipse models the different flow rate periods individually, rather than taking longer timesteps with an average rate. There is also an option to limit the timestep to the end of every month irrespective of whether the rate changes or not. This ensures that there is at least one entry in the `SUMMARY` file for each month. In Eclipse 100, timesteps limited to the end of the month can be identified by the mnemonic 'EMTH' in the `PRINT` file.

The field's production rate is constrained to meet the month's target by one of the two standard methods for group control - guide rate control or prioritization. By default, Eclipse automatically places the `FIELD` under guide rate control on reading the `GASYEAR` or `GASPERIO` keyword, and set its target gas rate each month from the DCQ and the swing or profile factors. You need to enter `GCONPROD` only if the field has limits on other phases, or if any groups have rate limits or guide rates. If on the other hand you wish the field's target rate to be met by prioritization instead of guide rate control, you must indicate this by entering the keyword `GCONPRI`. Place the `FIELD` under prioritization and give it a gas rate target. Any valid rate will do if you are going to use `GASYEAR` or `GASPERIO`, as the target is reset at each month in the Contract Period to the DCQ multiplied by the month's swing or profile factor. The default priority coefficients must also be set with keyword `PRIORITY`. If at a later date you wish to revert to guide rate control, place the `FIELD` under 'GRAT' control in keyword `GCONPROD`, giving it a nominal gas rate target.

The profile factors can be changed, if required, at the start of any Contract Period by entering the keyword `SWINGFAC` again.

Normalizing the profile factors

The Annual Contracted Quantity (ACQ) is related to the Daily Contracted Quantity by:

$$ACQ = 365.25 \times DCQ \quad \text{Eq. 9.3}$$

The number 365.25 is the average number of days in a year, with a leap year every fourth year.

In order to produce the required ACQ for a given DCQ, the profile factors should be normalized such that the sum over all months of {profile factor $\times$ number of days in month} = 365.25, with February allocated 28.25 days. If this "normalization requirement" is obeyed, then when leap years are taken into account

every fourth year, the average annual production at a constant DCQ is $DCQ \times 365.25$, which corresponds with the definition of ACQ given above.

When the [SWINGFAC](#) keyword is entered, Eclipse checks that the profile factors obey this normalization requirement, and issue a warning if they do not.

Operations and reports during the contract period

The keyword [GASYEAR](#) or [GASPERIO](#) advances the simulation through one or more complete contract periods. Well operations can be performed in the usual manner at the end of a contract period, and a printed report is generated at the end of each contract period depending on the [RPTSCHED](#) or [RPTPRINT](#) switches. But in order to perform well operations or request printed reports during a contract period, an "Annual Scheduling File" must be set up.

To set up an annual scheduling file, enter the keywords defining the required operations before the [GASYEAR](#) or [GASPERIO](#) keyword on the input data file, bracketed with the keywords [GASBEGIN](#) and [GASEND](#). When Eclipse reads the keyword [GASBEGIN](#) during the simulation, it copies the subsequent keywords up to [GASEND](#) to a separate file that it sets up automatically. The times within each year at which the operations should be performed are set with the keyword [GASMONTH](#).

During each subsequent contract period, Eclipse reads the keywords from this file and performs the required operations at the appropriate times. The same operations will be performed each year, until you set up a new Annual Scheduling File. See the keyword [GASBEGIN](#) for further details of this facility.

Swing factor and DCQ adjustment

A flag in the keyword [GASYEAR](#) or [GASPERIO](#) can be set to allow the DCQ to be adjusted automatically at the start of each Contract Period to obey a Swing Factor Requirement, that is, that the field should be able to produce at a rate equal to the DCQ multiplied by the swing factor. If the field cannot produce at this rate, the DCQ is reduced to a value that enables the requirement to be obeyed.

If automatic DCQ adjustment is requested, each contract period is solved in two passes.

- The first pass is a test calculation, which has the purpose of deciding how much the DCQ needs to be reduced in order to obey the swing requirement for the current contract period. If requested, Eclipse may perform a number of iterations of the first pass, with different DCQ values, to determine the highest DCQ that obeys the swing requirement for the period, to a specified accuracy.
- The second pass simulates the actual performance of the field, with the sales gas rate target set each month to the new value of the DCQ multiplied by the monthly Profile Factor. The [RESTART](#) and [SUMMARY](#) files are written as required during this pass.

The swing factor is specified separately for each month for flexibility. The monthly swing factors are entered along with the profile factors using the keyword [SWINGFAC](#). For example, the monthly swing factors could be set equal to the profile factors, in which case the Swing Factor Requirement causes the DCQ to be adjusted each period so that the seasonality profile fits inside the delivery capacity decline curve. Alternatively the swing factors could all be set to the same value each month. A check is made to ensure the profile factor does not exceed the swing factor during any month, otherwise obeying the Swing Factor Requirement may not guarantee the field could produce its target rate profile.

The keyword [FGDCQ](#) (which stands for Field Gas DCQ) outputs the current value of the DCQ to the [SUMMARY](#) file.

Eclipse 100

By default, Eclipse 100 adjusts the DCQ downwards. If the field has more capacity than is required to meet the swing requirement with the current value of the DCQ, Eclipse 100 does not automatically increase the DCQ. If you wish to increase the DCQ during a simulation, this may be done manually by resetting the

initial DCQ in item 2 of the `GASYEAR` keyword (or item 3 of `GASPERIO`) at the appropriate time. Alternatively, the accelerated iteration scheme may be used by setting item 2 in the `GASFIELD` keyword. This permits the DCQ to increase again if appropriate, up as far as its initial value.

Eclipse 300

Eclipse 300 has the accelerated scheme enabled by default, and so after any initial reduction, the DCQ may increase up as far as its initial value should the gas delivery capacity of the field permit this. This scheme cannot be switched off in Eclipse 300.

Interpretations of the swing requirement

To provide flexibility to handle different forms of contract, there is a choice of interpretations of the swing requirement. The chosen interpretation is selected by entering its associated mnemonic as Item 3 in keyword `GASYEAR` or Item 4 in keyword `GASPERIO`. If the contract period is less than a year, keyword `GASPERIO` must be used.

The choices within `GASYEAR` for contract years are:

- YEAR or YES

“The field must be able to produce at the DCQ multiplied by the monthly swing factor, for the whole contract year.”

During the first pass, the field is produced with a target rate of $DCQ \times SWING$ for the whole contract year. If this target cannot be met, Eclipse determines how much the DCQ should be reduced in order to meet the target.

This is a suitable interpretation for cases where the monthly swing factors are the same as the profile factors, as it ensures the production profile fits underneath the delivery capacity decline curve. But for cases where the swing factors exceed the profile factors, this interpretation may be too stringent because it would require the field to be able to produce more than the Annual Contracted Quantity.

- PRO

“The field must be able to raise its production rate from $DCQ \times \text{profile}$ to $DCQ \times \text{swing}$ for an instant at any time in the contract year.”

During the first pass, the field is produced with a target rate of $DCQ \times \text{PROFILE}$ for the whole contract year. In addition, the delivery capacity is calculated and compared against a target of $DCQ \times \text{SWING}$. If either of these targets are not met, Eclipse determines how much the DCQ should be reduced in order to meet them both. (The delivery capacity is calculated using the grid block conditions at the beginning of the timestep, that is, at the end of the previous timestep. Therefore it is compared against the $DCQ \times \text{swing}$ factor for the previous timestep.)

This interpretation ensures the field is able to produce its ACQ (that is, $DCQ \times \text{profile}$ for each month), and has the potential to increase its production rate to $DCQ \times \text{swing}$ for a short period. But it does not test whether the increased production rate can be sustained for any length of time.

- ACQ

“The field must be able to produce at the DCQ multiplied by the monthly swing factor continuously until it has produced its Annual Contracted Quantity.”

During the first pass, the field is produced with a target rate of $DCQ \times \text{SWING}$, but only until it has produced its ACQ ($= 365.25 \times DCQ$). Eclipse simulates through the contract year until the cumulative sum of {monthly swing factor $\times$ number of days in the month} equals 365.25. When accumulating this sum, February is allocated 28.25 days. If the target of $DCQ \times \text{SWING}$ cannot be met, Eclipse determines how much the DCQ should be reduced in order to meet the target.

This interpretation ensures the field is able to produce its ACQ, and would be capable of operating at the increased rate of $DCQ \times \text{swing factor}$ continuously until its ACQ had been produced.

- JAN, FEB, MAR, APR, MAY, JUN, JUL, AUG , SEP, OCT, NOV, DEC

The field must be able to produce at the DCQ multiplied by the monthly swing factor continuously until it has produced a specified portion of its Annual Contracted Quantity”.

This is similar to the ACQ option above, but the first pass is only calculated until a specified portion of the ACQ has been produced. The portion is defined as

$ACQ \times \text{sum over months from start of contract year to end of specified month of } \{\text{profile factor} \times \text{number of days in month}\}$

$\div \text{sum over all months in the contract year of } \{\text{profile factor} \times \text{number of days in month}\}.$

When accumulating these sums, February is allocated 28.25 days.

- NO

No automatic adjustment of DCQ.

The DCQ is held constant at its current value.

The first pass of the contract year is omitted, and Eclipse goes immediately into the second pass.

In all options, the second pass of the contract year is calculated with the field producing at the (perhaps reduced) $DCQ \times \text{PROFILE}$.

The choices within [GASPERIO](#) for general contract periods are:

- PER or YES

The field must be able to produce at the DCQ multiplied by the monthly swing factor, for the whole contract period.

- PRO

“The field must be able to raise its production rate from $DCQ \times \text{profile}$ to $DCQ \times \text{swing}$ for an instant at any time in the contract period.”

- PCQ

“The field must be able to produce at the DCQ multiplied by the monthly swing factor continuously until it has produced its contracted quantity for the period (= no. of days in the period $\times$ DCQ).”

In this option February has either 28 or 29 days depending on whether or not it is a leap year.

- NO

No automatic adjustment of DCQ.

Calculating the DCQ value

The DCQ value is calculated in the first pass of each contract period. As the field depletes, the DCQ must be reduced each period in order to obey the swing requirement. Several iterations of the first pass of the contract period may be needed to provide an accurate value for the largest DCQ able to meet the swing requirement, if the DCQ reduces substantially from the previous period. The reason is as follows. If the initial estimate of the DCQ is too high, too much gas is extracted during the first pass, making the delivery capacity decline faster than it should. The calculated value of the DCQ that can fit under this steeper

decline curve is therefore an underestimate. Further iterations of the first pass are needed to obtain a more accurate answer.

An "accelerated iteration scheme" is available for converging the DCQ to a desired accuracy. An initial guess is obtained, based on the previous period's DCQ reduction. The first pass of the contract period is simulated and a new DCQ is calculated. Successive iterations of the first pass are performed until the change in DCQ over an iteration lies within a user-defined tolerance. The convergence is accelerated with a secant method. A maximum number of iterations can be imposed (item 7 in keyword [GASYEAR](#) or item 8 in [GASPERIO](#)), and a warning is issued if the DCQ has not converged by then. The accelerated iteration scheme is the only scheme available in Eclipse 300. For Eclipse 100 however, it must be enabled by setting the 2nd item of the RUNSPEC section keyword [GASFIELD](#) to 'YES'. This is the recommended method, although to keep back-compatibility with pre-96A Eclipse 100 simulation runs by default the accelerated scheme is not enabled.

Eclipse 100

If the accelerated iteration scheme is **not** selected, the pre-96A scheme is employed. In this scheme, the DCQ can only decrease from one iteration to the next, and underestimates cannot be corrected. Thus, to obtain an accurate answer, the amount by which the DCQ can decrease from one iteration to the next should be constrained. You can supply a limiting reduction factor (item 5 in keyword [GASYEAR](#), item 6 in [GASPERIO](#)). If at the end of the first pass the DCQ has to be reduced to a fraction that is smaller than the limiting reduction factor, it is reduced only by the limiting reduction factor and the first pass is repeated again. The process continues until either the DCQ reduction lies within the limiting factor, or the maximum number of iterations has been reached ([GASYEAR](#) item 7, [GASPERIO](#) item 8) at which the DCQ reduction is no longer restricted. The nearer the reduction factor is to 1.0 the more accurate is the result, but at the expense of requiring more iterations of the first pass to converge it. Supplying an "anticipated reduction factor" ([GASYEAR](#) item 6, [GASPERIO](#) item 7) could help reduce the number of iterations required to converge the DCQ, by bringing the starting value nearer to the final value. Before the decline from plateau has begun, this factor is ignored. But once the DCQ has been reduced automatically below its initial value, the anticipated reduction factor is applied to the DCQ at the start of each contract period. Of course, you ought to have a reasonable idea of the rate of DCQ reduction before using this option. An anticipated reduction that is too fast results in unduly pessimistic DCQs, since Eclipse does not increase the DCQ automatically. A warning message is printed if the DCQ has not been reduced by more than the anticipated rate at the end of a contract period, as this would imply that the anticipated rate is too fast.

Control and output of sales gas rate

Eclipse 100

In Eclipse 100 the sales gas rate is defined as the **gross** gas production from the formation, **minus** the gas consumption, **plus** any gas imports. Gas can be consumed or imported at a specified rate at any group, as requested using the keyword [GCONSUMP](#). Gas can also be consumed by an automatic compressor when it is operating (see "[Automatic compressors](#)").

Eclipse 300

In Eclipse 300 the sales gas rate is defined as the **gross** gas production from the formation, **minus** the gas consumption. Gas can be consumed as fuel using the keyword [GRUPFUEL](#). Gas may only be imported via a gas producing satellite group (see keyword [GSATPROD](#)). Note that advanced and make-up gas (see keywords [GADVANCE](#) and [GINJGAS](#)) is only available for reinjection and thus does not alter the sales gas rate.

Normally the group and field gas rate targets apply to the gross production rate of gas from the formation. But when the gas field operations model is in use (triggered by the keyword [SWINGFAC](#)), these targets are automatically adjusted to allow for the specified consumption and, for Eclipse 100, import rates. (Gas injection is assumed not to occur, so no adjustment is made for it.) Thus the targets now apply to the sales gas rate, whether they are set manually using keyword [GCONPROD](#) or [GCONPRI](#), or set automatically according to the DCQ.

Eclipse translates a sales gas target for a group into a gross gas target by adding the rate of gas consumption at and below the group and, for Eclipse 100, subtracting any gas import rate at and below the group. For Eclipse 100, the gas consumption rate may depend on whether or not the automatic compressors are operating. The gas consumption rate (with or without compression) must be a known fixed value. Thus the option to make the consumption rate a given fraction of the production rate (a negative value in item 2 of keyword [GCONSUMP](#) or a non-zero value in item 3 of keyword [GRUPFUEL](#)) is not allowed when running the gas field operations model.

The production reports show the sales gas rate, and also the gross gas rate and consumption/import for any group that has gas consumption or import. The sales gas rates and totals can be output to the SUMMARY file using the keywords:

FGSR Field Gas Sales Rate

GGSR Group Gas Sales Rate

FGST Field Gas Sales Total (cumulative)

GGST Group Gas Sales Total (cumulative)

The keywords FGPR, GGPR, FGPT, GGPT retain their original meanings as the field and group **gross** gas production rates and totals.

Do not use the keywords [GCONSALE](#) or [GRUPSALE](#). These apply to the control of gas production from an oil field by reinjecting the surplus gas. It is not appropriate to do this when running the gas field operation model.

Estimation of delivery capacity

At the start of each timestep, Eclipse calculates the delivery capacity of the field. The delivery capacity is defined as the sales gas production rate that would ensue immediately after the removal of all group and field gas rate targets and limits, at the current grid block conditions. You can specify (with the keyword [GASFDELC](#)) whether any group or field oil or water rate limits, and any rate limits on the wells themselves, should remain in operation.

The delivery capacity is calculated at every timestep after the first entry of the [SWINGFAC](#) keyword (which effectively turns this facility on), irrespective of whether [GASYEAR](#), [GASPERIO](#), [DATES](#), [TIME](#) or [TSTEP](#) is used to advance the simulation. When the option for automatic DCQ adjustment is in use (see "[Swing factor and DCQ adjustment](#)") the delivery capacity is calculated during the second pass of the contract period. In Eclipse 100 the delivery capacities of the field and each group are printed in the Production Reports in the right hand column. The well delivery capacities are also printed in Eclipse 100 if their drainage radii are set negative in item 7 of keyword [WEL SPECS](#) (otherwise their productivity indices will be printed instead). The group and well delivery capacities are respectively the sales gas rate of each group and the gas production rate of each well when the field operates at its delivery capacity. The field, group and well delivery capacities can also be output to the SUMMARY file using the keywords FGDC, GGDC and WGDC.

The delivery capacity calculated at each timestep is not exact; in fact it is somewhat optimistic. There are two reasons for this. Firstly, it is calculated at the start of the timestep, using the grid block conditions existing at that time. It does not take into account the changes in grid block pressure and saturation that occur during the timestep. Thus at the end of the timestep (when the delivery capacity is in fact reported) it is actually smaller than the value calculated at the start of the timestep. Secondly, the grid block conditions used in the calculations are those that exist when the field produces at the required offtake rate, not when it produces at full capacity. If the offtake rate were actually to increase to full capacity, the grid block

pressures would decrease and thus decrease the delivery capacity. In fact, you may notice the delivery capacity increasing slightly when the offtake rate decreases in spring and summer. This is due to the buildup of grid block pressure that follows a decrease in production rate. The rate-dependence of the reported delivery capacity is more pronounced at low permeabilities, or if the grid blocks containing the well connections are small. The delivery capacity should therefore be regarded as an instantaneous transient value rather than a sustainable one.

Eclipse 100

When the gas field operations model is used in conjunction with the network option, the delivery capacity calculation takes account of the higher flows along the network branches when the field is producing at full capacity. Thus the network is balanced twice at the start of each timestep: first with the field operating at full capacity, then with the field producing at the required offtake rate. If a balancing interval > 0.0 is entered in keyword `NETBALAN` it is ignored, because the network has to be balanced at each timestep.

The delivery capacity should not be confused with the production potential. The potential of a production well is defined as the flow rate it would immediately have if, at the current grid block conditions, its only constraints were its THP and BHP limits. The field and group potentials are just the sum of the well potentials multiplied by their efficiency factors. The main differences between the delivery capacity and the potential are

- A well's production potential honors only its THP and BHP limits. The delivery capacity will honor all the well's rate constraints too, unless keyword `GASFDELC` has been used to disable them.
- The group production potential takes no account of any limits on group flow rate. The delivery capacity honors all group and field limits except on the gas production rate, unless keyword `GASFDELC` has been used to disable them.
- The delivery capacity is adjusted to allow for gas consumption and, in Eclipse 100, gas imports, while the potential is not.
- When using the network option, the delivery capacity is calculated with the network balanced at the capacity flow rates. The potential is calculated using the THP limits obtained from the nodal pressures of the network balanced at the required offtake rate. Thus the potentials do not account for the increased wellhead pressures that would result when the flows increase to capacity.

Eclipse 100

Automatic compressors

Eclipse 100

The network option can be used with the gas field operations model to take account of the pipeline pressure losses in the gathering system (see "[Network option](#)"). The delivery capacity is determined by the need to supply the gas at a specified onshore pressure.

Compressors can be placed in the pipelines to boost the gas flow. In the network option, the effects of turning on a compressor can be modeled either by switching to an alternative VFP table to calculate the pressure losses in the pipeline, or by changing the value of the Artificial Lift Quantity that is used to interpolate the table. One compressor is allowed in each branch of the network.

When the network option and the gas field operations model are used together, a class of compressors called "automatic compressors" can be declared. The basic compressor facility available with the Standard Network model (keyword `GNETPUMP`) is not versatile enough for the requirements of the gas field operation model, and you must use the keyword `GASFCOMP` instead. This defines a more versatile compressor facility, which was designed specifically for the gas field operations model. However, the automatic compressor facilities associated with the "extended network model" (keyword `NETCOMPA`) **are** suitable for use with the gas field operations model. Thus, when using the extended network model, automatic compressors should be defined with keyword `NETCOMPA` instead of `GASFCOMP`. In either of the two cases, Eclipse switches on compression whenever necessary to enable the field to meet its target rate. Compression is turned off when not needed (for example, during the summer "low" in the seasonality profile).

Each compressor, when operating, can consume a specified rate of gas for power. The gas consumption rate of the automatic compressors is set in the compressor data keyword (`GASFCOMP` or `NETCOMPA`).

Whenever the automatic compressors are operating, their gas consumption rate is added to the fixed consumption rate (if any) specified in keyword `GCONSUMP`, and subtracted from the gross production to give the sales gas rate.

Simple and extended compressor options with `GASFCOMP`

This section describes the options available for compressors defined with keyword `GASFCOMP` when using the Standard Network model. If you are using the Extended Network model, you should define the compressors with keyword `NETCOMPA` instead. These compressors have common functionality with the extended compressor option described below. See the chapter on the Network option for details of the Extended Network model and its associated compressor keyword `NETCOMPA`.

The simple option for automatic compressors has just two states: “all compressors on” and “all compressors off”. The compressors are turned on whenever the field cannot meet its target rate without them, and turned off whenever the target rate decreases to see if the field can meet it without them. The automatic compressors are also turned on while calculating the field’s delivery capacity.

The extended compressor option offers a more versatile set of facilities, firstly by giving each compressor up to five distinct levels of compression, and secondly by allowing them to be turned on or switched to a higher compression level individually in a predefined order until the field can make its production target. The parameters of each compression level (VFP table number, ALQ value and gas consumption rate) for each compressor are entered in separate records of keyword `GASFCOMP`.

In the extended compressor option, individual compressors are turned on, or switched to a higher level of compression, in a predefined sequence whenever necessary to maintain the production target. Each compression level of each compressor is given an “action sequence number”, specified by item 6 in the `GASFCOMP` keyword. A particular sequence number can be assigned to the compression levels of more than one compressor, if required, but the levels of any one compressor must all have different sequence numbers. When the field cannot make its target, the actions with sequence number 1 are performed first, that is, all compression levels having an action sequence number 1 are turned on. If the target still cannot be met, compression levels with action sequence number 2 are turned on, and so on until the target is met or no more compressors can be switched to a higher compression level. In common with the simple compressor option, all compressors are turned off whenever the target rate decreases, to see if the target can be met without them. Also in analogy with the simple option, all automatic compressors operate at their highest compression level while Eclipse is calculating the field’s delivery capacity.

To keep track of the state of each compressor, there is a `SUMMARY` section keyword `GMCPPL` that writes the current compression level of the compressor in each group’s pipeline to the `SUMMARY` file. The compression level is represented in the `SUMMARY` file by a number between 0 (for no compression) and 5 (for the highest allowed level).

The extended compressor option requires a small amount of extra storage, which must be reserved by setting the first switch in the `GASFIELD` keyword in the `RUNSPEC` section to YES.

Specifying production rates for the second pass using `GASFTARG` and `GASFDECR`

The `GASFTARG` keyword can be used to specify a rate at which the field is to be produced during the second pass of the Gas Field Operations Model, independent of the calculated DCQ.

The `GASFTARG` keyword is used in conjunction with `SWINGFAC` keyword. During the first pass of the model the DCQ will be calculated using the swing factors provided. However, on the second pass the field

is produced at the lesser of $DCQ \times PROFILE$ and the rate specified in the `GASFTARG` keyword. For months defaulted in the `GASFTARG` keyword the field is produced according to the `PROFILE` factors.

The `GASFDECR` keyword enables the field gas production rate for the second pass to be adjusted through monthly decrements. On the second pass the field is then produced at $DCQ \times PROFILE - DECREMENT$.

The `GASFTARG` and `GASFDECR` keywords are restricted to single contract simulations.

Applying separate gas supply contracts to individual groups

The gas field operations model by default applies a single set of swing factors and profile factors to the `FIELD` group at the top of the grouping hierarchy. There is, however, an option to allow individual groups to have their own independent gas supply contracts. Each "contract group" has its own set of swing factors and profile factors, and its own `DCQ` which Eclipse can adjust independently to meet its swing requirement.

Defining multiple contract groups

Eclipse 100

In order to use the multiple contract groups facility, the flag 'YES' must be set in item 1 of the `RUNSPEC` section keyword `GASFIELD`. This reserves the extra storage needed by various extensions of the gas field operations model. This step is not necessary in Eclipse 300.

To define multiple contract groups, use the keyword `GSWINGF` instead of the `SWINGFAC` keyword to activate the gas field operations model. The `GSWINGF` keyword is used to identify the contract groups and to give each one its own set of swing and profile factors. (In comparison, the `SWINGFAC` keyword reads in a single set of swing and profile factors, and applies it to the `FIELD` group.)

The set of contract groups must obey the following restrictions:

- One contract group must not be subordinate to another contract group.
- The set of contract groups must cover the whole field, that is, every open production well must be subordinate to a contract group.
- There must be no gas consumption or, for Eclipse 100, import at any node downstream of a contract group (otherwise the contract group's sales gas targets would not be set correctly).

Simulating with multiple contract groups

The keyword `GASYEAR` or `GASPERIO` is still used to advance the simulator through an integral number of contract periods. Eclipse assumes that all contract groups start their contract periods at the same time, and the contract period length is the same for each group. However, each contract group generally has a different `DCQ`. So item 2 in keyword `GASYEAR` or item 3 in `GASPERIO` (the initial `DCQ`) is ignored, and instead the initial `DCQ` of each contract group must be entered with the keyword `GDCQ`. The remaining items in keyword `GASYEAR` or `GASPERIO` are applied to all contract groups.

When you set the initial `DCQ` of each contract group with keyword `GDCQ`, you should state whether the `DCQ` is variable or must remain fixed at the specified value. Variable `DCQ`s are adjusted automatically as necessary to obey the group's swing requirement, depending on the option selected in item 3 of keyword `GASYEAR` or item 4 of `GASPERIO`. Fixed `DCQ`s remain at their specified values until reset in a subsequent `GDCQ` keyword.

Fixed-DCQ contract groups

If gas produced by other operators is fed into the same network that you use, you may wish to determine the DCQ which your group is able to supply against assumed production profiles from the other operators' groups. In this situation, you could model the other operators' groups as fixed-DCQ contract groups, and your own group as a variable-DCQ contract group. The group DCQ values can be written to the SUMMARY file with the SUMMARY section mnemonic GGDCQ (Group Gas DCQ).

When there is only one variable-DCQ group in the problem, the simulation process is exactly the same as for the single contract group case described in Section 2, except that the swing requirement and DCQ adjustment is applied to the variable DCQ group instead of the FIELD, and gas from the other fixed-DCQ groups also enters the network. All the options in item 3 of the GASYEAR keyword or item 4 of GASPERIO are available for the variable-DCQ group.

The flow targets applied to the fixed-DCQ groups are equivalent to the target rate applied to the variable-DCQ group. In the first pass of the contract period each group has a target rate equal to its $DCQ \times \text{Swing Factor}$; or its $DCQ \times \text{Profile Factor}$ if the 'PRO' option is selected. In the second pass of the contract period, the flow target is its $DCQ \times \text{Profile Factor}$. But in contrast to the variable-DCQ group, if a fixed-DCQ group cannot achieve its flow target its DCQ is not reduced. A missed flow target indicates that its specified DCQ is too high, so you should either give it a lower DCQ or model it as a variable-DCQ group. The group sales gas rates and delivery capacities can be written to the SUMMARY file with the mnemonics GGSR and GGDC.

Fixed-DCQ groups have two choices for calculating delivery capacities. The first choice (FIX1) has each contract group operating at its full delivery capacity. The delivery capacity of the variable-DCQ contract group is therefore determined at a time when the other groups are also producing at their full capacities into the network. The second choice (FIX2) imposes a maximum rate of $DCQ \times \text{Swing Factor}$ on each fixed-DCQ group. Then the delivery capacity of the variable-DCQ contract group will be determined at a time when the other groups are producing at $DCQ \times \text{Swing}$. The network pressures in the FIX2 case are lower than those in the FIX1 case, so the delivery capacity of the variable-DCQ group is higher. Again, you should ensure that the fixed-DCQ groups are able to make their delivery capacity targets, as otherwise it implies that their specified DCQs are too high.

Multiple variable-DCQ contract groups

When you wish to predict the future DCQs of two or more groups with independent gas supply contracts, you should model them as variable-DCQ contract groups. Eclipse will test each variable-DCQ contract group against its own swing requirement, and reduce its DCQ as necessary. The DCQ values of the contract groups can be written to the SUMMARY file with the SUMMARY section keyword GGDCQ (Group Gas DCQ).

The simulation method for multiple variable-DCQ contract groups is equivalent to the method for a single contract at the FIELD group. If automatic DCQ reduction is requested in keyword GASYEAR or GASPERIO, each contract period is solved in two passes. During the first pass, the contract groups aim to produce at their $DCQ \times \text{Swing Factor}$. If a contract group cannot make its swing requirement, its DCQ is reduced accordingly. During the second pass of the contract period, the contract groups have their flow targets set equal to their DCQ values multiplied by their monthly profile factors. The group rate targets are met by Guide Rate Control, unless Prioritization is requested with the keyword GCONPRI.

When there is more than one contract group with variable DCQ, item 3 in keyword GASYEAR or item 4 in GASPERIO (the swing requirement) is restricted to two choices:

- YEAR or PER or YES

For automatic DCQ reduction, with the production target set at $DCQ \times \text{swing factor}$ for the whole contract period in the first pass

- NO

For no automatic DCQ reduction.

The other options (ACQ and PRO for example) are disabled to avoid problems with the contract groups interacting through the network. (The delivery capacity of each group varies with the flow rates of the other groups, so the flow rates of all the contract groups must be defined for the whole contract period while the swing requirement is being tested.) The YEAR or PER option is most suitable for cases where the monthly swing factors are the same as the profile factors. For cases where the swing factors exceed the profile factors, you could reduce them to the profile factor value in all but two months - at the end of the peak flow period and at the end of the contract period.

The delivery capacity is calculated in the usual way, by removing the constraints on gas production rate. The delivery capacity reported for the individual contract groups will be their production capacity while all other groups are producing at their full capacity. Because of the coupling through the network, a group may possibly achieve a higher rate than its reported delivery capacity when the others are producing at the smaller rate of $DCQ \times \text{Profile}$.

Automatic compressors with multiple contract groups

Eclipse 100

When there are multiple contract groups, compressors may be situated either at (or subordinate to) specific contract groups, or further downstream at a node common to all contract groups. If a contract group G1 fails to meet its target, then turning on a compressor either at G1 or at a common node helps, but turning on a compressor located within a group that is still meeting its target does not help. Thus, when deciding which compressor to turn on next with the extended compressor option (keyword [GASFCOMP](#)), Eclipse ignores any compressors located within contract groups that are still meeting their target. Note that compressors located at a common node cannot have a gas consumption rate set, as gas consumption is not allowed downstream of a contract group.

Using the gas field operation model

RUNSPEC section

Eclipse 100

Recommendation: use the accelerated iteration scheme for calculating the DCQ. Accordingly, in the RUNSPEC section enter the [GASFIELD](#) keyword with item 2 set to YES. Set item 1 to YES if you are using either multiple contract groups (keyword [GSWINGF](#)) or the extended compressor option in keyword [GASFCOMP](#).

```
GASFIELD
YES    YES /
```

SUMMARY section

Useful quantities to include in the SUMMARY section are shown in the following table.

E100	E300	Field	Group	Description
x	x	FGSR	GGSR	Field/Group Gas Sales Rate
x	x	FGST	GGST	Field/Group Gas Sales Total (cumulative)
x	x	FGDC	GGDC	Field/Group Gas Delivery Capacity
x	x	FGDCQ	GGDCQ	Field/Group Gas DCQ
x			GMCPPL	Group Multi-level Compressor Level.

E100	E300	Field	Group	Description
x	x		GPR	Group or node PPressure in network
x			GPRDC	Group or node PPressure in network at Delivery Capacity
x			GPRFP	Group or node PPressure in network from end of First Pass
x			GGPRNBFP	Gas flow rate along Group's or node's outlet branch in network, from end of First Pass.

The other useful gas field operations keywords are:

E100	E300	Keyword	Description
x	x	NGOPAS	Number of iterations to converge DCQ in first pass
x		WGPRFP	Well Gas Production Rate from end of First Pass
x		WTHPFP	Well Tubing Head Pressure from end of First Pass.
x		WBHPFP	Well Bottom Hole Pressure from end of First Pass.

Remember that the keywords FGPR, GGPR, FGPT and GGPT refer to the **gross** gas produced from the formation, without allowing for consumption. Note also that the mnemonics WGPRFP, WTHPFP, WBHPFP, GPRFP and GGPRNBFP do not refer to continuous quantities but rather the values of these quantities at the end of the first pass.

Specifying the wells

Specify the wells in the usual way with [WEL SPECS](#), [COMP DAT](#) and [WCONPROD](#) (and [WCONINJE](#) for any water injectors). The optional keywords (for example [WECON](#), [WEFAC](#)) are not mentioned explicitly in this example.

If there is a multilevel grouping hierarchy (more than one level of groups below the FIELD), set it up with [GRUPTREE](#). For the purposes of this example, consider a grouping hierarchy as given by:

```
GRUPTREE
-- CHILD      PARENT
  PLAT-A      FIELD  /
  PLAT-B      FIELD  /
  G-B1        PLAT-B /
  G-B2        PLAT-B /
/
```

Setting up the network

If the standard network model is to be used, set up the network with [GRUPNET](#). If using the extended network model, set up the network with keywords [BRANPROP](#) and [NODEPROP](#) instead. In this example:

```
GRUPNET
-- GROUP      FIXED VFP ALQ S-SEA LIFT
-- NAME       PRESS TAB      MAN  GAS
  FIELD       500
  PLAT-A      1*      1
  PLAT-B      1*      2
  G-B1        1*      3
  G-B2        1*      4
/
```


[NETBALAN](#) may be used to alter the convergence tolerances and iteration limits, and set restrictions on the allowed balancing error at the end of each timestep. But the balancing interval set here is ignored because when the gas field operation model is in use, the network is balanced at the beginning of every timestep.

Eclipse 100

Declare any automatic compressors with [GASFCOMP](#) (or [NETCOMPA](#) if using the Extended Network model). Do not use [GNETPUMP](#). For example, to place a single-level compressor in the pipeline from PLAT-A in the Standard Network model:

```
GASFCOMP
-- GROUP      NEW VFP      NEW      GAS      COMP      SEQUENCE
-- NAME      TAB NO      ALQ      CONSUMP      LEVEL      NUMBER
  'PLAT-A'    1*          50      1000      /
/
```

Eclipse 100

Alternatively, to declare a pair of two-level compressors in the pipelines from PLAT-A and PLAT-B in the Standard Network model, and have them turned on in sequence:

```
GASFCOMP
-- GROUP      NEW VFP      NEW      GAS      COMP      SEQUENCE
-- NAME      TAB NO      ALQ      CONSUMP      LEVEL      NUMBER
  'PLAT-A'    1*          50      1000      1          1      /
  'PLAT-A'    1*          100     1500      2          2      /
  'PLAT-B'    1*          50      1000      1          3      /
  'PLAT-B'    1*          100     1500      2          4      /
/
```

This uses the extended compressor option, so remember to enter 'YES' in item 1 of keyword [GASFIELD](#) in the RUNSPEC section.

Group control instructions

If you are going to use [GASYEAR](#) or [GASPERIO](#) to advance the simulation, Eclipse will calculate the field's gas production target each month from the DCQ and the swing or profile factors. By default, Eclipse automatically places the FIELD under guide rate control to meet this target. You need to enter [GCONPROD](#) only if the field has limits on other phases, or if any groups have rate limits or guide rates. If on the other hand you wish the field's target rate to be met by prioritization instead of guide rate control, you must indicate this by entering the keyword [GCONPRI](#). Place the FIELD under prioritization and give it a gas rate target. Any valid rate will do if you are going to use [GASYEAR](#) or [GASPERIO](#), as the target is re-set at each month to the DCQ multiplied by the month's swing or profile factor. The default priority coefficients must also be set with keyword [PRIORITY](#).

For example, to use prioritization where the lowest water-gas ratio wells have the highest priority, with well priorities being updated after 100 days:

```
PRIORITY
100  0.0  0.0  0.0  1.0  0.0  0.0  1.0  0.0  /

GCONPRI
-- GROUP      OIL  PROC  WAT  PROC  GAS  PROC  LIQ  PROC  RESV
-- NAME      LIM  OIL  LIM  WAT  LIM  GAS  LIM  LIQ  LIM  LIM
  'FIELD'    4*                1.0  'PRI'
/
```

Eclipse 100

If any groups have a fixed rate of gas consumption or import (independent of the automatic compressors), this can be entered using [GCONSUMP](#):

```
GCONSUMP
-- GROUP      CONS  IMPRT
-- NAME      RATE  RATE
```

```
'PLAT-*' 5000 /
/
```

Eclipse 300

If any groups have a fixed rate of gas fuel usage, this can be entered using [GRUPFUEL](#):

```
GRUPFUEL
-- GROUP  CONS
-- NAME   RATE
'PLAT-*' 5000 /
/
```

Advancing the simulation

The gas field operations model is turned on when the keyword [SWINGFAC](#) is entered. The keyword is used to set the monthly swing factors and profile factors for the FIELD. For example, consider a case where the swing factors are the same as the profile factors, which are in ratio 1.0, 1.5, 1.0 and 0.5 for Autumn, Winter, Spring and Summer respectively. Their normalized values are 1.0024, 1.5036, 1.0024 and 0.5012. Since the swing and profile factors are the same, the swing factor requirement is equivalent to checking that the seasonality profile fits inside the delivery capacity decline curve.

```
SWINGFAC
-- JAN  FEB  MAR  APR  MAY  JUN  JUL  AUG  SEP  OCT  NOV  DEC
2*1.5036 3*1.0024      3*0.5012      3*1.0024      1.5036
2*1.5036 3*1.0024      3*0.5012      3*1.0024      1.5036 /
```

Once the [SWINGFAC](#) keyword has been entered, the features labelled 3, 4 and 5 in the Introduction are activated. This means that all group and field gas targets will refer to sales gas, the delivery capacity will be calculated at each timestep, and any automatic compressors are turned on and off as required.

If you wish to disable all rate limits for wells and groups when calculating the delivery capacity, the keyword [GASFDEL](#) must be used to indicate this requirement. The delivery capacity is then the sum of the well production rates when operating at their BHP or THP limits, minus any gas consumption.

```
GASFDEL
2 /
```

To use features 1 and 2, that is, seasonality profile generation and DCQ adjustment, the keyword [GASYEAR](#) or [GASPERIO](#) must be used instead of [TIME](#), [TSTEP](#) or [DATES](#). This advances the simulation through an integral number of contract periods, starting at the current date (which must be the first day of a month). [GASYEAR](#) may be used if the contract period is a year, otherwise [GASPERIO](#) must be used. At the end of each contract period, reports are printed and restarts are written as requested with [RPTSCHE](#) or [RPTPRINT](#). Reports can also be printed during the contract period if an Annual Scheduling File has been set up requesting them. For example, to request a printed report for the timestep ending 1st March each year:

```
GASBEGIN
GASMONTH
-- MONTH  REPORT
'MAR'    'YES' /
GASEND
```

The Annual Scheduling File can also contain keywords defining well operations to be performed at the same time each year. More detailed examples can be found in the documentation for keywords [GASBEGIN](#) and [GASMONTH](#).

With the first entry of the [GASYEAR](#) or [GASPERIO](#) keyword, the initial DCQ must be specified. For example, to advance the simulation through four contract years, with the DCQ initially 150,000 but reduced when necessary to obey the swing requirement:

```
GASYEAR
-- No of      Initial   Swing   Monthly   Lim DCQ   Ant DCQ   Max no   Conv
-- years      DCQ       req     t/steps   reduc     reduc     DCQ its  tol
--          4      150000  'YEAR'   /
```

The interpretation of the swing requirement most appropriate for cases where the swing factors are the same as the profile factors has been chosen (this is in fact the default option). Defaulting the remaining items forces a new timestep to start on the first of each month, applies no limit to the DCQ reduction in a single first pass, and applies no anticipated reduction factor to the DCQ at the start of a new contract year.

If you wish to model separate gas supply contracts for two or more groups, instead of a single contract for the [FIELD](#), you must use the keyword [GSWINGF](#) instead of [SWINGFAC](#). [GSWINGF](#) identifies the groups that have gas supply contracts, and sets their individual swing and profile factors. You must also set the initial DCQs of the contract groups individually, using keyword [GDCQ](#). Keyword [GASYEAR](#) or [GASPERIO](#) is still used to advance the simulation through one or more contract periods, but the initial DCQ in item 2 or 3 will be ignored, so leave it defaulted. For example, to specify swing and profile factors and initial DCQs of two contract groups, PLAT-A and PLAT-B, and advance the simulation through four contract years, reducing their DCQs as necessary to obey their swing requirements:

```
GSWINGF
-- GROUP      JAN  FEB  MAR  APR  MAY  JUN  JUL  AUG  SEP  OCT  NOV  DEC
-- 'PLAT-A'    2*1.5036 3*1.0024      3*0.5012      3*1.0024      1.5036
--              2*1.5036 3*1.0024      3*0.5012      3*1.0024      1.5036 /
-- 'PLAT-B'    2*1.6551 3*1.0031      3*0.3511      3*1.0031      1.6551
--              2*1.6551 3*1.0031      3*0.3511      3*1.0031      1.6551 /
/

GDCQ
-- GROUP      INITIAL  VAR/FIX
-- NAME        DCQ      DCQ
-- 'PLAT-A'    60000    'VAR' /
-- 'PLAT-B'    40000    'VAR' /
/

GASYEAR
-- No of      Initial   Swing   Monthly   Lim DCQ   Ant DCQ   Max no   Conv
-- years      DCQ       req     t/steps   reduc     reduc     DCQ its  tol
--          4          1*    'YEAR'   /
```

Eclipse 100

If there are multiple contract groups, remember to enter ' YES ' in item 1 of keyword [GASFIELD](#) in the [RUNSPEC](#) section.

Examining the results

The set of output reports requested in [RPTSCHED](#) is written to the print file at the end of each contract period. Output reports can also be written in the middle of a contract period if requested in an Annual Scheduling File. Summary information is written to the [SUMMARY](#) file each timestep as usual (only during the second pass, of course, if the automatic DCQ adjustment option is on).

Restrictions

- Do not use the keywords [GCONSALE](#), [GNETPUMP](#) or [GRUPSALE](#) none are appropriate for use with the gas field operations model.

- A variable consumption rate (a negative value in item 2 of keyword [GCONSUMP](#) or a non-zero value in item 3 of [GRUPFUEL](#)) is not allowed.
- Flux boundary conditions (see "[Flux Boundary Conditions](#)") should not be used in conjunction with automatic DCQ reduction.
- The gas fields operations model is not compatible with the parallel option in Eclipse 100.

Summary of keywords

RUNSPEC

Restriction	Keyword	Description
Eclipse 100	GASFIELD	Enables special options in the gas field operations model: the extended compressor option, multiple contract groups, and the accelerated iteration scheme for calculating the DCQ. The keyword may be omitted if the gas field operations model is used without these options.

SUMMARY

Refer to [Gas field operations SUMMARY keywords](#) in the *Eclipse Reference Manual*.

SCHEDULE

Restriction	Keyword	Description
Eclipse 100	COMPOFF	Switches off automatic compressors defined by the GASFCOMP keyword (or NETCOMPA if you are using the extended network model).
	GASBEGIN	Denotes start of keywords to go into the annual scheduling file. This is required only if you want to change well controls or request printed reports in the middle of a contract period, when using GASYEAR or GASPERIO to advance the simulation
Required only if there is an annual scheduling file.	GASEND	Denotes end of keywords to go into the annual scheduling file.
Eclipse 100	GASFCOMP	Defines automatic compressors in network branches. Omit this keyword if you do not require automatic compressors. If you are using the extended network model (see " The extended network model "), automatic compressors must be defined with keyword NETCOMPA instead.
	GASFDEL	Specifies the definition of the delivery capacity. If this keyword is omitted, the default definition is used, in which all group and well limits are honored except those on the group and field gas production rate.
	GASMONTH	Denotes the times within each year that operations on the annual scheduling file should be performed, and requests printed reports. Omit this keyword if there is no annual scheduling file (see GASBEGIN and GASEND).

Restriction	Keyword	Description
	GASPERIO	Advances the simulation through one or more complete contract periods, with the field sales gas target set equal to the DCQ multiplied by the monthly Profile Factor. Offers the option of automatic DCQ reduction to obey the Swing Requirement. TIME , TSTEP or DATES may be used instead, but the field target rate would have to be set manually each month and there would be no option of automatic DCQ reduction.
	GASYEAR	Advances the simulation through one or more complete contract years. This may be used instead of GASPERIO if the contract period is exactly one year.
	GDCQ	Sets the initial DCQs for individual contract groups. This keyword should only be used if there are multiple contract groups (keyword GSWINGF must be entered previously), in which case it must be entered before the first GASYEAR or GASPERIO keyword, and optionally again at later times if the initial DCQs need to be reset.
Eclipse 100	GDCQECON	Defines a minimum economic DCQ value for a contract group. All the producers in this group are shut when the DCQ falls below its minimum economic value. If the group is the FIELD , the run is terminated.
	GSWINGF	Turns on the gas field operations model in the multiple contract groups mode. It identifies the groups having gas supply contracts, and sets their swing and profile factors. Use this keyword only if you wish to apply separate gas supply contracts to two or more individual groups, in which case the keyword should be used in place of SWINGFAC
Eclipse 100	NETCOMPA	Defines automatic compressors in the extended network model. use this instead of GASFCOMP if you are using the extended network mode
Eclipse 100	OPTIONS	Item 34 prevents wells being drilled automatically in the second pass of the contract period, in the same month that they were drilled in the first pass, if they are not needed to meet the second pass production target.
Eclipse 100	SWINGFAC	Specifies the swing factors and profile factors for each month, which are applied to the FIELD . Use this keyword if you wish to apply a single gas supply contract to the FIELD . (If you wish to apply separate gas supply contracts to two or more groups, use keyword GSWINGF instead.) Use of either SWINGFAC or GSWINGF is compulsory, as they both turn on the gas field operations model. They activate the following features: • Calculation of the delivery capacity • Automatic compressor switching • Control of sales gas rate instead of gross production rate.
Eclipse 100	GASFTARG	Specifies a gas rate at which the field is to be produced in the second pass of the gas field operations model.
Eclipse 100	GASFDECR	Specifies a decrement to the gas rate at which the field is to be produced in the second pass of the gas field operations model.

Reservoir coupling

x	Eclipse 100
	Eclipse 300

The reservoir coupling facility is a special extension to Eclipse 100; it is not available in Eclipse 300. It provides a means of coupling a number of separate Eclipse 100 simulation models to account for constraints on overall production and injection rates, and optionally the sharing of a common surface network.

The individual simulation models are run as separate processes, each having its own standard Eclipse data file. One model is chosen to be the “master”, and the other models are defined as “slaves”. The slave processes are activated by the master, and run in synchronization with it. The master process imposes production and injection constraints on the slaves to meet overall targets. On suitable systems, the slave processes may be run in parallel with each other on separate workstations. Communication between the processes is handled by the message-passing system MPI, which must be present on the system.

A typical situation in which the reservoir coupling facility would be useful is as follows. Imagine you are responsible for an area that contains a number of separate independent reservoirs. For each reservoir, you have a separate simulation model, which you have used to history match the reservoirs independently. The simulation models may have different characteristics. For example, some may be three-phase while others are two-phase. One model may perhaps use the Vertical Equilibrium option, while the others use dispersed flow. There are now plans to produce these reservoirs into common surface facilities, and constraints on overall production and injection rates must be observed. Also, perhaps, the network model is to be used to determine the pressure constraints in a common surface network linking the reservoirs.

Without the reservoir coupling facility, the above scenario would require the various simulation grids to be amalgamated into a single, huge model. It would have to be three-phase if each phase was active in one or more of the reservoir models, and if one of them used the VE option then the whole amalgamated model would also have to use it. The task of amalgamating the models would be time consuming, and the resulting single model would be slow and expensive to run. The reservoir coupling facility offers an efficient solution to this problem, allowing the reservoirs to be represented by their original simulation models, but these are now coupled through their production and injection targets and network nodal pressures.

Master and slave reservoirs

Each slave reservoir consists of an Eclipse 100 simulation model with its own input data file. They may be restarts, if required, of simulations that have previously been run independently. Each must have one or more slave groups, which act as the path for communicating flow rates and constraints with the master process. For example, a reservoir may have its FIELD group as a single slave group, or alternatively all its Level 1 groups may be slave groups. The slave groups need not all be on the same level, but one slave group cannot be subordinate to another slave group.

The master reservoir is an additional Eclipse 100 simulation model, again with its own input data file. It must contain the group hierarchy of the whole system down as far as the slave groups in the slave reservoirs. The slave groups are represented in the master reservoir by master groups, which have no subordinate wells or groups. These are similar to satellite groups in concept (see keyword [GSATPROD](#)), except that the flows of satellite groups are user-defined, whereas the flows of master groups are determined by the coupled slave reservoir simulations.

The master reservoir may represent one of the reservoirs to be simulated. If so, its grid data, fluid properties and scheduling data for example, will be entered in the normal way in its data file. Alternatively, all the reservoirs to be simulated may be represented by slave reservoirs. In this case, the master reservoir should just be a dummy reservoir for example a single grid block), containing no wells but having the group hierarchy down to the master groups.

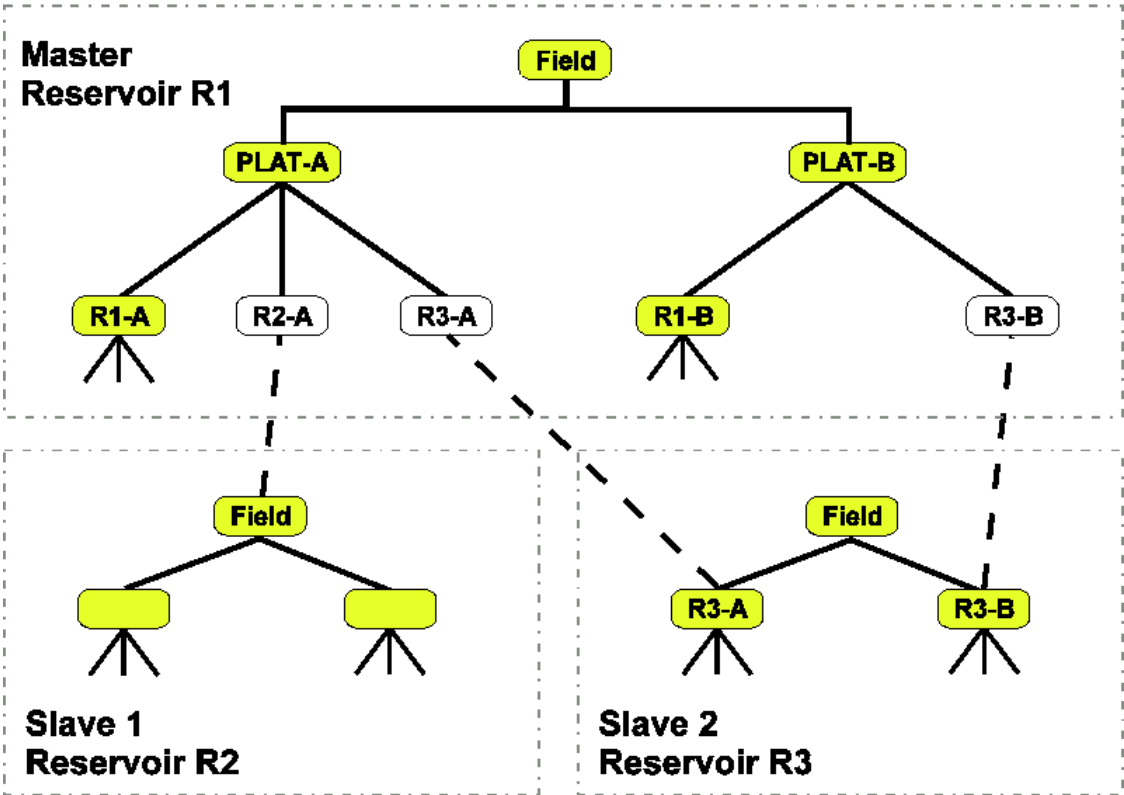


Figure 9.3. Three coupled reservoirs; one is the master

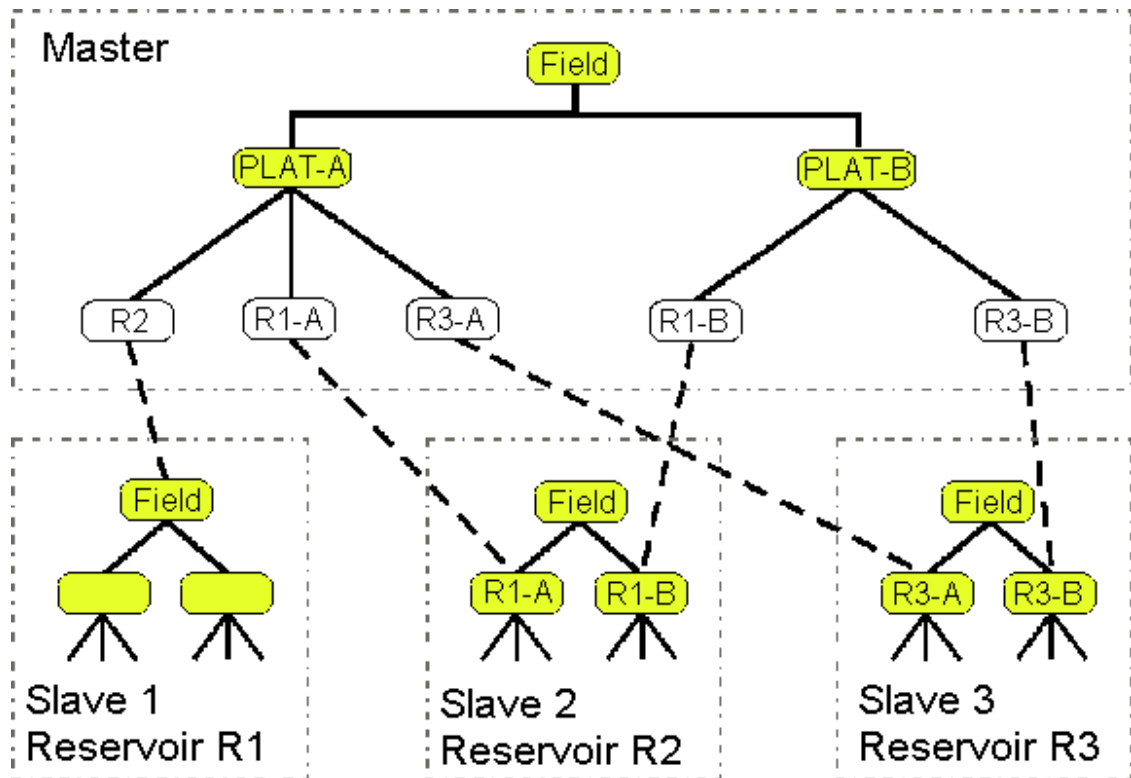


Figure 9.4. Three reservoirs coupled by a dummy reservoir master

Figures 9.3 and 9.4 illustrate these two cases applied to three coupled reservoirs. In figure 9.3, one of these reservoirs R1 is chosen as the master. Its own wells are under groups R1-A and R1-B. The other two reservoirs are represented as slaves. The master reservoir controls the slave reservoirs through the groups R2, R3-A and R3-B. These are declared as master groups in the master reservoir, and do not have any subordinate wells. The master group R2 is coupled to the FIELD group in the slave reservoir R2. The master groups R3-A and R3-B are coupled to the level 1 groups of the same name in the slave reservoir R3. In the slave reservoir R2, the FIELD group is declared as a slave group, coupled to group R2 in the master reservoir. In the slave reservoir R3, the groups R3-A and R3-B are declared as slave groups, coupled to the groups of the same name in the master reservoir. This example shows that a given slave reservoir may be coupled to the master by more than one group, which need not necessarily belong to the same parent group in the master reservoir.

Figure 9.4 shows the same set of reservoirs, but this time all three of them are represented as slave reservoirs. The master reservoir is a dummy reservoir with no wells. It contains the group hierarchy down to the master groups. Note that groups R1-A and R1-B are now master groups, being coupled to slave groups of the same name in reservoir R1 which is now a slave reservoir.

Coupling and synchronization

At the start of the master run, the master enquires the start date of each slave run. A slave may, if necessary, be set to start later than the master. In this case the slave will be “dormant”, not contributing any production or injection, until its start date is reached in the master run.

The events that couple the master and the slave runs are synchronized with the timesteps of the master run. Note that these are the **actual** timesteps, rather than the user-defined **reporting** steps. Thus the coupling

events can be made to occur more frequently (for greater accuracy) by reducing the maximum timestep size in the master reservoir (keyword [TUNING](#)).

For each timestep in the master run:

1. The master must first decide the length of the timestep. This is done in the usual way, but subject to two additional constraints:
 - a. The timestep is limited so as not to overshoot a reporting time in any of the slave runs. (Changes to wells may be scheduled to occur at these times.) The master asks each of the slaves when its next report time is. Timesteps limited in this way are identified by the timestep control mnemonic `SLVR` in the timestep summary written to the Print file.
 - b. The timestep is limited to keep the expected change in flow rate from the slave groups occurring over its length to within certain bounds. This limit is estimated from the changes that occurred over the previous timestep. (See keywords [GRUPMAST](#) and [RCMASTS](#).) Timesteps limited in this way are identified by the timestep control mnemonic `SLVC` in the timestep summary written to the Print file.
2. At the beginning of the timestep, the master enquires from the slaves what their slave groups' production and injection potentials are. These are determined from the reservoir conditions at the beginning of the timestep. They may be used to calculate the guide rates of the master groups, for apportioning the group flow targets.
3. The master allocates rate targets to all its master groups, based on their guide rates. This is done according to the usual procedure for guide rate group control. When a master group is given a rate target, it sends it to its equivalent slave group in the slave reservoir. The slave run calculates the flows of the group under this constraint, at the conditions prevailing at the beginning of the timestep, and tells the master.
4. The master completes its own timestep, assuming that the production and injection rates of the slave groups remain constant over the timestep.
5. When the master has completed its timestep, all the slave reservoirs are advanced to the end of the master's timestep, using the rate targets allocated to the slave groups as constraints. The slaves may take one or more of their own timesteps to reach this time. The timestep that is limited to reach this time is identified by the timestep control mnemonic `MAST` in the timestep summary written to the Print file. `RESTART` and `SUMMARY` files are written at the appropriate times by each slave process, as they would be if the slaves were being run independently. The slave simulation processes can be run in parallel with each other, if they are located on different workstations. Note that events occurring in a slave reservoir cannot influence the master or the other slaves until the next master timestep.
6. When the slave reservoirs have advanced to the new time, they send the new cumulative production and injection totals of the slave groups to the master. From these the master computes the average flow rates of each master group over its timestep. The master sums these flows up to its `FIELD` level, and updates its cumulative totals. The master writes its own `SUMMARY` and `RESTART` files as required, at the end of its timestep.

This procedure continues until the master run finishes. Its final task is to send a signal to the slave runs to terminate. If a slave run finishes before the master run, the master run will continue without any production or injection from the slave. However, the master run can be made to stop in these circumstances by setting the "end run" flag of its master groups in item 8 of keyword [GECON](#). Of course, if a slave run stops due to an error condition, the master and all the other slave runs will also stop.

Group control

The reservoirs are coupled by allocating production and injection rate targets to the slave groups, to meet global production and injection targets specified in the master reservoir. The allocation is made in proportion to the guide rates of the master groups, following the normal procedure of guide rate group control. Thus the master groups must be given guide rates in the master reservoir's data file for production control and any injected phases. The group guide rates may optionally be set as a function of their potential rates (for production), or equal to their voidage replacement needs (for injection).

In each slave reservoir, the target rates of the slave groups are communicated from the master run. These targets can be allocated to subordinate wells or groups by either guide rate control or prioritization. Additional constraints may be applied to subordinate groups in the usual way. However, any groups in the slave reservoir that are superior to the slave groups must not have any constraints on them, as these would interfere with the targets of the slave groups.

If the slave reservoirs were previously simulated independently, they may already have rate targets for the slave groups set in their data files. Accordingly, an option is provided either to ignore any limits specified for the slave groups in their data files, or to take the minimum of these limits and the targets set by the master. The option is independently selectable for each produced and injected phase.

If a rate constraint is applied to a phase that is not an active phase in the slave run, Eclipse transforms it into a constraint on an associated phase, using the latest available production ratios. This may happen, for example, if gas is an active phase in the master run, but not in the slave run, which instead has a constant R_s specified with keyword `RSCONST` or `RSCONSTT`. If the master imposes a gas rate target on a slave group in this reservoir, it is transformed into an oil rate target using the previously calculated GOR of the group.

Overall rate targets applied to the coupled system in the master run will not be met exactly. This is because the master run apportions the rate targets among the master groups according to their slave groups' flow capabilities at the **start** of the timestep, but the flows are reported at the **end** of the timestep. If the flow rate of a slave group changes over the length of the master's timestep, it causes the flow totals in the master reservoir to drift away from their targets. Obviously, the amount of drift increases with the length of the master run's timesteps. There is an option to limit the master run's timesteps to keep the expected drift within a specified tolerance (see item 4 of keyword `GRUPMAST`). The drift also occurs with reinjection and voidage replacement targets applied in the master run. If you wish to impose reinjection or voidage replacement targets on a slave reservoir, they will be met more accurately if they are imposed directly on the slave groups in the slave reservoir's data file than if they are imposed in the master reservoir.

Flowlines and surface networks

If the coupled reservoirs produce into the same flowline or surface network, the reservoir coupling facility can be used in conjunction with the network option to include the network conditions in the constraints that couple the reservoirs (see "[Network option](#)"). Specifically, the networks in each of the reservoirs are coupled by including the **nodal pressures** of the master groups in the set of constraints passed by the master to the slave reservoirs.

The global network should be constructed as follows:

- The master reservoir should contain a representation of the global network down to the master groups (and its own well groups also, if the master represents an actual reservoir).
- In each slave reservoir, the slave groups should be defined as fixed pressure nodes, and the network extended down from them to the well groups. Any groups superior to the slave groups should not be part of the network in the slave reservoirs.

The master and each slave may use either the Standard Network model or the Extended Network model, which allows the network structure to differ from the `GRUPTREE` hierarchy (see "[The extended network model](#)"). The Extended Network structure and the `GRUPTREE` hierarchy must however coincide at the master groups (in the master reservoir) and the slave groups (in the slave reservoir). Specifically, the requirements for the Extended Network structure are:

- If the master reservoir uses the Extended Network model, the master groups must coincide with source nodes in the network. For each master group there must be a corresponding network source node with the same name.
- If a slave reservoir uses the Extended Network model, the slave groups must coincide with fixed-pressure nodes in the network. For each slave group there must be a corresponding network fixed-pressure node with the same name.

The networks in each reservoir should be set up to be balanced at each timestep (which is the default in keyword `NETBALAN`). Thus all networks will be balanced simultaneously at each timestep of the master process. At each iteration of the network balancing calculation in the master:

1. The master sends the nodal pressures of the master groups to the respective slave groups in the slave reservoirs.
2. The slave reservoirs balance their own networks, using the nodal pressures passed from the master as the “fixed” pressures of the slave groups.
3. The slave processes send back to the master the flows of the slave groups at the current conditions.
4. The master sums the flows up to its `FIELD` level, and updates the pressures in the nodes of its network in the usual way.

Note that at each balancing iteration in the master run, the slaves balance their own networks to convergence.

If the slave reservoirs take more than one timestep to advance to the end of the master timestep, they will rebalance their own networks at each step. However, the “fixed” pressures of the slave groups will remain constant until they are updated again at the next master timestep.

Reservoir coupling files

During the simulation of the coupled system, the master run can be made to write the flow rates and coupling constraints of the master groups to a **reservoir coupling file**. At each timestep of the master run, the following information is written to this file for each master group:

1. The group’s coupling constraints. These include the rate target applied to the group by the master run. Other rate limits are included, to prevent the group increasing its flow sufficiently to violate any rate limits imposed on higher level groups, assuming the flows of all other groups remain constant. If the group is part of a pipeline network, its network nodal pressure is also included.
2. The group’s potential flow rates. These are the flow rates that the coupled slave group’s subordinate wells would achieve, at the current grid block conditions, if all rate limits were ignored and only their THP and BHP limits were observed. The potentials may be used to determine guide rates for group control.
3. The group’s production and injection rates. These are the rates that the coupled slave groups have achieved, averaged over the length of the master run’s current timestep. These rates are reported in the master run.

The reservoir coupling file will be written by the master run if the keyword `DUMPCUPL` is present in its data file. There is a choice to make this either a formatted or an unformatted file. Formatted files occupy

much more disk space than unformatted files, but should be chosen if the reservoir coupling file is going to be used on a system that does not have binary compatibility with the one on which it was created.

The reservoir coupling file may subsequently be used for either of two purposes:

- To substitute for one or more of the slave reservoirs in the master run. The flow rates of the slave groups will be read from the file and incorporated into the master groups' flows.

This would allow the master to be run with a reduced number of coupled slaves. In fact, if all the slaves were replaced by the data on the Reservoir Coupling file, the master could be run on its own without requiring MPI to activate the slave runs.

Obviously, the slave group flows read from the file cannot be influenced by any changes made to the coupled system. So be careful about making changes that would have significant repercussions on the slave reservoirs. Use of the reservoir coupling file for this purpose has similarities with the flux boundary conditions option.

- To drive a slave reservoir simulation on its own, supplying the rate (and perhaps network pressure) constraints that were originally dictated by the master. MPI is not required for this task.

At a time corresponding to the start of each timestep of the original master run, the rate (and network pressure) constraints of the reservoir's slave groups are read from the file. These are applied to the groups until the next master timestep. For additional flexibility, you can choose to ignore limits for particular produced or injected phases. For each produced and injected phase, there is an option to take the rate constraints as specified in the slave's data file, or to take the rate constraints from the Reservoir Coupling file, or to take the minimum of the two constraints. (See keyword [GRUPLAV](#).)

A set of SUMMARY section keywords is provided to output the rate limits the master groups are subject to, as a function of time. This offers a useful way of displaying the rate limits imposed by the reservoir coupling file.

When a Reservoir Coupling file is used for either of the two purposes described above, the timesteps of the simulation are limited so as not to overshoot the timesteps of the original master run that produced the file. At these times the flow rates and the constraints on the file may change. Timesteps limited in this way are identified by the timestep control mnemonic MAST in the timestep summary written to the print file.

If the original coupled runs were performed over a series of restarts, there should be a sequence of reservoir coupling files, each having the root name of the data file of the master run that wrote it. Each file has a header that contains the root name of the data file from which it was restarted. Thus Eclipse is able to establish the chain of reservoir coupling files to be used, provided that they are all present.

Current restrictions

- The global constraints that couple the reservoirs are limited to group rate control (using group guide rates) and network pressures. Other features (for example prioritization, economic limits, automatic drilling and workover facilities, group production rules) can be used within individual reservoirs, but cannot be employed globally. The master run does not have information about the individual wells in the slave reservoirs, and cannot directly command drilling, workover, shut-in or production rule actions ([PRORDER](#)) on these wells. In particular, [GECON](#) should not be used to apply economic limits to master groups or their superiors, and no procedure other than RATE should be specified for these groups in [GCONPROD](#). Prioritization (keyword [GCONPRI](#)) cannot be applied to these groups.
- Global rate targets and limits will not be met exactly, due to the flows of the slave groups changing over the length of the master timestep.
- It is not possible to guarantee that all current and future Eclipse facilities may be used in coupled reservoirs. In particular, Gas Lift Optimization cannot be used at present (see "[Gas lift optimization](#)").

However, individual reservoirs may use Local Grid Refinement, Vertical Equilibrium, the Flux option, tracers, and polymer or surfactant floods. The Gas Field Operations Model may be used to a limited extent with Reservoir Coupling (see ["Using the reservoir coupling facility"](#)).

- The master timestep cannot be influenced by events happening in the slave reservoirs, other than observing their report times. For example, automatic drilling or workovers happening in a slave reservoir will not cause the master to chop or repeat its timestep.
- The master run and all the slave runs must have the same unit convention. The slave runs need not all have the same set of active phases, but all phases active in any slave must be active in the master.
- The order of solving the different types of wells (for example producers, water injectors, gas injectors) must be the same in all the coupled reservoirs. Thus, all reservoirs must have the same top-up phase for voidage replacement or reservoir volume rate injection targets, as these injectors are solved last. For example, one reservoir cannot have water voidage replacement targets if another has gas voidage replacement targets. Similarly, either none or all of them must use production balancing, which requires the producers to be solved after the injectors.
- Automatic compressors in an extended network (see keyword [NETCOMPA](#)) cannot respond to falling production from a group in a different reservoir. For example, compressors in a slave reservoir cannot be activated by a group in the master reservoir failing to meet its [GCONPROD](#) target.

Using the reservoir coupling facility

The reservoir coupling facility uses the MPI message passing system to send messages and data between the master and the slave runs.

Reservoir coupling using MPI

To run a coupled reservoir simulation MPI must first be installed on your system. Refer to the "Eclipse Suite Installation Guide" for instructions on how to install MPI.

By using MPI as the message passing system slave reservoirs may be run either as serial processes or in parallel. To run a slave reservoir in parallel the number of processors on which it is to be run must be specified in the master data set using the [SLAVES](#) keyword. The [PARALLEL](#) keyword should also appear in the slave data set with the number of processors on which the slave is to be run as the first argument.

It is not possible to run a master reservoir in parallel.

Starting a reservoir coupling simulation using MPI can be done using `eclrun`. For example, `eclrun eclipse <dataset>`.

The master reservoir

The master reservoir is represented by a standard Eclipse data file. It may, if required, be a restart from a previous run. It must have as active phases all the phases that are active in any of the slave runs. It must contain the group hierarchy down to the master groups that couple it to the slaves, and of course the hierarchy must include its own well groups if it has any. The group hierarchy is defined in the usual way with keyword [GRUPTREE](#).

The master does not necessarily have to represent a real reservoir. It may just be a "dummy reservoir" whose purpose is to couple the slave reservoirs through its production and injection constraints (and optionally a pipeline network). In this case, the data file should still contain enough information to run Eclipse. At least one grid block is necessary. Fluid properties should be provided as required for its active phases. No wells should be defined, but the group hierarchy must be defined down to the master groups.

The **SLAVES** keyword must be used in the **SCHEDULE** section to activate the coupled slave runs. The keyword data specifies the machine on which each slave process is to be run, and the directory where their data files are located. For example, for figure 9.3:

```
SLAVES
  'RES-R2' 'base'      'rios'      '/usr/models/res2' /
  'RES-R3' 'base'      'sg-indigo'  '/usr/models/res3' /
/
```

If you wish the master run to create a Reservoir Coupling file containing the rates and constraints of all the master groups, enter the keyword **DUMPCUPL**:

```
DUMPCUPL
  'U' /
```

After the group hierarchy has been set up with keyword **GRUPTREE**, the master groups must be identified with keyword **GRUPMAST**. The keyword also specifies the slaves to which the master groups are coupled, and sets a limiting fraction for flow rate change in a single timestep. For example, for figure 9.3:

```
GRUPMAST
  'R2'      'RES-R2'      'FIELD'      0.1 /
  'R3-A'    'RES-R3'      1*          0.1 /
  'R3-B'    'RES-R3'      1*          0.1 /
/
```

If limiting rate change fractions have been set for the master groups, it is advisable to set a minimum length to which the timestep can be reduced by this mechanism, using keyword **RCMASTS**:

```
RCMASTS
  10.0 /
```

Set the global production rate targets and limits with keyword **GCONPROD**, with action 'RATE' selected to activate group control if limits are exceeded. Other actions, for example 'WELL' or 'CON' will not work with master groups, because the master run cannot request workovers and shut-ins for wells in the slaves. For similar reasons, prioritization group control (keyword **GCONPRI**) cannot be used for master groups and any of their superior groups. The master groups must have their production guide rates set, so that higher level rate targets can be apportioned among these groups (rather than their wells). The group guide rates can either be set to a fixed value, or calculated as a function of their production potentials (see keyword **GCONPROD** items 9 and 10).

```
GCONPROD
  'FIELD' 'ORAT' 4.E5 /
  'PLAT*' 'NONE' 3*          3.E5 'RATE' /
  'R1-*'  'FLD' 6*          100 'LIQ' /
  'R3-*'  'FLD' 6*          100 'LIQ' /
  'R2'    'FLD' 6*          200 'LIQ' /
/
```

If there are any global injection targets or limits to be applied to the coupled system, set them with keyword **GCONINJE** in the usual way. Any master group subordinate to a group with an injection target or limit must be given an injection guide rate for that phase, so that the higher level target can be apportioned down to it. The guide rate can optionally be set equal to the group's voidage replacement requirements (**GCONINJE** item 10). If reinjection or voidage replacement is to be performed independently in each slave, and there are no global injection constraints, the targets will be met more accurately if they are specified in the slave data files rather than the master, because of the "drift" caused by flow rate changes over the master timestep.


```

GCONINJE
'PLAT*' 'WAT' 'REIN' 2*          1.0 /
'R*'    'WAT' 'FLD'  6*          'VOID' /
/

```

If there is a pipeline network coupling the reservoirs, specify it in the usual way with keyword [GRUPNET](#) (or [BRANPROP](#) and [NODEPROP](#) for an Extended Network), defining it down to the master groups. If the Extended Network model is used, the master groups must coincide with source nodes in the network; for each master group there must be a corresponding network source node with the same name. If using keyword [NETBALAN](#), ensure it is set up to balance the network at every timestep (which is the default).

If you wish the master run to stop when a slave run finishes for any reason, set its master group's end-run flag in item 8 of keyword [GECON](#). Do **not** set any of the other [GECON](#) limits, as the master groups effectively contain no wells, and the master run cannot control workovers in the slaves. Economic limits can, of course, be set in the slave data files.

Set the report times or dates in the usual way. For easier synchronization it is best to use [DATES](#), especially if the slave runs start on different dates. If this is the case, it is also advisable to use the [DATE](#) keyword in the SUMMARY section. Aim to set common report dates in each coupled simulation. This is especially important for restarts, as the coupled simulations can only be restarted correctly from RESTART files written on the same date.

The slave reservoirs

The slave reservoirs are represented by standard Eclipse data files. These may be restarts from previous runs. You should ensure that they are set up to start at the required date. If they are restart runs, ensure also that the correct file name and report number are entered in the [RESTART](#) keyword.

A slave run may be set to start at a date later than the start of the master run. In this case the slave will be “dormant”, having no production or injection, until its start date is reached in the master run. Note that the simulation time will not be the same as in the master run, so it is best to use the [DATES](#) keyword to specify report times, and also to include the [DATE](#) keyword in the SUMMARY section to present the results by calendar date rather than simulation time. An error message will be issued if the slave run is set to start earlier than the master run.

The only additional keyword required in a slave run's data file is [GRUPLAV](#). This identifies the slave groups, and the associated master groups to which they are coupled. You can also set instructions to ignore any rate limits set in the slave run's data file, or take the minimum of these limits and the limits imposed by the master run. For example, for slave reservoir R2 in Figure 9.3, using the default option of observing only the rate limits communicated from the master for all phases:

```

GRUPLAV
-- slave  assoc  oil    wat/liq  gas    resv    oil    wat    gas
-- group  master prod    prod    prod    prod    inj    inj    inj
-- name   group  limits limits limits limits limits limits limits
'FIELD'  'R2'    /
/

```

The slave groups' production rate can be controlled by either guide rate control or prioritization. If you wish to use guide rate control, reference the slave groups with keyword [GCONPROD](#), selecting 'RATE' as the action to be performed on violating any limits:

```

GCONPROD
-- group  ctl  oil    wat    gas    liq    limit  field  guide  g-r
-- name   mode rate  rate  rate  rate  action ctrl?  rate  phase
'FIELD'  'ORAT' 1.0    3*          'RATE' /
/

```

The default option in GRUPSLAV will overwrite any rate limits specified here. Alternatively, if you wish to use prioritization, reference the slave groups with keyword **GCONPRI**, selecting 'PRI' as the action to be performed on violating each limit (the **PRIORITY** keyword must previously have been entered):

```
GCONPRI
-- GROUP      OIL      OIL      WAT      WAT      GAS      GAS      LIQ      LIQ      RESV  PBAL
-- NAME      RATE     ACTN    RATE     ACTN    RATE     ACTN    RATE     ACTN    RATE  FRAC
-- 'FIELD'    1*     'PRI'    1*     'PRI'    1*     'PRI'    1*     'PRI'    /
/
```

It is not strictly necessary to set up injection controls for the slave groups with keyword **GCONINJE**, as injection rate limits imposed by the master will automatically be handled by guide rate control. If injection constraints are already specified in the data file, you can choose either to ignore them or take their minimum with the limits imposed by the master run (see keyword **GRUPSLAV**). Reinjection and voidage replacement targets applied in the master reservoir to groups superior to the master groups are translated into surface rate and reservoir volume rate targets respectively at the slave groups. If you wish to apply reinjection or voidage replacement targets directly to the master groups, they will be observed more accurately if they are applied instead directly to their corresponding slave groups in the slave reservoir.

Additional production and injection constraints may be applied to subordinate groups in the usual way. But any groups in the slave reservoir that are superior to the slave groups (for example 'FIELD' in slave reservoir R3) must not have any constraints on them, as these would interfere with the targets of the slave groups.

If there is a pipeline network in the slave reservoir, specify it in the usual way with keyword **GRUPNET** (or **BRANPROP** and **NODEPROP** for an Extended Network), defining it from the slave group(s) down to the well groups. The network in the slave reservoir must not extend higher than the slave groups. To couple the network to the master reservoir's network, the slave groups must be defined as fixed-pressure nodes in **GRUPNET** or **NODEPROP**. In an Extended Network, for each slave group there must be a corresponding network fixed-pressure node with the same name. The value of the fixed pressure specified here does not matter, as it will be over-written by the nodal pressure communicated from the master. If using keyword **NETBALAN**, ensure it is set up to balance the network at every timestep (which is the default).

There should be sufficient **DATES** keywords to advance the simulation up to the end of the run. If you “overshoot”, the slave run is automatically terminated when the master run ends. If you “undershoot”, the slave run will stop at its last report date, and the master run will continue without any production or injection from this slave (unless told to stop with an “end-run” flag in keyword **GECON**).

Choose the report dates carefully. The timesteps in the master run are synchronized so as not to overshoot a report date in any slave. Thus if the report dates differ slightly between the slaves and the master run, this will restrict the timestep size throughout the set of coupled reservoirs. So it is best to have a common set of report dates. You will certainly need to synchronize the report dates on which **RESTART** files are to be written, if you wish subsequently to restart the coupled simulation.

Reading a reservoir coupling file in the master reservoir

A reservoir coupling file can be used to take the place of one or more of the slave reservoirs. Instead of activating their slave runs, Eclipse reads their flow rates from this file. To use this option, make the following changes to the master reservoir's data file:

- Add the keyword **USECUPL**, to specify the name and format of the Reservoir Coupling file to be read. If the run spans a sequence of Reservoir Coupling files generated by a series of restarts of the master run, enter the name of the last file in the sequence.
- Remove the particular slave reservoir's data record from the **SLAVES** keyword. If there are no more slave reservoirs to be actively coupled to the master, the **SLAVES** keyword should be removed.

- In keyword **GRUPMAST**, the master groups associated with the substituted slave reservoir must have their slave reservoir name in item 2 defaulted.

For example, to substitute slave reservoir R3 in figure 9.3 with a Reservoir Coupling file written by the run 'master.data':

```

USECUPL
'master' 'U' /
SLAVES
-- slave   datafile   machine   directory
-- name    root       hostname  of data file
--
-- 'RES-R2' 'base'     'rios'   '/usr/models/res2' /
/
GRUPMAST
-- master   slave      associated  limiting
-- group    reservoir  slave     rate change
-- name      name       group     fraction
-- 'R2'      'RES-R2'   'FIELD'   0.1 /
-- 'R3-A'    1*         1*        /
-- 'R3-B'    1*         1*        /
/

```

Driving a slave reservoir with a reservoir coupling file

A slave run may be performed on its own, instead of being activated by the master run. It will read the Reservoir Coupling file to obtain the constraints that the master run originally applied to its slave groups over the course of the simulation. MPI is not required for this option.

To use this option, set up the slave's data file as though it were going to be activated by the master, but in addition include the keyword **USECUPL** as described above. The slave groups are identified with the keyword **GRUPSLAV**, and you can choose whether to take their rate constraints from the Reservoir Coupling file, the slave's own data file, or the minimum of both.

You must set up the slave groups for production control (with keyword **GCONPROD** for guide rate control or keyword **GCONPRI** for prioritization), and for injection control of the injected phases (with keyword **GCONINJE**). Reinjection and voidage replacement targets applied in the master reservoir to groups superior to the master groups are translated into surface rate and reservoir volume rate targets respectively at the slave groups.

If the master and slave reservoirs are coupled by a pipeline network, the nodal pressures of the slave groups that were set by the master in the original coupled run will also be read from the Reservoir Coupling file. The slave groups must of course be defined as fixed-pressure nodes in keyword **GRUPNET**.

A useful way of monitoring the production and injection targets applied to the slave groups is to write them to the SUMMARY file using the SUMMARY section keywords **GOPRL**, **GOIRL**, **GWPR**, **GWIRL**, **GGPRL**, **GGIRL**, **GLPRL**, **GVPR**, **GVIRL**. See the **SUMMARY** section description in this manual. Note that the value of these SUMMARY quantities will differ between master and slave groups. The value reported in the slave group is the overall target applied owing to the combination of all limits on the group. The value reported by the same SUMMARY quantities for the master group will be the rate limit set in the **GCONPROD**, **GCONPRI** or **GCONINJE** keywords.

As an example, to drive slave reservoir R3 in figure 9.3 from the Reservoir Coupling file written by the run 'master.data', but imposing an additional limit on the gas production:

```

USECUPL
'master' 'U' /
GRUPSLAV
-- slave  assoc  oil    wat/liq  gas    resv    oil    wat    gas
-- group  master prod   prod    prod    prod    inj     inj     inj

```

```

-- name      group      limits      limits      limits      limits      limits      limits      limits
-- 'R3-A'     1*         1*         1*         'BOTH'      /
-- 'R3-B'     1*         1*         1*         'BOTH'      /
/
GCONPROD
-- group      ctl      oil      wat      gas      liq      limit      field      guide      g-r      .....
-- name      mode     rate     rate     rate     rate     action     ctrl?     rate     phase     .....
-- 'R3-*'     'ORAT'  1.0      1*      2.E5     1*      'RATE'     /
/
GCONINJE
-- group      inj      ctl      surf     resv     reinj     vrep      field      guide      g-r
-- name      phase    mode     rate     rate     frac      frac      ctrl?     rate     defn
-- 'R3-*'     'WAT'   'RATE'   10      /
/

```

Using the gas field operations model in coupled reservoirs

The gas field operations model (see "[Gas field operations model](#)") may be used to a limited extent in coupled reservoirs. The master run handles all the decisions concerning the calculation of DCQs, while the slaves take the more passive role of communicating to the master their production rates over each pass of the contract year, and their delivery capacities. The master run synchronizes with the slaves to start each pass of the contract year, and sets the target rate of each slave group every month for calculating the swing, the profile, and delivery capacities.

The restrictions on running the model are as follows.

- All contract groups must be situated in the master reservoir; there must be no contract groups in the slaves. The master run should contain the keyword [SWINGFAC](#) (for a single contract group) or [GSWINGF](#) (for multiple contract groups). These keywords must not be used in the slave runs.
- Each slave must use the [GASYEAR](#) or [GASPERIO](#) keyword in synchronization with the master. Each contract period must start on the same date and be of the same length in all the coupled runs. In the slave runs, all [GASYEAR](#) keyword items except item 1, or all [GASPERIO](#) keyword items except items 1 and 2, are ignored and should be defaulted, as there are no contract groups that require their DCQ to be calculated. In the master run, the timesteps will be limited to one month, regardless of the setting of [GASYEAR](#) item 4 or [GASPERIO](#) item 5.
- If the [PRO](#) option is selected in item 3 of keyword [GASYEAR](#) or item 4 of [GASPERIO](#), the master run cannot activate the drilling queue in a slave reservoir to keep the delivery capacity above DCQ x swing factor. This is because the master has no information about individual wells in the slave reservoirs.
- All automatic compressors (see keyword [GASFCOMP](#)) must be situated in the master reservoir. Any in the slave reservoirs will be ignored.
- In the master reservoir, there must be no gas consumption or import rate specified for the master groups (keyword [GCONSUMP](#)). This is because their consumption or import rates are set equal to those of their corresponding slave groups in the slave reservoirs. This also applies to consumption by automatic compressors. An automatic compressor (keyword [GASFCOMP](#)) located at a master group should not be given a gas consumption rate.
- Reservoir Coupling files should only be used with care. They store the master groups' flows during the second pass of the contract year, when producing at DCQ x profile factor. But the flows during the first pass (when producing at DCQ x swing factor) are **not** recorded. Thus if you replace a slave reservoir by a reservoir coupling file, its production rate in the first pass will be the same as in the second pass. This will only be correct when the swing factors are the same as the profile factors. A Reservoir Coupling file cannot be used to drive a slave on its own when it is using the gas field operations model.

Sharing licenses with the slaves

There was a major change in the licensing model used with Reservoir Coupling in 2005A. Prior to 2004A a reservoir coupling job would check out only one `eclipse` license and one feature license no matter how many SLAVES there were.

From 2005A this policy was modified so that each SLAVE requires an Eclipse license as well as the master. Each SLAVE also requires one feature license for every feature it uses.

The following example indicates how works:

Master

SLAVE 1 uses LGR and Multisegwells

SLAVE 2 uses LGR and networks

SLAVE 3 uses networks

Prior to 2004A the following licenses would have been required to run this model:

- 1 `eclipse`
- 1 `lgr`
- 1 `network`
- 1 `multisegwells`
- `rescoupling`

Since 2005A this is:

- 4 `eclipse`
- 2 `lgr`
- 2 `network`
- 1 `multisegwells`
- 1 `rescoupling`

Reservoir coupling file structure

The reservoir coupling file structure is similar to the structure of other files that Eclipse reads and writes. This is described in "[File handling in Eclipse](#)". The file consists of a series of data blocks, each headed by a descriptor record. But there is an additional type of block descriptor record that is not mentioned in the File Handling chapter: the "message record". The format of this type of record is:

- An 8-character keyword
- The integer zero
- The 4-character string 'MESS'

This type of record is **not** followed by a data block - the only information the record communicates is the 8-character keyword itself.

The file begins with the descriptor record

```
'RESTART' 9 'CHAR'
```

that is followed by the root name of the input `RESTART` file, divided into 9 8-character elements. If the run is a base run and not a restart, the 9 elements are all blank.

The next descriptor record is

```
'INTEHEAD' 5 'INTE'
```

that is followed by the integer header block containing:

- 1 = the internal version number of the program release that wrote the file (for example 9501). This is used in future releases to identify files from previous releases.
- 2-4 = the day (1-31), month number (1-12) and year (for example 1994) of the start of the master base run (as set in `RUNSPEC` section keyword `START`).
- 5 = the unit convention used (1 = metric, 2 = field, 3 = lab).

Next, for every timestep of the master run, the following records are written:

```
'Timestep' 2 'DOUB'
```

followed by the start and end time of the timestep, in days from the master base run's start date.

For each master group, the following records are written:

```
'GRUPNAME' 2 'CHAR'
```

followed by the name of the master group and the name of its associated slave group in the corresponding slave run.

```
'RATES' 22 'DOUB'
```

followed by the group's production and injection rates of each phase and other information to account for free and dissolved gas, gas consumption and lift gas:

- 1-4 = oil, water, gas and reservoir volume production rates
- 5-8 = oil, water, gas and reservoir volume injection rates
- 9-11 = separate reservoir volume rates of oil, water and gas injection
- 12-13 = total gas consumption and import rates at and below the group
- 14-15 = vaporized oil and dissolved gas production rates (used for `SUMMARY` section output generated by the keywords `GOPRF`, `GOPRS`, `GGPRF` and `GGPRS` for example)
- 16 = lift gas supply rate, reduced by well efficiency factors
- 17 = lift gas supply rate, **not** reduced by well efficiency factors (Items 16 and 17 are intended for future use in lift gas accounting.)
- 18-22 = oil, water, gas, resv rates and gas consumption at delivery capacity (Gas Field Operations Model)

23-27= energy production and injection rate, total energy consumption rate at and below this group, energy production rate at delivery capacity and energy consumption rate at delivery capacity (Gas Calorific Value Control).

```
'POTNS ' 10 'DOUB'
```

followed by the group's potential production and injection rates of each phase (that is the sum of the well potentials):

1-4 = oil, water, gas and reservoir volume production potentials

5-6 = oil surface rate and reservoir volume rate injection potentials

7-8 = water surface rate and reservoir volume rate injection potentials

9-10 = gas surface rate and reservoir volume rate injection potentials.

11 = energy production rate potential (gas calorific value control)

12 = free gas production potential

```
'CTLMODE ' 4 'INTE'
```

followed by integers defining the group's acting production and injection control modes:

1 = production control mode

(0 = NONE, 1 = ORAT, 2 = WRAT, 3 = GRAT, 4 = LRAT, 5 = RESV, 6 = PRBL, 7 = ERAT)

2-4 = oil, water and gas injection control modes

(0 = NONE, 1 = RATE, 2 = RESV, 3 = REIN, 4 = VREP)

```
'LIMITS ' 21 'DOUB'
```

followed by the limiting production and injection rates of each phase, and the group's nodal pressure if there is a network:

1 -5 = oil, water, gas, liquid and reservoir volume production rate limits (default = 1.0E20)

6 = limiting production balancing fraction (default = 1.0E20)

7-8 = oil surface rate and reservoir volume rate injection limits (default = 1.0E20)

9-10 = oil reinjection fraction and voidage replacement fraction limits (default = 1.0E20)

11-12 = water surface rate and reservoir volume rate injection limits (default = 1.0E20)

13-14 = water reinjection fraction and voidage replacement fraction limits (default = 1.0E20)

15-16 = gas surface rate and reservoir volume rate injection limits (default = 1.0E20)

17-18 = gas reinjection fraction and voidage replacement fraction limits (default = 1.0E20)

19= nodal pressure in production network (default = 0.0)

20-21 = nodal pressure in water and gas injection networks (default = 0.0).

22 = Energy production rate limit (default = 1.0E20)

```
'ENDSTEP' 0 'MESS'
```

to signify the end of the information for the current timestep.

The **RATES** and **POTNS** keyword data are required when the reservoir coupling file is used in the master run to substitute for a slave reservoir. The potentials are used to calculate group guide rates if a superior group has been given a guide rate of 0.0 in **GCONPROD**.

The `CTRLMODE` and `LIMITS` keyword data are required when the reservoir coupling file is used to drive a slave run on its own. These keywords are not written to the file for groups whose slave reservoirs are “dormant” (that is, whose start date has not yet been reached by the master run).

The structure and contents of the file are likely to evolve as the Reservoir Coupling facility is developed further, but back-compatibility is maintained wherever possible.

Reservoir coupling files can be written either formatted (as a text file) or unformatted (as a binary file). Formatted files occupy much more disk space than unformatted files, but this type should be chosen if the file is going to be used on a system that does not have binary compatibility with the one on which it was created. The Convert utility can convert reservoir coupling files from one format to the other (select the “General” file type).

The reservoir coupling file that is created by the master run has the same root name as the master run's data file. The filename's extension is `.CPL` for an unformatted file, and `.FCP` for a formatted file.

Example section of a reservoir coupling file

'RESTART'		9	'CHAR'										
'INTEHEAD'		5	'INTE'										
	9402		1		1		1994				2		
'TIMESTEP'		2	'DOUB'										
0.000000000000000D+00			0.100000000000000D+01										
'GRUPNAME'		2	'CHAR'										
'R3-A'	'R3-A'												
'RATES'		27	'DOUB'										
0.99995623191871D+04			0.43768081294321D+00			0.14973184139827D+05							
0.11192063433604D+05			0.00000000000000D+00			0.00000000000000D+00							
0.15000000000000D+05			0.87697823644459D+04			0.00000000000000D+00							
0.00000000000000D+00			0.87697823644459D+04			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.14973184125069D+05							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.00000000000000D+00							
'POTNS'		11	'DOUB'										
0.68819213068446D+05			0.00000000000000D+00			0.10322881960267D+06							
0.77047083871076D+05			0.00000000000000D+00			0.00000000000000D+00							
0.00000000000000D+00			0.00000000000000D+00			0.67192104926539D+05							
0.39284009120984D+05			0.00000000000000D+00										
'CTLMODE'		4	'INTE'										
	4		0		2		1						
'LIMITS'		22	'DOUB'										
0.100000000200409D+21			0.100000000200409D+21			0.55036185820205D+05							

```

0.100000000000000D+05 0.10000000200409D+21 0.10000000200409D+21
0.10000000200409D+21 0.10000000200409D+21 0.10000000200409D+21
0.10000000200409D+21 0.10000000200409D+21 0.87697823644459D+04
0.10000000200409D+21 0.10000000200409D+21 0.15000000000000D+05
0.10000000200409D+21 0.10000000200409D+21 0.10000000200409D+21
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00

'GRUPNAME'          2 'CHAR'
'R3-B'              'R3-B'
'RATES'             27 'DOUB'
0.99989832425748D+04 0.10167574252128D+01 0.14963814179795D+05
0.11190869653077D+05 0.00000000000000D+00 0.91405301592612D+04
0.00000000000000D+00 0.91431416015625D+04 0.00000000000000D+00
0.91431416015625D+04 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.14963814179716D+05
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00

'POTNS'             11 'DOUB'
0.35029663767571D+05 0.00000000000000D+00 0.52544495651356D+05
0.39217731821364D+05 0.00000000000000D+00 0.00000000000000D+00
0.36486043687679D+05 0.36496467721760D+05 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00

'CTLMODE'           4 'INTE'
'LIMITS'             22 'DOUB'
0.10000000200409D+21 0.10000000200409D+21 0.55026815860173D+05
0.10000000000000D+05 0.10000000200409D+21 0.10000000200409D+21
0.10000000200409D+21 0.10000000200409D+21 0.10000000200409D+21
0.10000000200409D+21 0.10000000200409D+21 0.91431420260235D+04
0.10000000200409D+21 0.10000000200409D+21 0.00000000000000D+00
0.10000000200409D+21 0.10000000200409D+21 0.10000000200409D+21
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00

'GRUPNAME'          2 'CHAR'
'R2'                'FIELD'
'RATES'             27 'DOUB'
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00

'POTNS'             11 'DOUB'
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
0.00000000000000D+00 0.00000000000000D+00

'ENDSTEP'           0 'MESS'

```

Summary of keywords

SUMMARY section

Refer to [Reservoir coupling facility](#) in the *Eclipse Reference Manual*.

SCHEDULE section

Keyword	Description
DEBUG	Switch 39 turns on debug from the systems annex for MPI calls. Switch 42 turns on debug from the reservoir coupling facility.
DUMPCUPL	Instructs the master run to write a reservoir coupling file, containing the flows and applied constraints of the master groups.
GRUPMAST	Must be used in the master run to identify the master groups and to indicate which slaves they are associated with.
GRUPLAV	Must be used in each slave run to identify the slave groups and their associated master groups in the master run, and to set the filter flags that determining which rate constraints are to be imposed on them
RCMASTS	Sets a minimum length for timesteps restricted by a group's limiting fractional change in flow rate.
SLAVES	Must be used in the master run to identify the slave run data files and to indicate where they are located; their simulation processes will be started when this keyword is read
USECUPL	Instructs Eclipse to read group flows or constraints from a reservoir coupling file

Network option

x	Eclipse 100
x	Eclipse 300

The network option is designed to provide variable tubing head pressure (THP) limits to groups of wells, which depend on the groups' flow rates according to a set of pipeline pressure loss relationships. The option calculates the well THP limits dynamically by balancing the flow rates and pressure losses in the network.

Consider a group of wells flowing into a common manifold which is connected by a pipeline to an outlet at a fixed pressure (figure 9.5). For example, the group may represent the wells flowing into a particular subsea completion manifold, and the fixed pressure outlet may be a separator or stock tank. The pressure in the manifold will vary with the overall production rate of the group, according to the relationship between the pressure loss through the pipeline and the flow rate, water cut and GOR of the group. This relationship is represented schematically in figure 9.6 by the pipeline curve. However, the group's production rate will itself vary with the manifold pressure if at least one of its wells is on THP control, since the manifold pressure is applied to each well as a THP limit. The combined production rate of the wells versus the THP limit is represented by the wellbore curve in figure 9.6. The actual manifold pressure and flow rate of the group will be given by the intersection of these two curves.

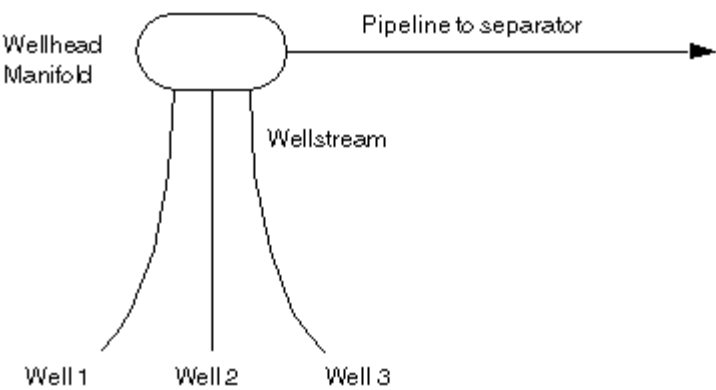


Figure 9.5. Common wellhead manifold and pipeline

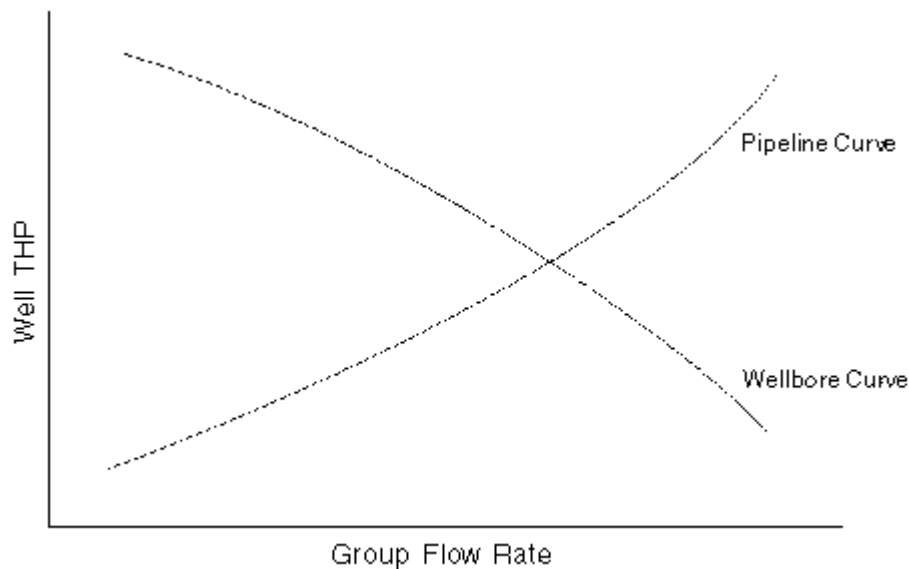


Figure 9.6. Well THP versus group flow rate for wellstream and pipeline

Using the Eclipse multilevel grouping hierarchy, a pipeline network can be built up, with a pipeline connecting each group to its 'parent' in the grouping tree. A group of wells connected to a common manifold passes its production stream through a pipeline to its parent group, which gathers the production from a number of common well manifolds and sends it along another pipeline to its own parent group, and so on up the grouping tree until a group with a fixed-pressure separator or stock tank is encountered. Each group is a 'node' in the network. The group with the separator is a 'fixed-pressure node', and the groups subordinate to this have their own 'nodal pressures,' which are determined from the pipeline pressure losses and the production rates. The nodal pressures of each well-group at the bottom of the tree are applied to their wells as THP limits. (Note that any fixed THP limits specified for the wells will be overwritten by the calculated values.) The nodal pressure of each group is printed in the well reports in the THP column, and can also be written to the SUMMARY file using the mnemonic GPR.

For example, consider the grouping tree shown in Figure 9.7. The groups on the bottom level (GR-A1 for example) may be subsea completion manifolds, connected to their respective platforms PLAT-A and PLAT-B by pipelines. The two platforms can be declared as fixed-pressure nodes, with pressures set equal to their separator pressures. Alternatively the FIELD can be given a fixed pressure, and the two platforms connected to it by additional pipelines. It is not essential for the network to cover the whole grouping tree. For example group GR-A2 can be isolated from the network (by not giving it a pipeline connection to PLAT-A), and THP limits for its wells set or defaulted independently.

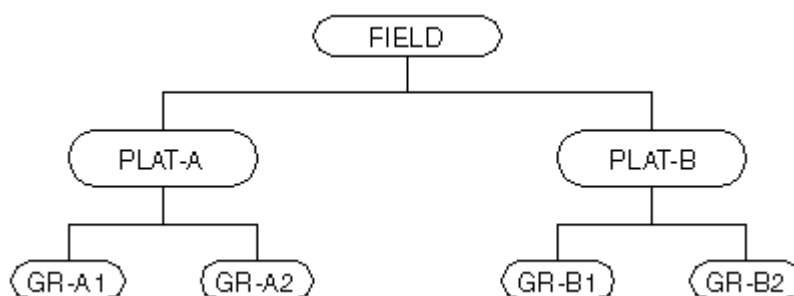


Figure 9.7. An example network

Note that the network structure coincides with the grouping hierarchy built using the `GRUPTREE` keyword. If you require the network to have a different topology from the group control hierarchy, then you must use the extended network model which is described in a later section (see "The extended network model").

Group production control in the network option

Group control targets and limits can be applied to any group regardless of whether it is part of a pipeline network. A production rate target applied to a group in the network will be met as long as its subordinate wells have sufficient potential. The wells' potentials will of course depend on the nodal pressures in the network, since these determine the wells' THP limits.

Consider, for example, a target production rate of 10,000 STB/DAY of oil applied to the group GR-A1. The THP limits of each of its wells are set equal to the group's nodal pressure in the network. The standard procedure for group control will apportion the rate target among the group's wells in proportion to their oil phase guide rates (which by default are set equal to their production potentials). If some wells cannot produce at their allocated rate targets, they will produce as much as they can under their own flow and pressure limits, while the remaining wells under group control make up the rest of the group's flow target.

When a well is under group control, it will be producing at the rate target allocated to it. Its THP has a value greater than the group's nodal pressure in the network (otherwise the well will have changed to THP control). The THP values of the group-controlled wells in this group are in general all different. However, if the group represents a sub-sea completion manifold, in reality all its wells are operating at the same THP, that is the pressure in the manifold. The standard method of group control by guide rates is therefore not appropriate for groups in which all the wells share a common manifold.

Eclipse 100

An alternative method of group control is available for groups representing subsea completion manifolds. This method calculates the THP required to make the group produce at the desired rate. Thus it allows the group to produce at a specified rate while its wells remain on THP control operating at a common THP. Physically this method is equivalent to applying a choke to the group's outlet flow line, to increase the upstream pressure and reduce the production rate, as opposed to choking back each well separately. The group is able to produce at its target rate as long as the calculated value of its wells' common THP remains above the group's (unchoked) nodal pressure in the network. If the calculated THP value falls below the group's nodal pressure in the network, the wells operate at the group's nodal pressure and the group is no longer able to produce its target rate. At this point the production rules (see keyword `PRORDER`) are activated to increase the productivity of the group and bring the group's production back to its target rate. This alternative method of group control may be invoked by declaring the group to be a subsea completion manifold in keyword `GRUPNET` or `NODEPROP`.

Eclipse 100

In the standard network model, this method of group control can only be applied to well-groups (that is the groups containing wells, not higher level groups that have subordinate groups). However, when using the extended network model, a similar treatment can be applied to production rate targets at any level below the

fixed pressure node by invoking the automatic choke option. (see "[Automatic chokes](#)".) In the standard network model a target rate applied to a higher level group will be apportioned among its subordinate wells, or groups with guide rates, according to the standard group control procedure. For example, consider a rate target of 20,000 stb/day of oil applied to group PLAT-A. This is apportioned among the wells in groups GR-A1 and GR-A2 in proportion to their guide rates. The wells under group control operate at their allocated rates, and in general they all have different THP values. Defining groups GR-A1 and GR-A2 as manifold groups has no effect, as the rate target applies to a higher level group. However, if the manifold groups are also given guide rates (in keyword [GCONPROD](#)), the alternative method of group control can be applied and their wells operate at their common manifold pressures. This is because the rate target given to PLAT-A is passed down to the groups GR-A1 and GR-A2 in proportion to their guide rates. If these two groups are given equal guide rates, for example, they both have a target of 10,000 stb/day of oil. In each of the two groups, the common THP required to produce this rate is calculated and applied to their wells as THP limits. If one group cannot make its target without falling below its network nodal pressure, the other group tries to make up PLAT-A's production to 20,000 stb/day. To summarize, if a rate target is to be applied to a group that is superior to a manifold group, the subordinate manifold groups should be given guide rates. A warning message is printed if this is not so.

Eclipse 300

Eclipse 300 does not use this method of applying rate constraints to manifold groups. Instead it uses automatic chokes to apply rate constraints at any level in the network, not just to well groups. (See "[Automatic chokes](#)".) The method is the same whether using the standard or extended network models. Eclipse calculates the pressure drop across the choke that is necessary to reduce the flow rate to the required target value. The choke must be represented by a branch with a VFP table number of 9999, so that the only pressure loss in this branch is that of the choke. If the branch represents a pipeline having its own pressure losses, an intermediate node must be inserted to distinguish the choke from the rest of the pipeline.

Pumps and compressors

Pumps and compressors in the production network's flow lines can be modeled using the Artificial Lift Quantity (ALQ), in a similar way to the modeling of a pump or gas lift in a well. The artificial lift quantity for each flow line can be set and reset at any time in the simulation.

Eclipse 100

There is an option to switch on a pump or compressor in the network if a group's oil or gas production rate falls below a specified limit. For example, consider the pipeline between PLAT-A and the FIELD. The production from PLAT-A declines when its subordinate wells are operating with their THP values equal to their groups' nodal pressures. When the production rate falls below a specified limit, a pump or compressor in the pipeline from PLAT-A can be switched on automatically. This reduces the pressure losses in the pipeline, which decreases PLAT-A's nodal pressure, consequently reducing the nodal pressures in the subordinate groups and boosting their wells' production.

Eclipse 100

Additional, more versatile, facilities for automatic pump or compressor switching are available when using the extended network model, which is described in a later section (see "[Automatic compressors](#)").

Eclipse 100

An alternative use for the ALQ is to define it to represent the average surface density of either the oil phase or the gas phase flowing along a pipeline. Eclipse then automatically sets it equal to this value when looking up the VFP table. The VFP table must have been prepared using the same definition of the ALQ, and should span the expected range of surface density values. This option is useful when a mixture of oil or gas with different surface densities flows into the network. Obviously the ALQ then cannot be also used to represent a pump or compressor, and the pipeline cannot contain any automatic pump/compressor.

Injection network

In addition to the production network, a separate network can also be defined for water or gas injection. Each network exists independently of the other. A particular group may be a node in both networks, if

required, but in this case the group will have two independent nodal pressures, one for each network to which it belongs.

Pumps can be modeled in the injection network flow lines, but the artificial lift quantity cannot be varied.

As an example of an injection network, consider a pump that takes water at a fixed inlet pressure and pumps it into a manifold supplying a number of water injectors. The outlet pressure of the pump, and thus the pressure in the manifold, varies with the total injection rate. The injection wells operate under THP control, with the THP equal to the pressure in the manifold. Using the grouping tree shown in figure 9.7, let the injection pump be situated on platform PLAT-B, and the injection wells belong to group GR-B2. Declare PLAT-B as a fixed-pressure node in the water injection network, with its pressure equal to the pump inlet pressure, and declare GR-B2 as a variable-pressure node connected to PLAT-A by a pipeline containing the pump. It makes no difference whether or not PLAT-A and GR-B2 are also nodes in the production network.

Eclipse 300

In Eclipse 300 it is possible to use the extended network model for injection networks, allowing group injection rate constraints to be handled by automatic chokes

Using the standard network option

This section describes how to use the Standard (non-extended) Network option, in which the network structure must be the same as the `GRUPTREE` hierarchy. If you require the network to have a different topology from the group control hierarchy, then you must use the Extended Network option which is described in a later section (see "The extended network model").

Eclipse 300

In Eclipse 300 you are recommended to use the Extended Network option as this is more versatile than the Standard option. The Standard option keywords, however, are available to assist the porting of Eclipse 100 data sets.

Constructing the network

The multi-level grouping tree must first be constructed using the keyword `GRUPTREE` (but this can be omitted if there is only one level of groups below the `FIELD`).

The production network is specified using the keyword `GRUPNET`. With this keyword, one or more groups (or the `FIELD` itself) can be declared as fixed-pressure nodes, while groups below them in the network are given pipeline connections to their parent groups. The pressure losses in each pipeline are obtained from production VFP tables, in a similar way to the treatment of THP calculations in wells. Thus each pipeline must be associated with a given production VFP table, which is supplied as input data to Eclipse in the usual way. The top node of each network must be a fixed-pressure node. It is also possible for subordinate groups to be fixed-pressure nodes; these form 'sub-networks' whose pressures are independent of the main network but whose flows will be added into it.

Water and gas injection networks are specified using the keyword `GNETINJE`. Again, one or more groups can be declared as fixed-pressure nodes, and groups below them in the network are given pipeline connections from their parent groups. Each pipeline must be associated with an **injection** VFP table for water or gas injection, from which the pressure losses will be calculated.

The pipeline VFP tables can be constructed using the program VFP *i*, which handles horizontal and undulating flowlines as well as vertical and deviated well bores. For a pipeline connecting a parent group and its child, the 'tubing head' end is the parent group and the 'bottom hole' end is the child group. Thus a pipeline VFP table gives the nodal pressures of the child group for selected values of the nodal pressure of the parent group and the flow rate through the pipeline (and also the water and gas fractions in a production network pipeline). The VFP tables should cover the expected range of these quantities in the pipeline. To

allow for the case when a group has zero flow rate, the lowest flow rate in the table should be small, for example 1 stb/day, since the VFP *i* program does not accept a zero flow rate.

Pumps, compressors and gas lift

The VFP tables for the production network pipelines may also include a non-zero value of the Artificial Lift Quantity (ALQ), to model the effects of a pump or compressor on the pipeline. In this case the quantity may be identified with the pump or compressor power. The program VFP *i* can include the effects of a pump or compressor in the pipeline when calculating the pressure losses. The acting pump or compressor power for the pipeline, with which the VFP table is to be interpolated, is set with the keyword [GRUPNET](#). The specified power should correspond with the respective Artificial Lift Quantity value in the VFP tables. An additional option allows a pump or compressor to be switched on automatically in a pipeline when its group's oil or gas production rate falls below a set limit. This option is selected using the keyword [GNETPUMP](#).

Injection network pipelines can also contain pumps, if their VFP tables are constructed to include them. But injection VFP tables do not have the Artificial Lift Quantity as a variable, so a pump cannot be turned on or off by varying this quantity. Instead, two VFP tables may be constructed for the pipeline, one which includes the effect of the pump, and one which does not. Thus the pump can be turned on and off manually during the simulation by changing the pipeline's VFP table number in keyword [GNETINJE](#).

If wells in a production network are subject to gas lift, the flow of lift gas can be included in the network branch pressure drop calculations by setting item 6 in keyword [GRUPNET](#) to 'FLO' or 'ALQ' for the pipelines that carry lift gas along with produced fluids. If 'FLO' is selected, the lift gas flows of the subordinate producers will be added to the gas flow rate along the pipeline. The lift gas flow is assumed to be equal to the sum of the ALQ values of the subordinate producers, multiplied by their efficiency factors. If 'ALQ' is selected, the gas flow rate will not be adjusted, but the ALQ value used to look up the pipeline's pressure drop will be set equal to the sum of the ALQ values of the subordinate producers, multiplied by their efficiency factors. If either option is selected, the Artificial Lift Quantity for the wells ([WCONPROD](#) item 12) and the pipeline (for the 'ALQ' option) must be defined as the rate of lift gas injection. This definition must be selected in the VFP *i* program if it is used to construct the VFP tables.

Balancing the network

The nodal pressures of the groups in the network are determined by a 'network balancing' calculation. This uses an iterative procedure to calculate the pressures at each node (or group) in the network consistent with the group flow rates and the pipeline pressure losses. At each iteration of the balancing calculation the well and group production rates are determined using the reservoir conditions existing at the beginning of the timestep, and the latest iterate of the nodal pressures. The pipeline pressure losses generated by these flow rates are then calculated, and the nodal pressures are updated at each level in the grouping tree from the fixed-pressure node(s) downwards using a secant method. The iterations are terminated when all the nodal pressures agree with the pipeline pressure losses, to within a certain tolerance. In Eclipse 100 the number of iterations required to balance the network each time will be written to the PRINT file if either the 10th or the 15th [RPTSCHE](#)D keyword switch is set (mnemonic VFPPROD or NEWTON). In Eclipse 300 the number of iterations coupling the network to the reservoir will automatically be printed in the timestep summary table in the column headed NIT.

At each network balancing iteration, the well and group production rates are determined at the latest iterate of the nodal pressures. If a group production target cannot be met, actions may be taken to increase production, such as drilling a new well (see keyword [PRORDER](#)). However, it is possible that an intermediate balancing iteration with high nodal pressures and consequentially low well productivities may cause such an action to be taken prematurely. This can be prevented in Eclipse 100 by using the [WAITBAL](#) keyword to request that these actions should not be performed until the network is properly balanced. Then

if more flow is required once the network has converged, an appropriate action is performed and the network must be balanced again. Using this option therefore requires more network balancing iterations in total. In Eclipse 300 there is no need to use WAITBAL as it is automatically enabled; the simulator always waits until the network is properly balanced before drilling a new well.

Eclipse 100

You can control the frequency with which the network balancing calculations are performed. The keyword `NETBALAN` is used to force a network balancing calculation to take place at the beginning of the next timestep, and to specify the ‘balancing interval’ for subsequent timesteps. Subsequent balancing calculations are performed at the beginning of each timestep that starts after the specified interval has elapsed since the previous calculation. The nodal pressures (which are used to set the well THP limits) are kept constant until updated by the next balancing calculation. The default value of the balancing interval is zero, which results in a balancing calculation being performed at the beginning of each timestep. The keyword `NETBALAN` can also be used to adjust the convergence tolerance and the maximum number of iterations in the calculation.

Eclipse 100

By default, the network is balanced using the grid block conditions existing at the beginning of the timestep. By the end of the timestep, the grid block conditions and well flow rates may have changed, and the flows in the network branches may no longer be exactly consistent with the nodal pressures. The longer the timestep or interval between balancing calculations, the greater the balancing error will be at the end of the timestep. Eclipse 100 checks the balancing error in each branch of the network at the end of each timestep, and prints out the largest error in any branch. The balancing error is defined as the difference between the pressure drop along the branch when the network was last balanced, and the pressure drop calculated using the current flow rates at the end of the timestep. The THP target of wells is also checked against the parent node pressure before the network is considered balanced.

Eclipse 100

By setting the balancing interval negative in keyword `NETBALAN`, Eclipse 100 can be made to balance the network in each of the first `NUPCOL` Newton iterations of every timestep. (The value of `NUPCOL` may be set using either keyword `NUPCOL` or keyword `GCONTOL`). This option, in general, results in much smaller balancing errors at the end of the timestep, as the reservoir conditions at each successive Newton iteration more closely resemble the converged conditions at the end of the timestep. However, the option may cause the timesteps to require more Newton iterations to converge, as the well THP limits vary at each balancing calculation.

Eclipse 300

In Eclipse 300 the network is always balanced in each of the first `NUPCOL` Newton iterations of every timestep. (This is equivalent to setting a negative balancing interval in Eclipse 100.)

Eclipse 100

You can supply a target value and a maximum permitted value for the largest branch balancing error, in the `NETBALAN` keyword. Eclipse 100 attempts to adjust the timestep length so as to keep the balancing error at approximately its target value, subject of course to any other constraints on the timestep. Timesteps limited by the target balancing error can be identified by the mnemonic ‘`NETW`’ printed in the timestep information line. If the maximum permitted balancing error is exceeded, Eclipse 100 chops the timestep and rebalance the network. Since timestep chops can severely increase the running time of a simulation, a maximum permitted balancing error should only be used with caution, and its value should be several times larger than the target error. A large balancing error is sometimes unavoidable during the timestep in which a well dies. You can set a minimum timestep size to prevent a large reduction in step size when wells die.

Sub-sea completion manifolds

Eclipse 100

When a group production control is going to be applied and some groups represent sub-sea completion manifolds in which the wells all operate at the same THP, then these groups should be declared as manifold groups in keyword `GRUPNET`. If a group control is going to be applied to any group superior to a manifold group, the subordinate manifold groups should be given guide rates in keyword `GCONPROD`. If a manifold group has a rate target (either imposed directly or allocated as a share of a superior group’s rate target in proportion to its guide rate), Eclipse calculates the manifold pressure required to make the group produce at

its target rate. The calculation is performed iteratively, using a secant method. The convergence tolerance and the maximum number of iterations allowed for this calculation can be set in keyword [NETBALAN](#). The manifold pressure is recalculated at each of the first [NUPCOL](#) Newton iterations of each timestep, and the wells' THP limits are set equal to the manifold pressure or the group's nodal pressure in the network obtained from the latest network balancing calculation, whichever is the greater. If the timestep takes more than [NUPCOL](#) Newton iterations to converge, the well THP limits are kept constant for the remaining iterations after the first [NUPCOL](#), and the group may not meet its flow target within the allowed convergence tolerance. This treatment corresponds to the standard group control method (see keyword [GCONTOL](#)).

Eclipse 300

The treatment of manifold groups in Eclipse 300 is different in that they are regarded as inlets to automatic chokes (see "[Automatic chokes](#)"). The group's outlet branch contains the choke and must be given a VFP table number of 9999 so its only pressure drop is that of the choke. Eclipse adjusts the pressure drop across the choke to keep the production rate at the required target, provided that it has enough potential.

Keywords required

The keywords needed in the [SCHEDULE](#) section to run the network option are summarized below:

Restriction	Keyword	Description
	VFPPROD	Enter the VFP tables for the production wells and the production network pipelines.
	VFPINJ	Enter the VFP tables for the injection wells and the injection network pipelines.
	WELSPEC	Enter the specification data for all wells.
	COMPDAT	Specify the connection data for all wells.
	WCONPROD	Set the control targets and limits for the production wells. THP limits should not be set for wells belonging to a group which is a node in the production network, as they will be determined at each balancing calculation.
	WCONINJE	Set the control targets and limits for the injection wells. THP limits should not be set for wells belonging to a group which is a node in the injection network, as they will be determined at each balancing calculation.
	GRUPTREE	Construct the grouping tree, if there is more than one level of groups below the FIELD . The example below shows the GRUPTREE records for the network in Figure 9.7 .
	GRUPNET	Define the production network. For example, in Figure 9.7 , let PLAT-A and PLAT-B have fixed pressures of 300 psia and the bottom level groups be sub-sea completion manifolds connected to the platforms by pipelines.
	GNETINJE	Define an injection network. For example, in Figure 9.7 , let PLAT-B have a fixed pressure of 14.7 psia and GR-B2 contain the water injectors. The water injection pipeline between these two nodes is represented by injection VFP table number 2, which includes the pump performance characteristics.

Restriction	Keyword	Description
	GCONPROD	You can, optionally, use this keyword to set production rate targets and limits for any group. For example, the FIELD has an oil production target of 50,000 stb/day, while the two platforms have water handling limits of 10,000 stb/day. Because the four well-groups are manifold groups subordinate to a group under rate control, they should be given guide rates. Here they are each required to produce the same amount of liquid, so their liquid guide rates are set to the same nominal value - say 100.
Eclipse 100	GNETPUMP	Use if automatic pump/compressor switching is required in the production network. For example, the pipeline from GR-B1 to PLAT-B will be interpolated with an artificial lift quantity of 50 if the oil production rate of GR-B1 falls below 500 stb/day.
	NETBALAN	Use of this keyword is optional. It sets the balancing interval, convergence tolerances and iteration limits, and instructions to limit the balancing error by reducing the timestep size. The example below sets a negative balancing interval so that Eclipse 100 balances the network in each of the first NUPCOL Newton iterations of every timestep (Eclipse 300 always does this).
Eclipse 100	WAITBAL	Optional keyword to prevent Production Rule actions being performed to maintain group production targets until the network is balanced (Eclipse 300 always prevents this happening).
	WNETDP	Optional keyword to apply an additional fixed pressure drop between a well's tubing head and its group's node in the network.

Table 9.1: SCHEDULE section keywords

Examples

[GRUPTREE](#) example:

```
GRUPTREE
'GR-A1' 'PLAT-A' /
'GR-A2' 'PLAT-A' /
'GR-B1' 'PLAT-B' /
'GR-B2' 'PLAT-B' /
/
```

[GRUPNET](#) example:

```
GRUPNET
-- GROUP    FIXED    VFP    PUMP    MANIFOLD    INCLUDE    ALQ=
-- NAME     PRESSURE  TAB    POWER   GROUP?     LIFT GAS?   DENS?
'PLAT-*'    300 /
'GR-A1'     1*       1     1*      'YES' /
'GR-A2'     1*       2     1*      'YES' /
'GR-B1'     1*       3     1*      'YES' /
'GR-B2'     1*       3     1*      'YES' /
/
```

Note: In Eclipse 300, the manifold groups must have VFP table numbers of 9999, so a set of intermediate groups must be declared whose outlet pipelines have their pressure drops determined by the VFP tables 1, 2 and 3

[GNETINJE](#) example:

```

GNETINJE
-- GROUP   NETWORK   FIXED       VFP
-- NAME     PHASE     PRESSURE   TAB
'PLAT-B'   'WATER'   14.7      /
'GR-B2'    'WATER'   1*        2   /
/

```

GCONPROD example:

```

GCONPROD
-- GROUP   CNTL   OIL   WATER   GAS   LIQU   LIMIT   AVAIL   GUIDE   G.R.
-- NAME     MODE  RATE  RATE   RATE  RATE   ACTION  FLD     RATE   PHASE
'FIELD'    'ORAT' 50000 /
'PLAT*'    'NONE' 1*   10000 2*
'GR*'      'NONE' 5*
/

```

GNETPUMP example:

```

GNETPUMP
-- GROUP   RATE   PHASE   NEW VFP   NEW LIFT   NEW GAS
-- NAME     LIMIT  TAB NO   QUANTITY  CONSUMPTION
'GR-B1'    500    'OIL'   1*        50
/

```

NETBALAN example:

```

NETBALAN
-1.0 /

```

WAITBAL example:

```

WAITBAL
'YES' /

```

WNETDP example:

```

WNETDP
-- WELL      PRESSURE
-- NAME      DROP
'PROD1'     20.0 /
'PROD2'     25.0 /
/

```

The extended network model

The extended network model contains several enhancements over the standard model for production networks:

- The surface network can be structured independently of the group control hierarchy defined with keyword [GRUPTREE](#). However, it must still have a “gathering tree” topology.

- The automatic compressor/pump facilities are more versatile.

Compressors are turned on whenever a nominated group fails to meet its production rate target. Multiple compression levels can be defined.

- An automatic choke facility can adjust the pressure loss across a choke in a designated network branch to meet a group’s production rate target. This provides an alternative method to guide rate control or prioritization for meeting group production rate targets.

- It provides more flexibility in handling well and group downtime.
- It is possible to remove a specified rate or fraction of the water flowing through a node.

Eclipse 100

In Eclipse 100 the extended network model applies to the production network only. However, it is possible to use it in conjunction with the Standard model for an injection network.

Eclipse 300

In Eclipse 300 the extended network model applies to both production and injection networks. The two types of model cannot be mixed, so either all networks must use the Extended model or all must use the standard model.

The network structure

Sometimes you need to impose a group control hierarchy that is different from the physical structure of the surface network. For example, figure 9.8 shows a case where production from several groups feeds into a main trunk line at different points. GA1, GB1, GA2 and GB2 are all well-groups feeding into the trunk line at nodes N1, N2, N3 and N4 respectively. FIELD is the fixed pressure node.

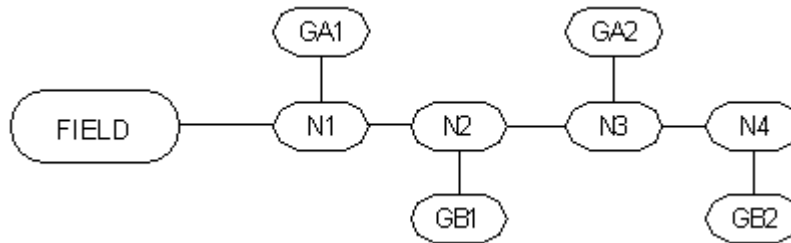


Figure 9.8. The network structure

In this example, groups GA1 and GA2 belong to Company A whilst groups GB1 and GB2 belong to Company B. Clearly both companies may need to apply targets or limits to their own total production rate. They can define a group hierarchy with `GRUPTREE` as shown in figure 9.9, which is different from the network structure.

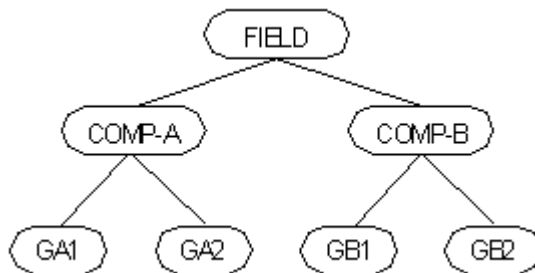


Figure 9.9. The group control hierarchy

The group control hierarchy is constructed in the standard model with keyword `GRUPTREE` and the relevant production targets set with `GCONPROD` or `GCONPRI`. But by using the Extended Network Model, a separate network structure can be constructed with keywords `BRANPROP` and `NODEPROP`. The network consists of a gathering tree structure of nodes and branches. The nodes at the bottom level of the tree are “source nodes”, which inject flow into the network. The source nodes must be well-groups or satellite groups (or master groups in Reservoir Coupling master runs), and all source nodes should correspond to groups in the `GRUPTREE` hierarchy; in Eclipse 100 the corresponding nodes and groups must have the same name in both structures but this is not necessary in Eclipse 300. The remaining network nodes and

GRUPTREE groups need not coincide. The top node of the network tree (the terminal node) must be a fixed pressure node.

In the example above, GA1, GA2, GB1 and GB2 are common to both the network structure and the GRUPTREE hierarchy. Also the fixed pressure node corresponds to the FIELD group (although this is not compulsory). But nodes N1, N2, N3 and N4 only appear in the network structure, and groups COMP-A and COMP-B only appear in the group control hierarchy.

The network may consist of two or more separate gathering trees, in which case each tree must have its own fixed pressure terminal node at the top. It is possible for subordinate groups also to be fixed-pressure nodes; these form 'sub-networks' whose pressures are independent of the main network but whose flows will be added into it. For example, in the network shown in figure 9.8 the nodes FIELD and N3 could both be fixed-pressure nodes. Node N3 then acts as a source into node N2 of the main network, while keeping its own pressure independent of the pressures in the main network.

A consequence of separating the group control hierarchy from the network structure is that the location of any gas consumption or import must be identified in both structures. Eclipse must know where consumed gas is extracted from the network (to compute downstream pressure losses correctly), and also where it is extracted from the GRUPTREE hierarchy (to perform the group production accounting). Accordingly the GCONSUMP keyword in Eclipse 100 has an additional item (Item 4) to identify the node at which the gas is extracted from or added to the extended network structure. Alternatively the keyword NCONSUMP can be used to define the consumption at a network node, and optionally to assign the consumption to a particular group. The corresponding keyword in Eclipse 300 for removing gas from the network is NGASREM.

Automatic compressors

Eclipse 100

Automatic compressors or pumps are defined with keyword NETCOMPA instead of GNETPUMP. Each is located at a specified branch of the network. They offer a much wider range of features than the GNETPUMP compressors of the standard network model. Their principal features are:

Activation by group target

Each compressor (or pump) is turned on whenever a nominated group fails to meet its gas (or oil) production rate target, which is specified with keyword GCONPROD or GCONPRI as usual. If the nominated group does not have a target set, but has a guide rate defined in GCONPROD and is under 'FLD ' control from a higher level group, then the compressor is turned on when the nominated group cannot make its share of the higher level group's production target.

Compressor switching sequence

It is possible for more than one compressor to support the same group. Each compressor can be given a sequence number, and when the group fails to meet its production target the compressor with the lowest sequence number that responds to the group will be activated. If the group still cannot meet its target, but has other compressors that respond to it, these other compressors are turned on in an increasing order of their sequence number. All compressors in the network that share the same sequence number are turned on simultaneously.

Production rule order

If there is a drilling queue (keyword QDRILL), by default Eclipse scans it for a suitable well to open in preference to switching on a compressor when a group cannot make its production target. However, by using the PRORDER keyword to define the order in which various actions are taken to increase group production, compressor switching can be prioritized with respect to the other possible actions. For example, if 'COMP' is placed before 'DRILL', Eclipse activates the automatic compressors before looking for a new well to open.

Multilevel compression

Multilevel compressors can be defined. Compression is increased one level at a time until the nominated group can meet its production target. The ALQ, and the gas consumption rate if there is any, increase linearly with the level number to their “full on” values. Any number of levels may be requested, but having a large number of levels slows the run down as each level must be tried in turn while rebalancing the network.

Turning compression off

Compressors can be turned off (or reset to zero compression) manually at any time by using the `COMPOFF` keyword. They are turned on automatically again as soon as they are needed. For example, the `COMPOFF` keyword can be entered whenever the target production rate is reduced, or when extra production capacity is added. (When using the gas field operations model, the compressors are turned off automatically whenever the contract group’s target rate decreases.)

Automatic chokes

Automatic chokes form an alternative means of controlling a group’s production rate when using the Extended Network option, which is distinct from guide rate control or prioritization. The choke is placed in a designated branch of the network, and at each network balancing iteration Eclipse adjusts the pressure loss across it to keep a specified group producing at its target rate. The operation of the choke varies the pressure in its inlet node, which in turn affects the productivity of the subordinate wells while they are operating under THP control.

Defining the choke

Eclipse 100

A branch is defined to be a choke by answering 'YES' to Item 3 of the `NODEPROP` keyword for the **inlet node** of the choke. A fixed-pressure node cannot be defined as a choke inlet node, since its pressure cannot respond to the choke’s pressure loss. Also, source nodes cannot be defined as choke inlet nodes. Source nodes that are well-groups can instead be designated to be ‘manifold groups’, whose operation is similar to choke inlet nodes in that Eclipse determines a common well THP that makes the group meet its target rate (see “Group production control in the network option”).

Eclipse 300

A branch is defined to be a choke in a **production** network by answering 'YES' to Item 3 of the `NODEPROP` keyword for the **inlet node** of the choke. For an **injection** network this should be the **outlet** node of the choke; in either case it should be the node at the end of the branch that is nearer the reservoir. A fixed-pressure node cannot be defined as a choke inlet node, since its pressure cannot respond to the choke’s pressure loss.

The only pressure drop allowed across a branch acting as a choke is that caused by the action of the choke itself. The branch should therefore be declared in the `BRANPROP` keyword with a VFP table of 9999. Any other table number is flagged as an error.

Eclipse 100

The wells subordinate to the choke respond to the choke pressure drop whenever they are operating under THP control. Thus, for the automatic choke to work correctly, its subordinate wells must be made to ignore any rate targets set for them by guide rate group control. Accordingly, Eclipse 100 automatically makes them unavailable for guide rate group control (this is not necessary in Eclipse 300), equivalent to setting Item 2 in the `WGRUPCON` keyword to 'NO'. This is done for all producers in source groups subordinate to the choke, as long as there is no intermediate fixed-pressure node between the source group and the choke. The wells can still be given individual flow rate and BHP limits with keyword `WCONPROD`, which causes them to come off THP control if they are violated.

Eclipse 100

Note: If a producer subordinate to a choke is subsequently turned into an injector which is subject to group injection control, you must make the well available for group control again by entering the `WGRUPCON` keyword with 'YES' in item 2 at the time that the well is redefined as an injector.

Setting the rate target

The operation of the choke is governed by the production rate target of a particular group, which is specified in item 5 of keyword `NODEPROP`. If the inlet node to the choke shares the same name as a group, as declared with the `GRUPTREE` keyword, then Item 5 will default to that group. Obviously the choke's inlet node in the network and the nominated group in the `GRUPTREE` hierarchy must have a common set of subordinate wells, in order for the choke to influence the group's production rate.

The group's production rate target is set in the same way as for guide rate group control, using keyword `GCONPROD` (or `GCONENG` for an energy rate target). Alternatively the group's target may be passed down, using guide rates set with the `GCONPROD` keyword, as a share of a superior group's target. Thus the group must either have its own production rate target or have a guide rate set to give it a share of a superior group's target.

Since guide rate group control will not operate below a choke, groups subordinate to the nominated production rate target group must not be given production guide rates.

Eclipse 300

Injection networks that use the Extended model may also have chokes, which are governed by the injection rate target of a particular group, for the phase that corresponds to the network's injection phase. The group's injection rate target is set in the same way as for guide rate group control, using keyword `GCONINJE`. Alternatively the group's target may be passed down, using guide rates set with the `GCONINJE` keyword, as a share of a superior group's target. Thus the group must either have its own injection rate target or have an injection guide rate set to give it a share of a superior group's target. Since guide rate group control will not operate below a choke, groups subordinate to the nominated injection rate target group must not be given injection guide rates for the same phase.

Calculating the choke's pressure drop

Having defined the choke and its corresponding group's production rate target, Eclipse proceeds as follows. At each network balancing iteration, it compares the group's flow with its target. If the flow is too large Eclipse increases the inlet pressure of the choke to reduce the flow. If the flow is below the target the pressure at the inlet will be decreased until either the target is met or the inlet pressure is the same as the outlet pressure (that is there is no pressure drop across the choke). In this latter case Eclipse then looks to the production rules to see if the flow may be increased by drilling a new well, for example (see keyword `PRORDER`). The choke's pressure drop can be written to the `SUMMARY` file using the `SUMMARY` section keyword `GPRB` and naming the choke's inlet node.

Eclipse 300

Chokes in injection networks operate in a similar manner. The pressure in the choke's outlet node (nearer the reservoir) is iterated until the injection rate of the subordinate wells matches the injection target of the corresponding group. If the wells do not have sufficient injectivity to meet the target, the outlet node's pressure is kept the same as the inlet node's pressure (so there is no pressure drop across the choke) and the injection rate will fall below the target. Eclipse then scans the drilling queue to see if there is a suitable injector to drill to increase the group's injectivity. The choke's pressure drop can be written to the `SUMMARY` file using the `SUMMARY` section keyword `GPRBW` or `GPRBG`, for a water or gas injection network respectively, and naming the choke's outlet node.

Eclipse 100

The group targets are met to within a tolerance specified in item 4 of the `NETBALAN` keyword. But unlike subsea manifold groups, the calculation of the pressure drops at chokes is calculated as a part of the main network iteration and hence the maximum number of iterations to solve for the choke is the same as for the

network as a whole, that is it is governed by item 3 of NETBALAN and **not** item 5. Whilst it is clearly dependent upon the relative sizes of the tolerances in items 2 and 4 of NETBALAN, in general substantially more iterations are needed if there are chokes in the network than if there were not. You should be prepared to increase item 3 of NETBALAN accordingly.

Note: Convergence problems may arise if a group is given a very small rate target, either manually or as a result of a relatively small guide rate. This is particularly true if the low flow required from the wells under that group results in some of them being unable to operate under THP control.

Well and group efficiency factors

When wells have efficiency factors set in keyword WEFAC, by default each well contributes its ‘time-average’ production rate to the network flows, that is its flow rate multiplied by its efficiency factor. The network branch pressure losses are thus calculated from the time-average rate sums of the subordinate wells. This is equivalent to the way in which the group production rates are summed, and is most appropriate when the wells go down individually for short ‘random’ periods.

But if you wish to model the ‘peak’ pressure losses in the network, when all the wells are flowing at the same time, then it is possible to specify that the network branch flows should reflect the full flow rates of each well. By setting Item 3 of keyword WEFAC, it is possible to specify for each well individually whether its production rate should or should not be multiplied by its efficiency factor when adding its contribution to the network flows. The **group** flow rates, however, are still the time-average sum of the well flows.

If all the wells in a group have their downtimes synchronized, so that the group either flows with its full rate or not at all, it is more appropriate to apply an efficiency factor to the group (with keyword GEFAC) rather than the individual wells. If there is a network node with the same name as this group, by default the group’s efficiency factor is applied to this node and its flow rate is multiplied by the efficiency factor when adding the node’s flow rate to the flow in its parent node. The pressure drop in a node’s outlet branch is calculated from the node’s flow rate. Thus the node’s outlet branch and the branches upstream of the node reflect the full flow rates, while the branches downstream of its parent node reflect the time-average flows. (This is equivalent to the Standard Network model’s treatment of group efficiency factors.) However, by setting item 3 of keyword GEFAC it is possible to specify that the group’s full flow rates should be used for calculating all downstream pressure losses, by not assigning the group’s efficiency factor to the corresponding network node. If there is no network node with the same name as the group, it is possible to apply an efficiency factor to another appropriate network node using keyword NEFAC.

This flexibility is only available when using the Extended Network model, since network node and branch flows are accounted separately from the group flows. In the Standard Network option the branch flows are always the same as the reported group flow rates.

Using the extended network model

Using the Extended Network model requires different keywords and settings compared to the Standard model (keywords not mentioned here for example NETBALAN) are common to both models).

RUNSPEC section

To enable the extended version of the network model, the keyword NETWORK must be entered in the RUNSPEC section. It has two items of data: the maximum number of nodes in the network and the maximum number of branches in the network. So, for the example network in figure 9.8:

```
NETWORK
  9 8 /
```

SUMMARY section

The same keywords apply to both the standard and extended network models. When used with the Extended version, names supplied to these keywords must correspond to network nodes instead of [GRUPTREE](#) groups.

Construct the network with BRANPROP and NODEPROP instead of GRUPNET

The network structure is defined in the SCHEDULE section by its branches. Each branch is identified by its inlet node and outlet node. The branches must first be defined with the keyword [BRANPROP](#). This also sets the branch properties, that is the VFP table number and ALQ value. An example of how the production network in Figure 9.8 may be constructed is:

```
BRANPROP
-- Inlet      Outlet      VFP      ALQ      ALQ=
-- node       node       table     value    dens?
'GB2' 'N4'          3 /
'GA2' 'N3'          2 /
'GB1' 'N2'          1 /
'GA1' 'N1'        9999 /
'N4'  'N3'          4 /
'N3'  'N2'          4 /
'N2'  'N1'          5 /
'N1'  'FIELD'       6 /
/
```

After the network structure has been defined with [BRANPROP](#), the properties of the network nodes must be defined with keyword [NODEPROP](#). Only the fixed pressure node(s), and other nodes with non-default properties need to be specified here. In the example in figure 9.8, the fixed-pressure node corresponds to the FIELD group and is thus named 'FIELD'. Another assumption is that the source node GA1 corresponds to an automatic choke inlet (or manifold group in Eclipse 100) and there is no gas lift. The [NODEPROP](#) keyword is then:

```
NODEPROP
-- Node      Fixed      Choke inlet or      Add      Choke control
-- name      pressure   manifold group?    lift gas?    group
'FIELD'      200.0                /
'GA1'        1*          'Y' /
/
```

Define any gas consumption or import with either GCONSUMP or NCONSUMP

Eclipse 100

Any gas consumption or import should be declared in both the network and the group hierarchy. This can be done with either keyword [GCONSUMP](#) or [NCONSUMP](#). [GCONSUMP](#) declares consumption/import at a group and allows you to specify the network node where this should be applied as well. [NCONSUMP](#) declares consumption at a network node and allows you to specify the group where this should be applied as well. For a constant consumption rate, the two keywords are equivalent and it does not matter which you use. However, there are some differences between the two keywords:

- [NCONSUMP](#) allows only a constant consumption rate, while [GCONSUMP](#) also allows a consumption rate that is a fraction of the group's production, and also a gas import rate.
- The two keywords have different defaults for assigning consumption to the other structure. If consumption is declared at a source group in [GCONSUMP](#), by default it will also be assigned to the corresponding source node of the network. However, if consumption is declared at a source node in [NCONSUMP](#), by default it will **not** be assigned to the corresponding group.
- [NCONSUMP](#) may only be used with the Extended Network model. [GCONSUMP](#) may be used with either the Extended or Standard models.

Remove any fuel gas from the network with NGASREM

Eclipse 300

Any fuel gas consumed at a group (see keyword [GRUPFUEL](#)) by default still remains present in the network flow and contributes to the downstream branch pressure drops. But it is possible to remove a corresponding amount of gas from the network at a particular node by using the [NGASREM](#) keyword. The gas removal maybe defined as a specified rate or a fraction of the gas flow through the node.

Define any automatic compressors with NETCOMPA

Eclipse 100

Compressors or pumps can be modeled ‘manually’ by re-entering the [BRANPROP](#) keyword with a different ALQ value or VFP table number. However, more versatile automatic compressors can be defined using the [NETCOMPA](#) keyword. (Do not use [GNETPUMP](#) for these compressors.) They will be turned on automatically when a nominated group fails to meet its gas or oil production target set in [GCONPROD](#) or [GCONPRI](#). If other actions to increase the group’s production rate may also be performed (for example, opening a new well from the drilling queue), the order in which the possible actions are performed can be defined with the [PRORDER](#) keyword.

The following example defines two compressors responding to the gas production rate of groups PL-A and PL-B respectively. The second one has three compression levels, and consumes gas from the network and from group G-B3 in the [GRUPTREE](#) hierarchy.

```
NETCOMPA
-- Inlet Outlet Target Phase VFP ALQ Cons Cons Type Num ALQ Seq
-- node node group tab rate rate group lev lev 1 num
'N4' 'N3' 'PL-A' 'GAS' 1* 50.0 1* 1* 'TEMP' /
'N6' 'N5' 'PL-B' 'GAS' 1* 100.0 1000 'G-B3' 'MULT' 3 40.0 /
/
```

Automatic compressors can be turned off manually at any time by using the [COMPOFF](#) keyword (except for those defined in [NETCOMPA](#) to stay on permanently). They are turned on automatically again as soon as they are needed.

Defining a choke

In the example network in figure 9.8 there is the requirement to control the flow of the FIELD group using a choke. To do this, insert a new node, NCHOKE, to represent the choke inlet, between the nodes FIELD and N1. The single entry in [BRANPROP](#) for the branch N1 - FIELD is now replaced with two records. The choke itself is represented by the branch NCHOKE - FIELD, and this must have a VFP table of 9999 so that its only pressure loss is that across the choke. The branch N1 - NCHOKE accounts for the pressure loss along the rest of the pipe:

```
BRANPROP
-- Inlet Outlet VFP ALQ
-- node node table value
. .
'N1' 'NCHOKE' 6 /
'NCHOKE' 'FIELD' 9999/
/
```

The node NCHOKE is declared to be a choke inlet in the [NODEPROP](#) keyword, and the choke is told to respond to the rate target of the group FIELD.

```
NODEPROP
-- Node Fixed Choke inlet or Add Choke control
-- name pressure manifold group? lift gas? group
'NCHOKE' 1* 'YES' 'NO' 'FIELD' /
/
```


The FIELD's rate target is set in the usual way as for guide rate group control.

The network model in Eclipse 300

There are some important differences between the implementations of the network model in Eclipse 100 and Eclipse 300. However, the Eclipse 300 model can emulate much of the Eclipse 100 model's capability and shares a common set of keywords.

Standard and extended network models

Eclipse 300 can run either the standard model or the extended model. When constructing a network model from scratch, you are recommended to use the extended network model as this is more versatile than the standard model. The standard model keywords, however, are available to assist the porting of Eclipse 100 data sets.

Unlike Eclipse 100, Eclipse 300 allows you to use the Extended model for injection networks as well as production networks. The two types of model cannot be mixed, so if you use the extended model for the production network then you must also use it for any injection networks. This requirement differs from Eclipse 100, where the injection networks have to use the standard model even if the extended model is used for the production network. Thus the data file for runs containing both production and injection networks cannot be ported directly between the simulators if the production network uses the extended model.

When using the extended network model for both production and injection networks, node names must be unique across all networks; an injection network node cannot have the same name as a node in the production network. However, a source node in the production network may represent the same group as a sink node in the injection network, if the group contains both producers and injectors. The corresponding nodes in the two networks must have different names; they cannot both have the same name as the well group (although one node may have the same name). Accordingly, Eclipse 300 allows source nodes (corresponding to well groups) to have a different name from their corresponding group.

Currently there are no automatic pump/compressor facilities in Eclipse 300.

Balancing the network

The network balancing calculation solves the branch flows and pressure losses across the network, and balances the flows into the network from the wells with the pressures at the source nodes, which are translated into THP limits for the wells. Eclipse 300 balances the combined network and reservoir system in each of the first NUPCOL Newton iterations at every timestep. (The value of NUPCOL may be set using keyword [NUPCOL](#) or keyword [GCONTOL](#).) During any subsequent iterations that are required to converge the timestep, the nodal pressures and well THP limits remain unchanged. Thus the balancing error at the end of the timestep reflects only the changes in reservoir conditions that have taken place after the NUPCOLth Newton iteration. Eclipse 100 allows you to reduce the frequency of the balancing iterations or to balance the network only during the first iteration of the timestep; either of these options would increase the balancing error and so they are not implemented in Eclipse 300.

The default convergence criterion (item 2 of keyword [NETBALAN](#)) is substantially tighter in Eclipse 300 because the balancing calculation is more advanced. The branch flows and pressure losses are solved in an inner iteration loop nested inside the iterations that balance the network with the reservoir. The inner loop is converged to a tolerance that is a factor of 10 tighter than the tolerance for the network/reservoir coupling iterations.

The total number of network/reservoir coupling iterations performed over each timestep is reported in the timestep summary table in the column headed NIT .

Another difference from Eclipse 100 is that Eclipse 300 automatically activates the [WAITBAL](#) instruction, so that a new well from the drilling queue is drilled only after the network/reservoir iterations have converged. In Eclipse 100 it is necessary to activate the instruction with the [WAITBAL](#) keyword, otherwise wells may be drilled prematurely before the network is properly balanced.

Rate constraints and automatic chokes

Eclipse 300 has the option of meeting group rate targets by applying a pressure differential across an automatic choke, similarly to the Extended Network model in Eclipse 100 (see "[Automatic chokes](#)"). However, there are some differences between the two simulators:

- Eclipse 100 uses a 'subsea manifold' model to handle rate constraints for well groups via the network (it iterates to find a common well THP that produces the required total rate). Eclipse 300, on the other hand, does not use this approach; it uses automatic chokes to handle rate constraints at all levels of the node or group hierarchy, including well groups. While this unifies the handling of rate constraints for all levels of the hierarchy, it may require some data changes to be made when porting an Eclipse 100 data set to Eclipse 300. If a well group acts as a subsea manifold in Eclipse 100, the pressure losses in its outlet pipeline can still be represented by a VFP table. When this becomes an automatic choke in Eclipse 300, its outlet branch must be given a VFP table number of 9999 so the only pressure differential along it is that of the choke. In order to include the pressure losses in the rest of the pipeline it will be necessary to create an intermediate network node to split the pipeline into two sections: one section to represent the choke with a VFP table number of 9999, and the other section to represent the rest of the pipeline with the well group's original VFP table number.
- Eclipse 300 can apply automatic chokes to handle rate constraints in injection networks using the Extended Network model. Extended injection networks are not available in Eclipse 100.
- When using the Standard Network model, Eclipse 100 can only handle production rate constraints via the network at well groups, by declaring them as subsea manifolds. In the Standard Network model of Eclipse 300, production rate constraints may be handled through the network at any level of the group hierarchy by automatic chokes.
- Automatic chokes in Eclipse 300 can apply constraints on the flow of oil, water, liquid (oil + water) and gas directly. Other constraints (reservoir fluid volume rate, wet gas rate) are converted into constraints on these quantities using the latest available ratios, and so may not be met exactly.

Summary of keywords

RUNSPEC

Keyword	Description
NETWORK	Indicates that the extended network model is to be used, and sets array dimensioning data. The keyword should not be entered if the 'standard' network model is to be used for the production network.

SUMMARY section

The following keywords control output of data specific to the network option.

When using the extended network model, the list of names following these keywords must correspond to network nodes instead of [GRUPTREE](#) groups.

Refer to [Network option](#) in the *Eclipse Reference Manual*.

SCHEDULE section

Restriction	Keyword	Description
	BRANPROP	Defines the branches in an extended network.
Eclipse 100	COMPOFF	Switches off automatic compressors defined by NETCOMPA in the extended network model or GASFCOMP in the gas field operations model.
Eclipse 100	GNETDP	Dynamically adjusts the pressure of a fixed-pressure node with the aim of maintaining its production flow within specified limits. If the network is an extended network, the automatic choke facility (item 3 of keyword NODEPROP) provides a better method of controlling the flow rate.
	GNETINJE	Defines the structure of a standard network model injection network.
Eclipse 100	GNETPUMP	Defines automatic compressors or pumps in the Standard model production network. If the network is an Extended Network, use keyword NETCOMPA instead.
	GRUPNET	Defines the structure of a standard model production network.
Eclipse 100	NCONSUMP	Specifies gas consumption rates at extended network nodes.
Eclipse 100	NEFAC	Specifies efficiency factors for extended network nodes.
	NETBALAN	Sets parameters for the network balancing calculation.
Eclipse 100	NETCOMPA	Defines automatic compressors in the extended network.
Eclipse 300	NGASREM	Removes gas from a node in the extended network.
	NODEPROP	Defines node properties in the extended network.
	NWATREM	Removes water from a node in the extended network
Eclipse 100	OPTIONS	Item 33 sets the number of network balancing iterations in which a well can be revived after being closed because it cannot operate against its current THP. Item 57 invokes an alternative procedure for handling automatic choke calculations when a low flow target has been set.
Eclipse 300	OPTIONS3	Item 47 sets the number of network balancing iterations in which a well can be revived after being closed because it cannot operate against its current THP. Item 51 controls the frequency of recalculating the well potentials, which depend on the THP limits obtained from the network. Item 63 prevents extrapolation of the VFP tables for the branch pressure drop.
Eclipse 100	WAITBAL	Prevents automatic drilling and other production rule actions being performed until the network is balanced.
Eclipse 100	WNETCTRL	Selects whether well THP limits or flow rate limits should be held constant after the network has been balanced.
	WNETDP	Applies an additional fixed pressure drop between the tubing head of a well and the corresponding network node of its group.

Gas calorific value control

x	Eclipse 100
	Eclipse 300

The gas calorific value control option is only available in Eclipse 100; it is not available in Eclipse 300, but the GASWAT option in that simulator (see "[GASWAT option](#)") offers some of the same functionality. The purpose of this option is to enable you to control the mean calorific value of gas produced from the field, while also controlling the gas production rate. Production rate targets are handled by guide rate group control (keyword [GCONPROD](#)), and Eclipse automatically adjusts the guide rates so that the produced mixture has, whenever possible, the required mean calorific value. As an alternative to controlling the gas production rate, you may elect instead to control the overall energy produced (that is the calorific value times volumetric production rate).

Calorific value

The calorific value is a measure of the energy content of gas, that is the energy released when a given volume of the gas is burned. Eclipse measures the calorific value in units of kJ/sm³ (METRIC), Btu/Mscf (FIELD), J/scc (LAB).

The term 'quality' is also used to denote the energy content of gas, with a distinction between the volumetric quality and the molar quality. Volumetric quality is the energy content per unit volume, while molar quality is the energy content per mole of gas. Molar quality has units of kJ/kg-M (METRIC), Btu/lb-M (FIELD), J/gm-M (LAB). Conversion between the two definitions of quality can be performed assuming ideal gas behavior:

- 23.6903 sm³/kg-M (METRIC)
- 0.379506 Mscf/lb-M (FIELD)
- 23690.3 scc/gm-M (LAB).

As Eclipse 100 is not a compositional simulator, the calorific value of reservoir gas cannot be computed from its composition; you must enter it. The calorific value of the produced gas may either be specified independently for each well, or equated to the concentration of a gas phase tracer.

Controlling the mean calorific value

When the production rate of a group (or the field) is under guide rate control (keyword [GCONPROD](#)), its rate target is shared between the subordinate wells in proportion to their guide rates. It is also possible to control simultaneously the mean calorific value of the gas produced by the group, by automatically adjusting the well guide rates to give an appropriate mixture of high and low calorific value gas. The set of subordinate wells currently under group control are divided into two categories: those producing gas with higher calorific value than the target, and those with a calorific value lower than the target. All the wells in the second category have their guide rates multiplied by the same adjustment factor, calculated during each of the first [NUPCOL](#) iterations of the timestep to give the required overall calorific value. This method works provided that there is at least one subordinate well under group control in each of the two categories.

There could ultimately come a point when the required calorific value cannot be met, when there are no group controlled wells left in one or other of the two categories. It may be possible to restore the required mean calorific value by drilling a new well or by performing one of the other actions associated with the group production rules (keyword [PRORDER](#)). Otherwise there is an option to reduce the group's flow target in stages until wells straddling the required calorific value are brought back under group control. This option would therefore attempt to honor the calorific value target at the expense of the flow rate target.

Controlling the energy production rate

Instead of supplying a target for the group's flow rate, you may supply a target for its energy production rate (that is the gas production rate multiplied by its calorific value). Eclipse shares a group's energy target among its subordinate wells in proportion to their energy guide rates (that is the gas guide rates multiplied by their latest calorific values).

This facility may be used with or without a control target for the mean calorific value. If there is no such control target, the mean calorific value depends on the guide rates of the wells.

Adjusting guide rates to control mean calorific value

Wells under group control have their production rate targets set equal to their guide rates for the controlled phase multiplied by the controlling group's guide rate multiplier. But if the group's mean calorific value is also controlled to meet a specified target, those wells under group control whose calorific values are less than the target will have their guide rates multiplied by an adjustment factor α . Thus the group's overall production rate of the phase p under guide rate group control is given by

$$Q_{pG} = \sum_{W \in i} Q_{pW} + M_G \sum_{W \in g^+} G_{pW} + M_G \alpha \sum_{W \in g^-} G_{pW} \quad \text{Eq. 9.4}$$

where:

Q_{pG} is group G's production rate of the controlled phase p

Q_{pW} is well W's production rate of the controlled phase p

G_{pW} is well W's guide rate of the controlled phase p

M_G is the guide rate multiplier for group G

α is the guide rate adjustment factor for the set $W \in g^-$

$W \in i$ represents the set of subordinate wells operating under their individual control modes

$W \in g^+$ represents the set of subordinate wells on group control whose calorific values exceed the target

$W \in g^-$ represents the set of subordinate wells on group control whose calorific values are less than the target.

The group's guide rate multiplier is obtained from equation 9.4 by setting

$$Q_{pG} = Q_{pG}^{\text{targ}} \quad \text{Eq. 9.5}$$

where

Q_{pG}^{targ} is group G's target production rate of the controlled phase p.

This gives

$$M_G = \frac{Q_{pG}^{\text{targ}} - \sum_{W \in i} Q_{pW}}{\sum_{W \in g^+} G_{pW} + \alpha \sum_{W \in g^-} G_{pW}} \quad \text{Eq. 9.6}$$

The guide rate adjustment factor α is calculated from the condition that the mean calorific value of the group's gas production must be equal to a specified target,

$$\frac{\sum_W C_{gW} Q_{gW}}{\sum_W Q_{gW}} = C_{gG}^{\text{targ}} \quad \text{Eq. 9.7}$$

where:

Q_{gW} is well W's gas production rate

C_{gW} is the calorific value of the gas produced by well W

C_{gG}^{targ} is the target mean calorific value for group G.

The flow rate of a well under group control depends upon its guide rate and the group's guide rate multiplier,

$$\begin{aligned} Q_{gW} &= M_G G_{pW} (Q_g / Q_p)_W & \text{for } (W \in g^+) \\ Q_{gW} &= \alpha M_G G_{pW} (Q_g / Q_p)_W & \text{for } (W \in g^-) \end{aligned} \quad \text{Eq. 9.8}$$

where

$(Q_g / Q_p)_W$ is the well's latest known phase production ratio of gas to the controlled phase p.

Substituting equation 9.8 into equation 9.7 yields

$$\begin{aligned} M_G \sum_{W \in g^+} (C_{gW} - C_{gG}^{\text{targ}}) G_{pW} (Q_g / Q_p)_W - \alpha M_G \sum_{W \in g^-} (C_{gG}^{\text{targ}} - C_{gW}) G_{pW} (Q_g / Q_p)_W = \\ \sum_{W \in i} (C_{gG}^{\text{targ}} - C_{gW}) Q_{gW} \end{aligned} \quad \text{Eq. 9.9}$$

Simultaneous solution of equations 9.4 and 9.9 yields the guide rate adjustment factor α

$$\alpha = \frac{a \sum_{W \in g^+} (C_{gW} - C_{gG}^{\text{targ}}) G_{pW} \left(\frac{Q_g}{Q_p} \right)_W - b \sum_{W \in g^+} G_{pW}}{a \sum_{W \in g^-} (C_{gG}^{\text{targ}} - C_{gW}) G_{pW} \left(\frac{Q_g}{Q_p} \right)_W + b \sum_{W \in g^-} G_{pW}} \quad \text{Eq. 9.10}$$

where:

$$\begin{aligned} a &= Q_{pG}^{\text{targ}} - \sum_{W \in i} Q_{pW} \\ b &= \sum_{W \in i} (C_{gG}^{\text{targ}} - C_{gW}) Q_{gW} \end{aligned} \quad \text{Eq. 9.11}$$

Once α has been obtained from equation 9.10, the group's guide rate multiplier M_G can be obtained from equation 9.6. α of course must be constrained to be positive. A negative solution for α implies that the target mean calorific value cannot be achieved at the target flow rate. When both M_G and α are known, the group controlled wells can be given their production rate targets. If a well cannot achieve its rate target, it drops out of group control and becomes independently controlled (at its own rate or pressure limit), and the calculations for M_G and α must be repeated.

If the group G has subordinate groups that have been given group guide rates, then these groups and their guide rates will be dealt with as if they were wells in the above equations. The wells subordinate to these groups with guide rates are not processed as individual entities. Essentially, the subscript W in the above equations refers to the set of wells or groups with guide rates that can come under direct control of group G.

Action when calorific value target cannot be met

In order to meet the mean calorific value target, the set of wells under group control must include at least one well producing gas with a calorific value greater than the target, and at least one producing with less than the target calorific value. Eclipse can take certain remedial actions when there is insufficient high or low calorific value gas available.

Not enough high calorific value gas

If the group is meeting its production rate target, but the numerator of equation 9.10 is zero or negative, there are two actions that Eclipse may take:

- Increase the productivity of high calorific value gas sources.

If group production rules have been specified (keyword `PRORDER`), Eclipse performs the requested actions with the aim of obtaining more high quality gas. The `DRILL` action opens the next suitable well from the drilling queue that has a calorific value greater than the target value. The `REPERF` action selects the well with the largest sum of {connection factor * gas phase mobility * $(C_{gW} - C_{gG}^{targ})$ } of its remaining connections on `AUTO`, and opens the next connection in its `AUTO` queue. The `THP`, `RETUBE` and `LIFT` actions select the well with the largest value of $\Delta Q_{gW}(C_{gW} - C_{gG}^{targ})$, where ΔQ_{gW} is the increase in gas production rate that would result from the action. `COMP` actions are not triggered by a calorific value target. If the `PRORDER` keyword is not present, only the `DRILL` action will be performed, if there are wells in the drilling queue.

- Cut back the group's gas rate target to bring more high calorific value gas wells under group control.

If the productivity of high calorific value gas sources cannot be enhanced by the production rule actions described above, but there are some high calorific value gas sources that have dropped out of group control because they are unable to meet their share of the target, it could be possible to bring them back under group control by decreasing the group's gas rate target. If this action is requested, Eclipse cuts back the group's gas rate target by a specified fraction each time this situation arises, until the rate target is low enough to bring them back under group control. The need to maintain the target calorific value takes precedence over the group's rate target.

Not enough low calorific value gas

If the group is meeting its production rate target, but the denominator of equation 9.10 is zero or negative, there are two actions that Eclipse may take:

- Increase the productivity of low calorific value gas sources.

The production rule actions as described above are invoked, but this time with the aim of making more low calorific value gas available. The `DRILL` action looks for a well with a calorific value less than the target. The `REPERF` action selects the well with the largest sum of {connection factor * gas phase mobility * $(C_{gG}^{targ} - C_{gW})$ } of its remaining connections on `AUTO`. The `THP`, `RETUBE` and `LIFT` actions select the well with the largest value of $\Delta Q_{gW}(C_{gG}^{targ} - C_{gW})$.

- Cut back the group's gas rate target to bring more low calorific value gas wells under group control. This is performed as described above, if the production rule actions are not able to restore the group's mean calorific value.

Not enough gas to meet the group's rate target

If there are no wells at all under group control, the group's rate target will no longer be met. Eclipse may take the following actions:

- Apply the production rules to enhance gas production.

The production rules are applied in the usual way to maintain the group's production target (as described in keyword [PRORDER](#)), but with the following difference. If the group's mean calorific value is below the target value, Eclipse initially only considers the wells that are producing gas with calorific value higher than the group's target; if no such wells are available for action then the low quality wells are examined also. Conversely if the group's mean calorific value is above the target value, Eclipse initially only considers the wells that are producing gas with calorific value lower than the group's target; if no such wells are available for action then the high quality wells are examined also.

- Cut back the group's gas rate target to bring some high and low calorific value gas wells under group control.

This is performed as described above, if the production rule actions are not able to restore the group's mean calorific value.

Controlling the energy production rate

The energy production rate of the group is given by

$$E_G = \sum_{W \in i} E_W + M_G \sum_{W \in g^+} G_{eW} + M_G \alpha \sum_{W \in g^-} G_{eW} \quad \text{Eq. 9.12}$$

where:

E_W is $C_{gW}Q_{gW}$, the energy production rate from the well

G_{eW} is

$$C_{gW}G_{gW} \quad \text{Eq. 9.13}$$

is the well's energy guide rate, which is set equal to its gas phase guide rate multiplied by its calorific value at the beginning of the timestep.

When the control target refers to the overall energy produced rather than the gas flow rate, the group's guide rate multiplier is obtained from equation 9.12 by equating E_G with the target value. If there is **no** simultaneous control of the mean calorific value, the group's guide rate multiplier is

$$M_G = \frac{E_G^{\text{targ}} - \sum_{W \in i} E_W}{\sum_{W \in g} G_{eW}} \quad \text{Eq. 9.14}$$

where:

E_G^{targ} is group G's target energy rate

$W \in g$ represents the set of subordinate wells on group control.

If in addition there is also a target mean calorific value for the group, its guide rate multiplier is

$$M_G = \frac{E_G^{\text{targ}} - \sum_{W \in i} E_W}{\sum_{W \in g^+} G_{eW} + \alpha \sum_{W \in g^-} G_{eW}} \quad \text{Eq. 9.15}$$

When the controlled quantity is energy, equation 9.9 becomes

$$\begin{aligned} M_G \sum_{W \in g^+} \left(1 - \frac{C_{gG}^{\text{targ}}}{C_{gW}}\right) G_{eW} - \alpha M_G \sum_{W \in g^-} \left(\frac{C_{gG}^{\text{targ}}}{C_{gW}} - 1\right) G_{eW} \\ = \sum_{W \in i} (C_{gG}^{\text{targ}} - C_{gW}) Q_{gW} \end{aligned} \quad \text{Eq. 9.16}$$

Simultaneous solution of equation 9.12 and 9.16 gives the guide rate adjustment factor required to meet the target mean calorific value

$$\alpha = \frac{e \sum_{W \in g^+} \left(1 - \frac{C_{gG}^{\text{targ}}}{C_{gW}}\right) G_{eW} - b \sum_{W \in g^+} G_{eW}}{e \sum_{W \in g^-} \left(\frac{C_{gG}^{\text{targ}}}{C_{gW}} - 1\right) G_{eW} + b \sum_{W \in g^-} G_{eW}} \quad \text{Eq. 9.17}$$

where:

$$\begin{aligned} e &= E_G^{\text{targ}} - \sum_{W \in i} E_W \\ b &= \sum_{W \in i} (C_{gG}^{\text{targ}} - C_{gW}) Q_{gW} \end{aligned} \quad \text{Eq. 9.18}$$

Action when energy target cannot be met

When the group's energy target cannot be met, Eclipse applies the production rules in the usual way to maintain the target (see keyword [PRORDER](#)). However, since the control is on energy rather than gas, the rules are applied to enhance energy production rather than gas production. The `DRILL` action opens the next suitable well from the drilling queue, as usual. The `REPERF` action selects the well with the largest sum of {connection factor * gas phase mobility * calorific value} of its remaining connections on `AUTO`, and opens the next connection in its `AUTO` queue. The `THP`, `RETUBE` and `LIFT` actions select the well with the largest value of $\Delta Q_{gW} C_{gW}$, where ΔQ_{gW} is the increase in gas production rate that would result from the action. `COMP` actions are triggered in the usual way. If the `PRORDER` keyword is not present, only the `DRILL` action is performed, if there are wells in the drilling queue.

When the group also has a calorific value target, the production rule actions described in "[Not enough high calorific value gas](#)" are applied in order to maintain it.

Using the facility

Defining the calorific value

The calorific value of the produced gas may be defined in either of two ways:

- it may be set independently for each well
- it may be equated to the concentration of a nominated gas phase tracer.

```
WCALVAL
-- Well      Calorific
-- name      value
'PRODA*'    30000.0 /
'PRODB*'    26000.0 /
/
```

Defining it for each well provides a simple approach that is suitable when gases of different calorific value do not mix within the reservoir, for example when the field consists of a collection of isolated formations containing gases of different calorific value. The value for each well is specified in the SCHEDULE section with keyword [WCALVAL](#), for example:

Changes in calorific value with time can be accommodated ‘manually’ by re-entering the [WCALVAL](#) keyword.

Equating the calorific value to the concentration of a gas phase tracer provides a more general approach, which is recommended when gases of different calorific values may mix within the reservoir. A gas phase passive tracer may be defined (keyword [TRACER](#) in the PROPS section), and initialized to equal the calorific value in units of kJ/sm^3 ([METRIC](#)), Btu/Mscf ([FIELD](#)), J/scc ([LAB](#)). (See "[Tracer tracking](#)" for an account of how to define tracers.) Eclipse should then be told which tracer to use for the calorific value, with the keyword [CALTRAC](#), for example:

```
CALTRAC
CAL /
```

Eclipse then takes each well's calorific value from its produced concentration of this named tracer, unless the well's calorific value has already been defined with keyword [WCALVAL](#). If keyword [CALTRAC](#) is **not** used to nominate a gas phase tracer, all production wells should have their calorific values set with [WCALVAL](#).

Controlling the mean calorific value

The target mean calorific value for a group (or the field) is set with keyword [GCONCAL](#). The keyword also contains a flag to specify the action to be taken when this target cannot be met. The options are either to take no action (‘[NONE](#)’), or to cut back the group’s gas rate target successively until the mean calorific value target is honored (‘[RATE](#)’). You set the fraction by which the rate target is reduced each time.

```
GCONCAL
-- Group      Target      Action      Rate reduction
-- name      Cal Val      fraction
FIELD        29000.0      RATE        0.9 /
/
```

The facility can only work when the group is operating under guide rate control. Thus it must have a target gas production rate set for it with keyword [GCONPROD](#), or alternatively a target energy rate set with keyword [GCONENG](#) (see below).

Calculating guide rates on the basis of calorific value

Another way of influencing the mean calorific value is to calculate the individual well guide rates (and any group guide rates) as a function of their calorific values. Well guide rates that are not set directly with keyword [WGRUPCON](#) are calculated at the start of each timestep as a function of their production potentials.

By default the guide rates are set equal to their production potentials, but the keyword `GUIDERAT` can be used to specify the coefficients of a more general function of production potentials.

The keyword `GUIDECAL` can be used to make the guide rate calculation take account of calorific values also. It sets the coefficients A and B of a function β_W which depends on the calorific value C_{gW}

$$\beta_W = (1 + AC_{gW})^B \quad \text{Eq. 9.19}$$

When a well W has its guide rate calculated from its production potentials, the resulting guide rate will be multiplied by β_W . Thus by setting appropriate values for the coefficients A and B in keyword `GUIDECAL`, it is possible to either increase or decrease the well guide rates with increasing calorific value. For example, setting a positive value for B will bias the production towards wells with higher calorific values. The coefficient A cannot be negative.

If **groups** are given guide rates that are calculated from the guide rate formula in keyword `GUIDERAT`, their guide rates will also be multiplied by β_G , which is calculated as for β_W but using the groups' mean calorific values.

Controlling the energy production rate

As an alternative to controlling the gas production rate with `GCONPROD`, a target may be supplied for the overall energy production rate. The energy rate is simply the gas production rate multiplied by its calorific value, and is measured in units of kJ/day (`METRIC`), Btu/day (`FIELD`), J/hr (`LAB`). The energy rate target for a group (or the field) is set with the keyword `GCONENG`,

```
GCONENG
-- Group      Target
-- name       energy rate
  FIELD      1.0E9 /
/
```

The `GCONENG` keyword may be used with or without a `GCONCAL` target for the mean calorific value. If there is no `GCONCAL` target the mean calorific value will depend on the well guide rates. Note that a `GCONENG` target requires the gas production rate to increase as the mean calorific value falls. Consequently if the `GCONCAL` keyword is also used, the action flag in item 3 should normally be set to `NONE` to leave Eclipse free to increase the flow rate as necessary.

Economic limits

The keyword `GCALECON` allows minimum economic limits to be applied to a group's energy rate and to its mean calorific value. If a group violates either of these limits its subordinate producers will be closed, and optionally the run may be ended.

```
GCALECON
-- Group      Min Economic      Min Economic      End run flag
-- name       energy rate       calorific value
  FIELD      1.0E9 /           20000.0
  'G*'       5.0E7
/
```

Use with the gas field operations model

The `GCONCAL` keyword may be used in combination with the Gas Field Operations Model, if required (see "Gas field operations model"). It should only be applied to contract groups, and the contract groups must be under guide rate control rather than prioritization. There is no need to use `GCONPROD` to place contract

groups under guide rate control; this is done automatically when their swing and profile factors are entered, as long as [GCONPRI](#) is not used.

If the delivery contract is expressed in terms of energy production rate, instead of gas flow rate, use the [DCQDEFN](#) keyword in the `SCHEDULE` section.

```
DCQDEFN  
ENERGY /
```

This tells Eclipse that the DCQ values entered in keyword [GASYEAR](#), [GASPERIO](#) or [GDCQ](#) refer to a daily contracted quantity of **energy** instead of flow rate. You may then enter the energy rate DCQ values in place of the gas rate DCQ values, in the units appropriate to energy production rate: kJ/day (`METRIC`), Btu/day (`FIELD`), J/hr (`LAB`). If the [DCQDEFN](#) keyword is used it **must** be entered before any other Gas Field Operations keyword, including the [SWINGFAC](#) and [GSWINGF](#) keywords. Eclipse gives the contract groups a [GCONENG](#) target equal to the energy DCQ multiplied by the month's swing or profile factor.

Satellite groups

Energy can be produced or injected by satellite groups by specifying a gas production or injection rate **and** defining a non-zero calorific value for the gas. See the keywords [GSATPROD](#) and [GSATINJE](#) for more details.

Energy injection

Eclipse does not allow direct control over injected energy rates or net (that is the produced minus injected) energy rates. However, energy sales rates can be reported in the `SUMMARY` file, which represent the energy produced minus energy injected minus the energy content of any consumed gas. Injected energy can be taken into account by giving the gas injection wells a specified calorific value with the [WCALVAL](#) keyword. Alternatively, if a tracer is used to describe the energy content of the gas, the [WTRACER](#) keyword can be used to set the tracer value in the injection wells. Gas injection should not be applied when using the Gas Field Operations Model.

Output

An additional table is printed in the Well Reports showing the calorific value, energy production rate and cumulative energy produced for each well and group. Note that if the calorific value is equated with the tracer ' `CAL` ', these quantities may also be obtained from the Tracer Reports.

Refer to [Gas calorific value control option](#) in the *Eclipse Reference Manual*.

The energy delivery capacity and energy DCQ are available only in conjunction with the Gas Field Operations Model.

Restrictions

- If [GCONCAL](#) is used in a reservoir coupling master run (see "[Reservoir coupling](#)"), the master cannot control the production of individual wells in the slave reservoirs. Instead it will apply adjustment factors to the guide rates of the master groups. Also, production rules applied in the master reservoir cannot affect individual wells in the slave reservoirs.

Summary of keywords

SUMMARY section

Refer to [Gas calorific value control option](#) in the *Eclipse Reference Manual*.

SCHEDULE section

Keyword	Description
CALTRAC	Specifies which tracer is to be equated with the calorific value.
DCQDEFN	When using the Gas Field Operations Model, use this to tell Eclipse whether the DCQ targets refer to energy or gas rate.
GCALECON	Allows minimum economic energy rates and mean calorific values to be defined for groups and the field.
GCONCAL	Specifies the target calorific value for a group or the field.
GCONENG	Specifies the target energy rate for a group or the field.
GUIDECAL	Scales the guide rates calculated for each well by a function of the well's calorific value.
WCALVAL	Specifies the calorific value of gas at a well.